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LOGISTICS MODULE OVERVIEWS

Version 3.5.2

Product Guide



FRONTISPIECE

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Chapter 1

Advanced Customer Scheduling

Overview



Introduction

The Advanced Customer Scheduling module aims to cater for the special requirements of repetitive supply chain industries, for example in the automotive or white goods sectors.

The application allows you to manage and monitor complex supply contracts and automates the receipt and validation of scheduled data enabling you to intervene on an 'exemption only' basis.

Advanced Customer Scheduling provides the following functions:

- Master File Maintenance
- Schedule manipulation for entering/maintaining schedule data
- Facilities to cater for different types of schedules:
 - Releases
 - DCIs
 - RANs
- Flexible Comparison Processes
- Multiple Authorisation Periods
- Reconciled Requirements
- Maintenance of Contract Cumulatives
- Contract Pricing (including a zero pricing facility).
- Sleeper processes to receive, validate, compare, and accept schedule data

Master File Maintenance

The main file maintenance functions are:

- Company
- Customer
- Contract

These are repetitive functions which filter the default values down from company to customer, and from customer to contract level. This minimises the amount of keying involved in setting up a new contract whilst allowing the flexibility of overrides at the lower levels.

A contract can have two sets of parameters: one which determines how the system should handle 'Release' type schedules; and secondly a second set of parameters which determine how the system should process DCI or RAN type schedules.

Advanced Customer Scheduling also provides maintenance facilities for Contract Pricing and Comparison Rules.

The Comparison Process

You can specify comparison rules for monitoring variances between subsequent versions of schedules against a contract.

Comparison rules can have five different types of rule element attached to them, with corresponding date ranges. The combination of element/date range is unique on the rule. The elements determine where the factor which is being compared should be measured, for example the quantity within the date range, or the quantity at the date range end.

Each combination of element/date range on a rule can have tolerance values (positive and/or negative) for: the quantity difference; the percentage difference; and the sales value difference between subsequent schedules.

Each rule element/date range set of tolerances has a corresponding severity level between 0 and 9. A variance which is outside the tolerance with a severity of 4 or over will cause the new schedule data to fail comparison and prevent acceptance. A variance of severity less than 4 will record the failure as an exception without preventing acceptance.

Rules may then be attached to Contract Release and /or contract DCI defaults via Contract Maintenance. Rules are attached to contracts in combination with comparison methods, which tell the system which previous version of a contract's schedule to use as a base for comparing the new data.

Reconciled Requirements

When new schedule data is accepted into the current schedule files, it goes through a process which creates reconciled requirements. In brief, this is the process of converting the raw data, as sent by the customer, into requirements to be fed into Advanced Shipping.

This involves adjusting the data by the following factors (as applicable), using the reconciliation and arrears processing methods linked to the contract:

- Goods in transit.
- Arrears from previous schedules
- Ship lead time adjusting (converting required by dates into ship by dates).
- Delivery frequency consolidation. This is consolidating deliveries according to customer and stock room calendars.
 - when do they require the goods?
 - what days can we ship to them?
 - what days can they receive shipments?
- Pack rounding. Contracts may be set up to perform pack rounding with an associated pack rounding percentage. This will round the shipment quantity up or down to the nearest pack size on the basis of the pack rounding percentage.
- After reconciling a new schedule, any shortfall or over resulting from pack rounding is held against the contract for reconciling with the next version of the schedule.



Note: You may also explode a single 'raw' requirement into multiple 'reconciled' requirements where the customer sends, for example, monthly requirements.



Note: RANs are only ship lead time adjusted, they are never pack rounded.

Contract Cumulatives

The system stores and monitors the following data:

- Cumulative despatches against the contract.
- High water cumulatives. These are the highest quantity of requirements by authorisation period that the customer has ever committed to. Particularly important for firm and material authorisation periods where there may be a contractual obligation against the customer to pay for the high water values.
- Engineering change levels. The system maintains a log of valid engineering change revision levels against the contract, with valid dates/quantities. The current engineering change level is held on the contract and there is a facility for 'holding' new schedules with a change of engineering levels.

Contract Pricing

You can choose between:

- Contract pricing, using effective and charge from dates, with a retrospective invoicing report facility.
- Normal pricing which will price despatches out of Advanced Shipping.

In addition, contract pricing will permit zero pricing for samples or free of charge shipping during pricing negotiations.



Note: The pricing is established at company level.



Note: If you have installed Advanced Pricing only, contract pricing can be installed for Advanced Customer Scheduling.

Schedule Manipulation

You can review and/or maintain new schedule data and send the resulting schedule through the appropriate sleeper processes.

Schedule manipulation may be used to manually enter schedule data which has arrived by, for example, fax. Alternatively, existing (current) schedule data may be maintained. In this case the system will copy the data to the new files to allow manipulation. The manipulated data may then be sent back through the sleepers to update the current schedule data.

You can also review comparison exceptions, i.e. see which comparison rule tolerances have been exceeded and with what severity. It will also allow you to rerun the comparison process on-line, following manipulation.

You can also produce reconciled requirements on-line and review them prior to acceptance.

Schedule Maintenance caters for new schedule data which may still be in the Application Interface (EDI) files, in addition to catering for data in the 'current' files. Hence schedules which have failed the EDI receive and validate sleeper may be reviewed and maintained.

Error Messages

Error messages are reported extensively during schedule manipulation. In many instances it will be possible to either amend the schedule data, or change some of the contract values to correct an error.

All error messages that arise from the customer schedule validation, comparison and acceptance processes have a default severity assigned to them. These default severities can be overridden using the Maintain Message Severities facility on the Utilities menu.

If the actual severity is less than 4, the error message is treated as a warning only. If it is greater than or equal to 4 then it is treated as a normal error message, and you will not be able to accept changes to the screen until such errors are corrected.

Error Correction

This section details the error conditions that are currently trapped by the acceptance process, and describes how schedule manipulation will be used to correct the error.

Error	Option	Action
Unable to match to an active contract.	Process EDI Error Transactions	Select the appropriate schedule, and use the contract match facility. Possibly enter a value in the link field.
Contract already has a new schedule, of this type.	Process EDI Error Transactions	Delete the schedule that is already in the new files, or take the appropriate action to allow it to

Error	Option	Action
		progress to 'current'.
Duplicate release.	Process EDI Error Transactions	Delete the duplicate.
Schedule amendment - previous issue details do not match.	Process EDI Error Transactions	Change the previous document information to match the current schedule, or leave this schedule until the version to which it relates does become current.
Schedule out of sequence.	Process EDI Error Transactions	Delete the schedule, since a later version has already become current.
Invalid effectivity dates.	Contracts Maintenance Process EDI Error Transactions	The invalid dates can be changed, or the contract duration can be extended to cover the specified period.
Manual review.	Process New Schedules Process EDI Error Transactions Schedule Maintenance	You will confirm that the schedule can progress to the next stage in the process.
Schedule amendment (RAN), the original RAN does not exist or has been completed.	Process New Schedules	Delete the RAN amendment and reconcile situation with customer, then update the schedule requirement.
Date validation.	Process New Schedules	Correct the date in error.
Time Validation.	Process New Schedules	Correct the time in error.
Date of new release.	Process New Schedules	Correct the date.
Unknown despatch note number.	Process New Schedules	Correct the customer quoted despatch note numbers. A list of despatches can be displayed.

Error	Option	Action
Invalid final delivery address code and additional destination, consignee, freight forwarder, or carrier.	Process New Schedules	Change Schedule Header Additional Details Screen.
Invalid authorisation period end dates.	Process New Schedules	Correct the date.
Invalid authorisation period cumulatives.	Process New Schedules	Correct the cumulatives.
Invalid article status code.	Process New Schedules	Change Schedule Header Additional Details Screen.
Design revision number.	Process New Schedules	Change Schedule Header Maintenance Screen.
Invalid engineering change number.	Process New Schedules	Change Schedule Header Maintenance Screen.
Invalid first date display format.	Process New Schedules	Change using Release Details Maintenance - More Keys Screen.
Invalid authorisation status.	Process New Schedules	Is only checked, and will only need correction, if the authorisation method on the contract is '01'. If correction is required, this can be amended using Release Details Maintenance - More Keys Screen.
Invalid call off reference.	Process New Schedules	Change the call off reference using Release Details Maintenance - More Keys Screen.
Any comparison errors.	Process EDI Error Transactions Schedule Maintenance	Correct by manipulating the schedule header in conjunction with the schedule detail requirement.

Sleeper Processes

There are three sleeper processes which are run as background tasks.

Validation Sleeper

The first sleeper is the validation sleeper. The data queue which feeds this program contains EDI records, and/or manual schedule manipulation records.

The sleeper receives data in its raw state, and performs fundamental validation (such as identifying the contract). Where it can, data from the Application Interface (EDI) files are transferred to the new Advanced Customer Scheduling data files.

Comparison Sleeper

The second sleeper is the comparison sleeper. This takes the new Advanced Customer Scheduling data and compares it to previous versions of schedule data for the same contract and schedule type.

This is performing more critical validation as specified by the comparison rules, methods and elements specified on the contract. Its primary function is to highlight inconsistencies.

An example of this type of check is frontloading. This is where the customer tries to shift an unforeseen increase in demand up the supply chain. As authorisation periods are constantly moving and data is generally stated as cumulatives, it could be quite difficult to spot such circumstances manually.

Acceptance Sleeper

The third sleeper process is the acceptance sleeper. This takes scheduled data which has passed the first two sleeper processes and updates the Advanced Customer Scheduling current schedule data.



Note: The current data will contain only the schedule details that are effective in the system. All previous schedules will be kept in the History files until purged by the relevant option.



Note: This means that for a contract operating releases and DCIs, there will be two sets of data in the current data files.

As new sets of records come through the first sleeper, they update the new schedule data files. The acceptance sleeper writes the current set of schedule records to the history files, then moves the new records into the current files.

The acceptance sleeper also takes the newly accepted raw requirements and creates reconciled shipping requirements out of them.

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Chapter 2

Advanced Shipping Overview



Introduction

Advanced Shipping provides the following functions, which are described more fully later:

- Master file maintenance.
- Build and maintain shipping consignments.
- Confirm despatches and consignments.
- Consignment processing.
- Consignment stocks and self billing.
- Fastpath shipping and confirmation.
- Automatic pack and pallet creation, with override capability.
- Sourcing from multiple stockrooms.
- Stock reservation.
- Special rules for shipping receipt advice notification (RAN) requirements.
- Load simulation facility.
- Multiple invoice consolidation levels.
- ASNs and ANSIX12 856 ASNs.

Advanced Shipping was designed as a separate module as it is integrated into other modules in addition to Advanced Customer Scheduling.

This module is an advanced business module and this product guide assumes that you are familiar with the following:

- A knowledge of how the server operates.
- A knowledge of how the JBA modules work.
- An understanding of Customer Scheduling in a repetitive manufacturing environment.
- Knowledge of how goods are collated and despatched.

Master File Maintenance

The Advanced Shipping module aims to cater for the special requirements of repetitive supply chain industries, for example in the automotive or white goods sectors.

The module allows you to manage and monitor complex supply contracts. Advanced Shipping provides maintenance functions which are explained in more detail later in this section:

- Company
- Customer
- Contract

These are repetitive functions which filter the default values down from company to customer, and from customer to contract level. This minimises the amount of keying involved in setting up a new contract whilst allowing the flexibility of overrides at the lower levels.

Advanced Customer Scheduling calls Advanced Shipping Maintenance Default screens when you enter new contracts and customers. These defaults are also maintained within Advanced Shipping. Other key Advanced Shipping master files are:

- Name and address. These define the EDI trading partner details for sending Advanced Shipping Notifications (ASNs) to be sent by EDI.
- Calendar files. These define calendars which may be attached to the shipping company stockrooms and the customer contract.
- Despatch profiles. These define a sequenced list of stockrooms for multi-sourcing of Advanced Customer Scheduling contracts.
- Pack and pallet information. These define the parameters of a particular pack type; a particular pallet type and the relationship of a customer / item with a specific pack and pallet.

Build and Maintain Consignments

This function allows you to create or maintain consignments on which to ship to a specified customer on a specified day. It is the main area of the Advanced Shipping processing function.

The Build Screen

This screen displays an item or RAN per screen, and the screen is divided into three distinct areas:

- The top section of the screen shows the required quantity to be shipped to the customer on or before the specified date. This is driven from the reconciled requirements file which is updated by the Advanced Customer Scheduling acceptance sleeper.
- This section of the screen also shows quantities which have already been reserved or built on to a consignment for the requirement, and what outstanding action quantities there are. It also highlights, in red, if there is not enough stock available to satisfy the requirement.
- The second portion of the 'build' screen covers sourcing the requirement. It lists either the single stockroom or despatch profile stockrooms associated with the relevant contract. It displays how much stock has been used for this requirement and how much is still available.
 - 'Allocated unbuilt', stock has been reserved but not yet built on to a consignment;
 - 'Allocated and built' has been allocated from the stockroom(s) and built on to consignment(s)
- Normally supply companies will not reserve stock without building it on to a consignment simultaneously, but this facility would be used to reserve stock against a RAN until sufficient stock is available to build a whole RAN onto a consignment.
- The third section of the 'build' screen lists consignments for the specified customer on the specified date. It will display any stock for the requirement being viewed which has already been built on the consignment(s).
- Default action quantities are suggested on this screen; sourcing is suggested against the available stock in priority sequence until either the requirement is satisfied or no stock remains. Suggested stock allocations are in pack rounded quantities where appropriate. Action quantities can be forced through which will make available stock negative (this is assuming the stockroom has been set up in Inventory Management to accept negative stock).

Confirm Despatches and Consignments

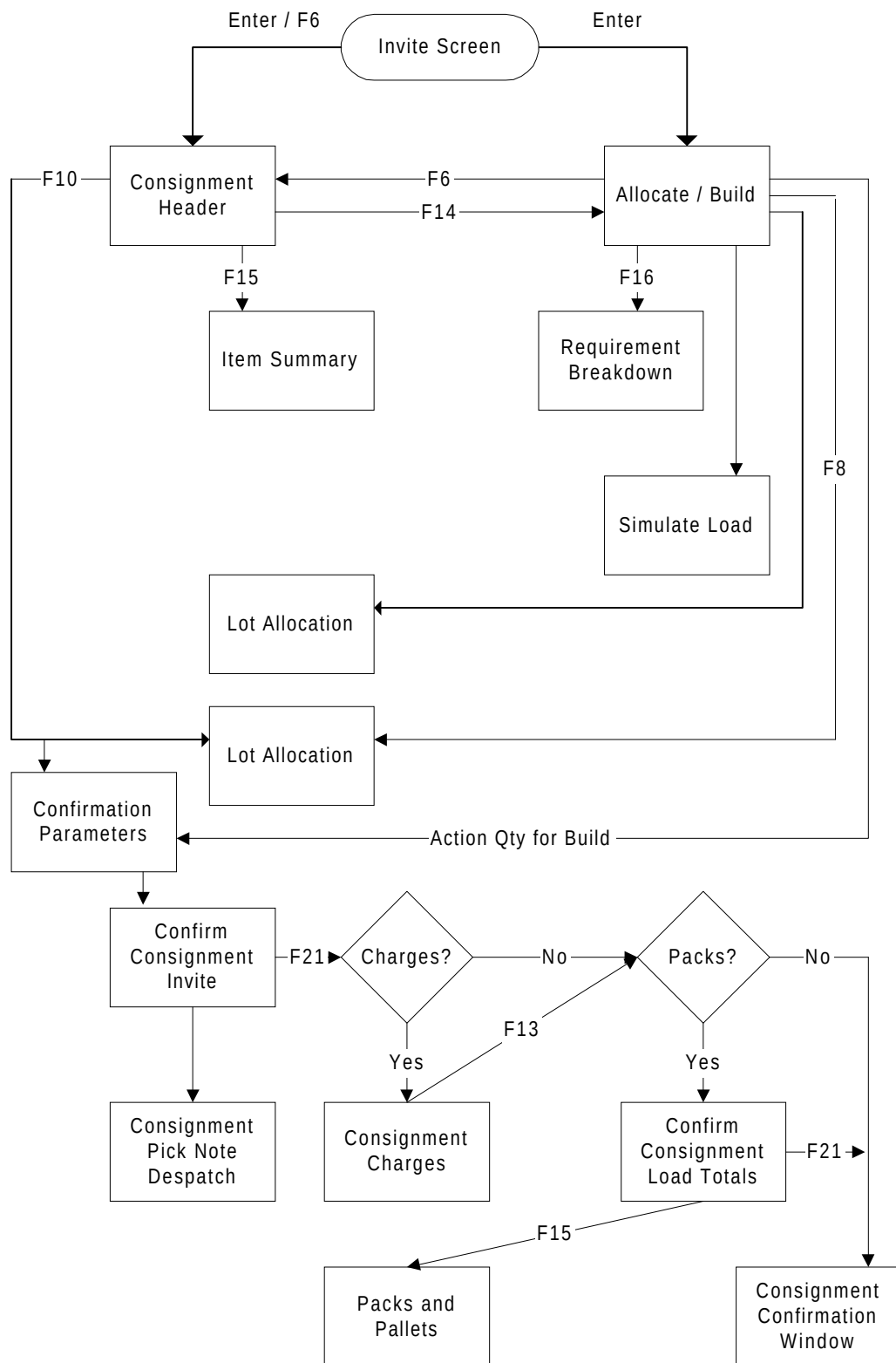
When you have built everything you want on a particular consignment, you can, as required, print picking documentation. You can then confirm despatch against the consignment. This function can be accessed through the same build function, or via a separate menu option: Consignment Confirm Despatch.

In despatch confirmation, the invite screen summarises the despatches on a consignment by stockroom and allows you to fully confirm all despatches for one or more stockrooms with a function key. Alternatively you can override despatch quantities by stockroom.

Once despatches have been confirmed, you can confirm the total consignment, and charges may be added at consignment level. Where pack and pallet information is required, this will be generated when you first confirm the consignment. Totals will be displayed and overrides are allowed.

You can drill down to review and, if necessary, maintain pack and pallet information or simply accept the calculated values. Final consignment confirmation creates the records necessary for producing Advanced Shipping Notes (ASNs).

Consignment Processing



Consignment Invoices

Shipments may be invoiced either according to normal consolidation rules (daily, weekly, monthly) or at one of the lower level Advanced Shipping consolidation levels:

- One invoice per consignment.
- One invoice per consignment / contract.
- One invoice per consignment / contract / RAN.

Consignment Notes

Confirm Consignment produces a consignment note which summarises despatches for the consignment regardless of the stockroom. (Pick notes, if required, will be produced for each stockroom).

The consignment note will give brief pack and pallet details where appropriate.

Consignment Stocks and Self Billing

Consignment Stocks

Consignment stock processing is used when a supplier transfers goods to a holding area, generally known as a consignment warehouse, instead of directly to a customer. The goods remain the property of the supplier and are referred to as consignment stock. At the Advanced Shipping confirmation stage no updates are made to inventory, invoice records are not created and the shipment is a transfer order and not a sales order.

The customer has direct access to goods in the consignment stockroom and takes them as they are required. The customer then notifies the supplier that they have consumed these goods and upon receipt of this notification inventory is updated and an invoice is raised.

Self Billing

Self billing allows the customer to reduce the cost of matching invoices against consignment notes by calculating the invoice they are expecting from the supplier. The customer notifies the supplier of the invoice details that it has raised against itself.

Consignment stock processing and self billing are often performed together as the customer notification of stock consumption can be used to match a self billed invoice.

Fastpath Shipping And Confirmation

Fastpath shipping allows you an alternative way of creating and confirming a consignment. The consignment build and confirmation processes provide a comprehensive system but if you do not require a wide choice of options then Fastpath shipping offers you a limited but more simplified process.

The Fastpath shipping option allows you to create and confirm a consignment in one step. The Fastpath confirmation option allows you to amend details and confirm existing Fastpath consignments.

When a consignment has been confirmed, inventory values are updated and invoice records are created.

However, before using the Fastpath shipping option, please note that:

- Fastpath shipping uses the single stockroom or first stockroom entered in the Despatch Profile option from the Contract Defaults screen, and not multiple stockrooms.
- Fastpath shipping uses a simple separate packaging mechanism and not multilevel packaging.
- Consignments created using Fastpath shipping are not suitable for use with the ASN options. It is recommended that the send ASN field in the customer defaults maintenance screen is set to no.
- Fastpath shipping does not integrate with the Warehousing module.

Pack and Pallet Details

- A pallet is an outer container; it may be a large crate which packs sit in, or a traditional pallet which packs sit on top of.
- A pack is the inner container an item is shipped in and the normal pack quantity is used in pack rounding.

The system uses some basic rules to produce pack and pallet details for the goods on a consignment. The system creates details for each combination of:

- Item
- Consignment
- Contract
- Stockroom
- RAN

The system assumes different lot numbers can be mixed on a pallet, but not in a pack, and that different RAN's cannot be mixed on a pallet, or in a pack. You can later manipulate the pack and pallet details.

Specifications can be made for pack quantities, number of packs per pallet layer, a minimum number of layers and a maximum number of layers, whether complete pallet layers must be shipped, and whether odd packs can be shipped.

The system can generate dummy packs (empty boxes) to fulfil the complete layers flag. There may even be complete dummy layers up to the minimum number of layers.

Pack, pallet and item weights and volumes are accumulated to calculate consignment totals, which you can amend before you confirm the despatch.

Shipping RAN Requirements

Advanced Shipping recognises Advanced Customer Schedule RAN requirements and applies different validation rules. They are displayed individually on the build screen where other requirements are summarised by item. They may be split or part sourced but they cannot be part built. The system will only allow a RAN to be built on to a single consignment, in its entirety. In addition, a warning message is presented when you put multiple RANs on the same consignment.

Load Simulation Facility

During the build process you can assess the impact of building a particular action quantity on to a consignment by using the simulate load function.

This facility runs through the packs and pallets routine normally performed by Consignment Confirm Despatch and summarises the weights, volumes, and numbers of pallets required by what is already built onto the consignment.

The system then simulates the pack and pallet requirements for the current action quantity and shows the resulting totals for the consignment. The action quantity may be manipulated and the results recalculated.

ASN and ANSI X12 856 Maintenance and Send

ASNs are electronically transmitted messages (usually via EDI) which provide advance information about the configuration of shipments or consignments. Normally an ASN will contain hierarchical information relating to the shipment itself; the orders or schedules on the shipment; the items contained in the shipment; and how those items are packaged.

Typically the ASN is transmitted to the delivery customer but in certain circumstances may be transmitted to other parties with an interest in the shipment, such as the carrier.

Original equipment manufacturers (OEMs) are increasingly placing a requirement upon their suppliers to transmit ASNs to them, often within a specified time span of the shipment actually leaving the suppliers plant. This time span is as small as fifteen minutes for some OEMs.

Many manufacturers wish to use the information contained in the ASN to streamline their goods receiving processes. The ASN information may include details of bar-coded packaging labels. Upon receipt, the bar-coded labels are read via hand held scanners and the goods contained within are automatically checked in to stock against expected receipts information provided by the ASN.

The screens relating to item details and their associated packaging details are optional. You can choose how deep into the maintenance options you wish to go by either accepting the packaging information already generated (or not if pack details are not required), or delving deeper and maintaining the packaging details manually.

The ANSI X12 856 ASNs is of increasing importance if you trade in, or with clients in, North America. It behaves in a similar way to a European ASN but differs in its use of the hierarchical levels and does not have a fixed structure. It therefore allows more specific data to be included. Hierarchical levels can be set up for each shipment in any sequence and levels can be omitted. The sequence is agreed between the customer and the supplier.

The ANSI X12 856 ASN is only available if your system has been specifically set up to use this facility.

Record Locking Considerations

- Company and calendar maintenance require exclusive use of the system.
- The following simultaneous updates are prohibited via record locking:
 - Maintaining contract shipping details whilst processing them in Build & Maintain Consignments.
 - Maintaining a customer whilst processing one of their contracts in shipping processing.
 - Maintaining despatch profile / pack /pallet / item whilst processing with them in shipping processing.



Note: The Advanced Shipping company, customer, and contract maintenance options will automatically be linked to the corresponding Advanced Customer Scheduling options if this is available, to prevent users having to enter details from two separate menu options when initially creating new customer and contract processing defaults.

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Chapter 3

Forecasting Overview



Introduction

Forecasting provides a method of avoiding costly stockouts and inventory excesses which inhibit cash flow. This is particularly true when the forecasts are used to drive the creation of supply plans within DRP and MPS/MRP modules.

The Forecasting module contains seven simple forecasting methods. They are methods which people use every day, such as:

- Whatever was sold in the last three periods divided by three is what will be sold in the next period.
- Whatever was sold in the last twelve or thirteen periods divided by twelve or thirteen is what will be sold in the next period.
- Whatever was sold in the corresponding next three periods of last year divided by three is what will be sold in the next period.

These are simple forecasting methods. Forecasting selects the one best method to forecast this item, or group of items, at this moment in time. This is done through a process of simulation, where the module goes back in time, evaluating each formula for each item or group of items.

The module projects these simulated periods, using each of the methods, pretending they did not happen. Whichever forecast method best projects these periods that have already happened, by comparing their forecasts with actual data, is the one chosen to predict the future.

Forecasting goes through this simulation every time you run the period-end forecast. It is a dynamic simulation. To pick the one forecast method that worked best, the module calculates a forecast error percentage. The closer to zero the error, the better the method. The actual sales are extracted from the Sales Analysis module.

Relationship to Other Modules

Forecasting is a fully standardised module, and as such operates under the control of System Manager. Forecasting is one of the Group of Advanced Modules which extend the functionality of the Base Modules.

The modules which are pre-requisite for the operation of Forecasting are:

- Inventory Management.
- General Ledger.
- System Manager.

Item information is extracted from the Inventory module. Sales History can be extracted for these items from the Sales Analysis module.

However, if these modules are not installed, it may be possible to obtain this information from equivalent modules. Contact your software support organisation for further details.

In addition to this, you can use Forecasting in conjunction with, and to enhance, the following modules:

- Sales Analysis.
- Distribution Requirements Planning.
- Material Requirements Planning.
- Master Production Scheduling.
- Service Management.

Once Forecasting has generated forecasts, and these have been reviewed/manipulated, you can transfer into DRP and/or Material Requirements Planning/Master Production Scheduling. This can be done at Minor Item Group or Item Level.

Alternatively, this Requirements file could be downloaded to a PC for further evaluation.

Forecasting Configuration

You can operate Forecasting for a number of companies, the characteristics of each being maintained on control files. Forecasting does not use control data stored for any other module. The policies which can be controlled for each company are:

- Period (month) descriptions.
- Backup parameters.
- Forecast Formulas descriptions.
- Interface defaults.
- Sales Analysis database details.
- 12 or 13 periods forecasting.

Forecasting can take place at two levels - Minor Item Group (as defined in Inventory Management) and Item. Every group or item must have a profile record in this module.

The Item code used in Forecasting must match the one in Inventory Management. If sales history is required then it is extracted from Sales Analysis files.

Forecasting enables the creation and transfer of forecasts for individual locations or groups of locations. This is achieved by defining Forecasting Models.

A Forecasting Model is any logical grouping of Inventory stockrooms. This may represent a Central Warehouse, a Distribution Centre, a Manufacturing facility, or any other meaningful association.

You can adapt Forecasting to use either 12 or 13 periods per year. Options exist within the Forecasting module to extract items from Inventory Management and sales quantities from Sales Analysis for selected Models. Depending on whether 12 or 13 period forecasting is selected, a thirty-six or thirty-nine period sales history is extracted and stored in the module. This history is also aggregated and stored at Minor Item Group level.

Forecasts are generated and maintained for selected Models.

Forecasts can be transferred into DRP and/or Production modules at Minor Item Group or Item level for selected Models.

In addition to the period spread profile in Forecasting, the spread profile can be extended to include seasonal profiles from annual to period, period to weekly, and weekly to daily when transferring forecasts into DRP and/or Production modules.

Backup routines are built into the Forecast Period-End, and may be selected, with relevant parameters on the Company Profile.

Forecasting Methods

Seven forecasting methods are available within this module. These methods vary slightly according to whether 12 or 13 periods per year have been selected. Below is a list of abbreviations used in the forecast analysis:

- NPF - Next Period Forecast.
- L3P - Sum of last 3 periods sales.
- L12P/L13P - Sum of last 12 or 13 periods sales respectively.
- 2NPLY - Two times next period sales last year.
- NPYBL - Next period sales year before last.
- N3PLY - Sum of next 3 period sales last year.
- L3PLY - Sum of last 3 period sales last year.
- LP-5 - Sales 6 periods ago.
- LP-4 - Sales 5 periods ago.
- LP-3 - Sales 4 periods ago.
- LP-2 - Sales 3 periods ago.
- LP-1 - Sales 2 periods ago.
- LP - Sales last period.

Each method corresponds to a separate computer program. Addition of further forecasting methods require new programs to be written. Contact your software support organisation for further details. However, the description held against any of the existing seven forecasting methods may be amended on the Company Profile. The following descriptions relate to 12 periods a year.

Forecast Method A

Formula: $NPF = L3P / 3$

Method A calculates the next period forecast as the sum of the last 3 period sales divided by 3.

For example: If the Item had the following sales history for the last 36 periods, and method A was chosen, the forecast would be as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	751	767	779	766	771	772	770	771	771	771	771	771



Note: The first forecast is a third of periods 10, 11 and 12 and the second forecast is a third of periods 11, 12 and 01.

From the above example,

NPF for Period 1 = $(702 + 733 + 818)/3 = 751$

Forecast Method B

Formula: $NPF = L12P/12$

Method B calculates the next period forecast as the sum of the last 12 period sales divided by 12.

For example: If the Item had the following sales history for the last 36 periods, and method B was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	864	840	815	806	804	801	805	809	796	799	808	814



Note: The first forecast is a twelfth of periods 1 to 12, and the second forecast is a twelfth of periods 2 to 12 and forecast 01.

NPF = $(1161 + 1137 + 926 + 820 + 841 + 761 + 752 +$

for $961 + 759 + 702 + 733 + 818)/12$
 Period
 1 $= 1037/12 = 864.25$



Note: If 13 periods have been specified, then the formula is based on L13P/13.

Forecast Method C

Formula: $NPF = (2NPLY + NPYBL)/FACTOR$

Method C calculates the next period forecast as twice next period sales last year, plus next period sales year before last, divided by a factor.

To determine factor, calculate $NPLY/NPYBL$.

If this is greater than or equal to 1.5 use 2 as the factor.

If this is less than or equal to 0.25 use 4 as the factor.

If the result falls between 0.25 and 1.5 use 3 as the factor.

For example, if the Item had the following sales history for the last 36 periods and formula C was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818

Forecast	1194	1042	849	777	874	726	722	873	799	644	776	911
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Note: The forecast for period 01 is calculated as follows:

NPLY = 1161
 NPYBL = 1259
 $NPLY/NPYBL = 1161/1259$
 NPF = 0.922 therefore the factor will be 3
 (that is, value between 0.25 and 1.5)
 $= ((2 \times 1161) + 1259)/3 = 1193.66$

Forecast Method D

Formula: $NPF = (2NPLY + NPYBL)/3$

Method D calculates the next period forecast as twice next period sales last year plus next period sales year before last divided by 3.

For example, if the item had the following sales history for the last 36 periods and method D was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818

Forecast	1194	1042	849	777	874	726	722	873	799	644	776	911
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Note: The forecast for period 01 is calculated as follows:

NPLY = 1161

NPYBL = 1259

NPF = $((2 \times 1161) + 1259) / 3 = 1193.66$

Forecast Method E

Formula: $NPF = N3PLY / 3$

Method E calculates the next period forecast as the sum of the next 3 period sales of last year divided by 3.

For example, if the item had the following sales history for the last 36 periods, and method E was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	1075	961	862	807	785	825	824	807	731	751	875	951

The forecast for period 01 is calculated as follows:

$$\text{NPF} = (1161 + 1137 + 926)/3 = 1074.66$$

Forecast Method F

Formula: $\text{NPF} = ((\text{N3PLY} / \text{L3PLY}) \times \text{L3P}) / 3$

Method F calculates the next period forecast as the sum of the next 3 period sales last year divided by the sum of the last 3 period sales last year, times the sum of the last 3 period sales divided by 3.

For example, if the item had the following sales history for the last 36 periods, and method F was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	974	778	653	602	553	576	589	589	518	515	564	624



Note: The forecast for period 01 is calculated as follows:

$$\text{N3PLY} = 1161 + 1137 + 926 = 3224$$

$$\text{L3PLY} = 528 + 861 + 1096 = 2485$$

$$\text{L3P} = 702 + 733 + 818 = 2253$$

$$\text{NPF} = ((3224/2485) \times 2253 / 3 = 974.336$$

Forecast Method G

Method G calculates the next period forecast as sales 6 periods ago times 1, plus sales 5 periods ago times 2, plus sales 4 periods ago times 3, plus sales 3 periods ago times 4, plus sales 2 periods ago times 5, plus sales last period times 6, all divided by 21 (the number of periods calculated).



Note: This formula equates to a simple single-level exponential smoothing approach with a smoothing factor of around 0.15.

For example, if this item had the following sales history for the last 36 periods and formula G was chosen, the forecast would be calculated as shown:

Period	01	02	03	04	05	06	07	08	09	10	11	12
3 Years ago	402	912	704	588	762	581	728	1004	698	412	802	879
2 Years ago	1259	853	695	690	939	655	661	697	880	528	861	1096
Current year	1161	1137	926	820	841	761	752	961	759	702	733	818
Forecast	778	775	770	773	776	776	775	775	775	775	775	775

The forecast for period 01 is calculated as follows:

$$\begin{aligned} \text{NPF} &= [(752 \times 1) + (961 \times 2) + (759 \times 3) + \\ &\quad (702 \times 4) + (733 \times 5) + (818 \times 6)] / 21 \\ &= 16332 / 21 = 777.71 \end{aligned}$$

Maintenance of Reference Data

This module uses minimal primary data from the Inventory Management module. You must, however, enter additional information to obtain maximum benefit from the module, such as:

- A generic search on Item code.
- Minimum use of codes and where they are employed a prompt is always available to assist in selecting the correct one.

Selection is normally from a displayed list. Information for the following may be maintained:

- Models.
- Group/Item Profiles.
- Demand History.
- Spread Profiles.

Forecasting Models

You can create and transfer forecasts for individual (or groups of) locations using Forecasting Models. For each Forecasting Model, the following details are maintained:

- Model Code.
- Model Description.
- Model Type.
- Target Model - Any DRP Distribution Centre, or Production Model, dependent on the entered Model Type.
- Simulation Period.
- Stockrooms - Any valid Inventory-defined stockroom directly attached to the entered DRP DC/Centre or Production Model.

Group/Item Profiles

To process a Model, a profile must be maintained for each Item or Group of Items in the Model. Group/Item profiles contain details including:

- **Model Code** - assigns the Item or Group of Items to a Forecast Model. Items and Groups of Items can be assigned to several Models. This enables forecasts created from the same Sales Analysis data to be maintained and transferred separately.
- **Simulation Period** - the number of months the month end run simulates when selecting the most accurate forecast method.
- **Locked in Forecast Method** - a specific forecast method can be locked in for the group or item. The lock in forecast will be used to calculate the next 12 or 13 periods respectively, (no simulation is performed). Alternatively, the item can be flagged as manually forecasted. In this case the Period-end will ignore the item. If no locked in or manual method is specified, all forecast methods will be evaluated.
- **Conversion Factor** - this is an equivalent units conversion factor which allows you to convert items sold in different pack sizes into equivalent units. The equivalent units are accumulated to Item group level on all reports that print forecast quantity.
- **Marketing Masks** - these direct the program to adapt the calculated forecast for a selected group of items or an individual item so as to correspond to a changing market situation which is not evident in the past. They are particularly useful for promotional or new items. The marketing masks consist of 12 or 13 sets of adjustment factors which correspond to the next 12 or 13 periods. These factors are multiplied by the original forecast.

Demand History

For each group of items, or individual item, the last three years' demand history may be amended for selected Models.

This is particularly useful for volatile items and new items just extracted from the Inventory Management module (which have no history).

Spread Profiles

Spread Profiles are required to download forecasts from this module into Distribution Requirements Planning (DRP) and Production modules. This is because these modules process forecasts on a weekly or daily basis whereas Forecasting deals with twelve or thirteen periods per year.

Processing

Processing, for each Model, can be divided into three distinct stages:

- Extracting Groups of Items and their sales.
- Applying Forecasting techniques.
- Downloading Forecasts.

Extraction

This involves the extraction of selected groups of items from the Inventory Management module, followed by sales history for these groups and items from Sales Analysis. You cannot create profile records until you have created the associated Model record.

You can select Items by Inventory Minor Item Group, or transfer all items together. This may be done as required. New items will be added to the Forecasting database, and existing item details will be updated.



Note: Group/Item profiles cannot be maintained in Forecasting until this combination has been extracted.

Full sales history, or last period sales only, can be selected for extraction from the Sales Analysis module. The extraction of last period sales should be done immediately after Sales Analysis Period-end has been run for the company, and before the Forecast Period-end run.

These sales quantities are extracted for every item which has a profile defined in Forecasting. Item Sales quantities are aggregated to Minor Item Group level and are also stored as part of the Model records.

The Transfer Full Sales History option will attempt to extract three years of sales demand from Sales Analysis.

Applying Forecasting Techniques

The Forecast Period-end job will attempt to calculate forecasts for the next twelve or thirteen periods respectively, for all item/item group profiles defined within Forecasting. Groups or items having a locked in forecast method immediately use the corresponding formula to create forecasts. Items with a locked in manual forecast are ignored.

Otherwise, the process of simulation is used. The module goes back in time evaluating each forecast method for each item and group of items. This period of time, or Simulation Period, is defined for each item/item group in its profile record.

The module predicts sales figures for the simulation period by applying each of the forecast methods. It then calculates the difference between the forecast sales figures and the actual sales figures for each period, that is, the error.

The Forecasting module decides which is the best method by calculating forecast errors for each method. The error is expressed in terms of a percentage.

The totals are accumulated over the Simulation Period. The absolute value of this error percentage is used to compare methods.

Simulation Example

Simulation Period=3, using 12 periods per annum. Forecast Method A=L3P/3.

Period	LP11	LP10	LP-9	LP-8	LP-7	LP-6	LP-5	LP-4	LP-3	LP-2	LP-1	LP
Actual	1162	1137	926	820	841	761	752	961	759	702	733	818
Forecast (Simulated)										824	807	731

Forecast Error %
=

for Method A =



Note: || around figures indicate that the absolute value should be used in the equations, that is, minus signs are ignored.

There are seven forecasting methods used in this period end processing. For specific details of these methods refer to Forecasting Configuration.

Each of the seven methods has the Forecast error % calculated and the method with the lowest error is selected to project forecasts for the particular item or group of items.



Note: Forecasts created by the period-end run can be manually changed by using the Forecast

Manipulation option. This is identical to the Forecast Enquiry but with update facilities.



Note: Item Forecasts may be optionally aggregated up to Group Level using the Forecast Aggregation option.

Downloading Forecasts

Forecasts can be downloaded into:

- The DRP and/or Production modules. This may be done at Minor Item Group or Item Level.
- A specified DRP Centre DC or a Production model. This will apply the DRP/Production Spread Profiles set up within Forecasting Maintenance and distribute the forecasts over working weeks/days.
- A requirements file. This may then be removed from this module for further evaluation and downloaded into your own modules. Software programs must be written to achieve this.

Enquiries

You can select an enquiry to see Forecasts for a specific item, or a group of items within a selected model. (A model is a grouping of Inventory stockrooms).



Note: Forecasts for the group are generated by the Period-end run using Sales History quantities stored for the group. Forecasting treats the group as just another set of figures to be evaluated.

The Forecast Enquiry displays the last 3 years sales demand figures, together with the forecasts for the next year.

Also shown is the code for the forecast method selected together with the calculated forecast error.

The forecast method code can be changed to one of the other seven. This results in the forecast quantities being recalculated and displayed. No update is performed, as this is for viewing the impact of changes without affecting the data.

A Bar Chart is also available for displaying these quantities pictorially.

Reports

As part of the general policy of modules, selection parameters are available where appropriate before a report is run.

In general, reports are printed on standard listing paper with 132 print positions and 6 lines to the inch.

All reports are printed with the procedure or program number in the top left corner and the page number, date and time in the top right.

The reports can be grouped into one of three categories:

Forecast Details

The Demand Forecast report provides a detailed listing of all forecasting items, showing the forecasts for the next 12 or 13 periods. Cost and/or sales values may also be output on the report. A range of models and groups, or all items, may be selected. Totals are accumulated and printed for groups.

The Detail Forecast report shows the last three years' sales history as well as the forecasts for the next year. This is by quantity only. Selection and totals are as per the Demand Forecast report.

Forecast Performance

The Forecast History report outputs details of historical forecast, and the actual sales demand for comparison. This shows only the latest forecast for each period, together with the actual sales.

The Forecast Evaluation report shows historical and predicted forecasts for each period, together with the actual sales for comparison. Up to 6 periods of historical and 12/13 periods of predicted forecasts are included.

Static Information

Item profiles, Group profiles and Marketing Mask details may be selected for printing.

Implementing Forecasting

After installing the software provided, the recommended steps for implementing Forecasting are outlined as follows:

1. Define and create the Company Profile
2. Define and create Forecasting Models
3. Run the Transfer Item Details processing option.
4. If necessary, review the Profile records created by Step 3, particularly those allocated to the dummy Group 999, as these have no Minor Item Group defined to them in the Inventory Items file.
5. Re-run the Transfer Item Details option, if necessary.
6. Run the Transfer Full Sales History processing option.
7. If necessary, review/maintain the demand history created by Step 6 via the appropriate maintenance option, or the Detail Forecasts report.
8. At the end of the current period after Sales Analysis period-end has been run, the Forecasting Transfer Last Period Sales processing option must be run.
9. If necessary, review/maintain the last period's demand quantity created by Step 8 via the appropriate maintenance option.
10. Run the Forecast period-end processing option.
11. Review/maintain the forecasts created by Step 10.



Note: Step 8-11 must then be followed for subsequent periods & can be scheduled to run automatically via Machine Manager

Preparing to Use Forecasting

Preparing to use the Forecasting module requires careful planning and preparation. Your existing Forecasting procedures and data need to be transferred to the module.

Entering Basic Data

You must initialise your module by entering the following basic data:

- **Company Profile** - This contains control and parameter information operating your company, it also defines any links to attached modules.

The primary information maintained per company is:

- Company name.
- Current period and year.
- Period names.
- Backup parameters.
- Activate Conversion Flags (Yes/No).
- Forecast History Save Period (in periods).
- Forecast method descriptions.
- Seasonal profiles flag.
- Interface Company Code.
- Sales Analysis database details.
- Default Forecasting model (optional).

Reference Data

Within each company you should also maintain the following Reference Data:

- Forecasting Models - These models enable the creation and transfer of forecasts for individual locations or groups of locations.
- Items - These must be defined in the Inventory Management Items file. Forecasting extracts items from this source to create Profile records in the Forecasting database.
- Minor Item Groups - These should be defined in the Inventory Management Parameters file and allocated to appropriate items in the same module. It must also be defined in the Sales Analysis module for the appropriate company.

Forecasting extracts:

- The selected Groups from Inventory and creates a Profile record for each Group within the Forecasting database.
- Invoiced sales from Sales Analysis at item level, and accumulates these quantities at Minor Item level before updating them in Forecasting at both levels.

The Forecasting module expects that the stockroom (depot) field, Minor Item Group and quantity fields exist within the Sales Analysis database. If this is not the case, then please contact your software support organisation.

- Demand History - The demand history for the last three years is created using the Transfer Full Sales History processing run.
- DRP Spread Profile - If Distribution Requirements Planning is installed, then a Spread Profile is used to split the period forecast quantities into weekly or daily quantities for DRP.

You have the option of specifying a weekly Spread Profile in Forecasting, or to use the weekly and daily Spread Profiles defined in DRP. You do not need to set up both - only one will be used (based on the Company profile Seasonal Profiles setting.)

When defining the Spread Profiles, Forecasting displays the appropriate DRP weekly calendar for each period. Against each week a percentage is entered.

The second option automatically recognises the spread profiles already defined and maintained in DRP. These Spread Profile records are used by the Transfer Forecasts to DRP option within Forecasting.

- MPS/MRP Spread Profile - If Master Production Scheduling and/or Material Requirements Planning are installed, Spread Profiles can be used for the same purpose as described previously.

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Chapter 4

Job Management Overview



Introduction

Job Management covers the requirements of the wide range of companies that need to capture costs and income on specified projects. Such jobs may range from a large project such as a multi-site building scheme which needs to be broken down into various jobs and tasks with stage payments, to a one-off commission where a consultant is required by a client to produce a time and material invoice.

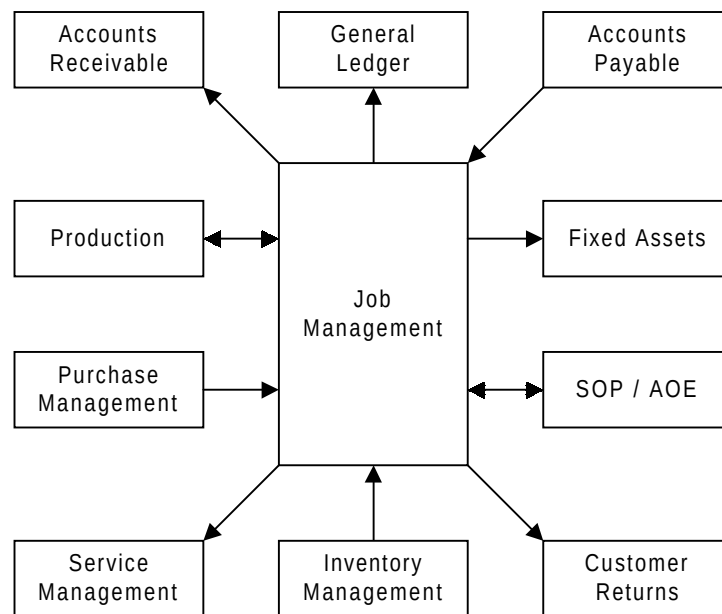
Job Management supports both multi-company and multi-currency processing, providing the necessary information and facilities in the following key areas:

- Quotations - once created, these can be converted into jobs.
- Job entry / amendment - set up fixed price, time & materials, or non-chargeable jobs.
- Flexible job hierarchy enables costs and income to be recorded at all levels.
- Costs and income can be posted to the General Ledger; assets can be created in Fixed Assets.
- Cost recording at any job level is by user-defined expense type.

Costs are captured via:

- Entry of employee timesheets.
- Entry of employee expenses.
- Direct entry of user-defined transactions.
- Entry of standing costs.
- Costs from other applications: Accounts Payable, Purchase Management, Inventory Management and Production.
 - Budgets at all job levels by period.
 - Automatic invoicing (for chargeable jobs), with the facility to review and amend prior to printing.
 - Flexible enquiry facilities .

The Environment



General Ledger

Detailed cost and income values are held in the Job Management system while a financial summary of job values is provided in the General Ledger (GL). Obviously, it is imperative that the two sets of values are balanced and can be reconciled.

A Job Management sub-ledger can be set up within the General Ledger with its own control account for work in progress control, as well as other holding accounts for costs entered within Job Management, Inventory Management and Accounts Payable. This is optional.

Each transaction (cost or income) is posted to a valid General Ledger account which must already be set up. Posting masks enable you to target a GL account without having to enter a code on every transaction.

Accounts Receivable

Job Management shares customer detail information with Accounts Receivable. Open items are created and relevant accounts are updated on printing of invoices.

Production

You can create production orders for jobs within Job Management.

Purchase Management

Purchase Management accumulates the value of outstanding purchase orders, with matching job numbers, against each job level. You can use Purchase Management to display the outstanding orders for a job.

Accounts Payable

Costs attributed to outside sources such as sub-contractor costs, bought out materials and services, can be entered into the Job Management system via Accounts Payable. This occurs when the invoice is accepted into the Accounts Payable system by a request for additional information, i.e. job / sub-job / task, plus a breakdown of the type of expense. Employee expenses can also be entered through Accounts Payable.

Inventory Management

Inventory Management is used for issuing physical stock on an ad hoc basis.

Fixed Assets

The interface to Fixed Assets enable fixed assets to be held against more than one job detail. Where projects using Job Management involve the construction or assembly of fixed assets, the transfer of the asset from Job Management to fixed assets can be completed via a transfer facility within Job Management.

Sales Order Processing / Advanced Order Entry

Job Management can interface with Inventory Management to capture the cost of inventory items. Job Management captures the necessary costs by defining a list of parts needed for a job detail. It interacts with Advanced Order Entry or Sales Order Processing to define the parts list.

Sales Order Processing progresses the list of parts, defined as an order, from entry through to despatch. This facilitates the production of acknowledgements, pick notes and 'clean' despatch notes for the parts list. Because the invoices are produced within Job Management, confirmation of despatch prevents the creation of an invoice – instead the inventory costs are posted to Job Management. Each job detail (i.e. any job / sub-job / task number) may be assigned to one sales order.

Customer Returns

Job Management allows access to the Customer Return functions of:

- adding or amending a return
- receiving a return

This enables returns to be linked to a job.

Service Management

Job Management will inform you if installation records deriving from a parts list need attaching to a contract, and it can provide access to Service Management to perform the attachment.

Two Core Principles

Job Hierarchy

To provide complete flexibility for diverse types of projects and reporting requirements, the definition of a job is divided into three levels:

- Main job - 6 characters long.
- Sub-job - 4 characters long.
- Task - 3 digits long.

The main job is mandatory, but the entries for sub-job and tasks are optional depending on the complexity of the project and the amount of detail required for control and analysis.

Roll-Up

Costs - Cost transactions are entered against a job by specifying the main job, the sub-job and task (if required), plus the expense type. Costs and income are 'rolled up' (i.e. transferred) to the next level. The system will hold the 'this level' costs and income, as well as the rolled-up costs, so that main job and sub-jobs show values of work carried out both at that level and at all subordinate levels.

Budgets - The system attaches budgeted costs at every level, with a breakdown of year and period. A budget type identifies whether the values are associated with budget or forecast costs.

Budgets can be set at any job level. The application will automatically roll up budgets from a lower level through the job hierarchy. For greater flexibility, it is possible to record a rolled-up budget at a higher level - and then protect it from being overwritten during the automatic roll-up.

Budgets can also be produced against an expense type at each job level, with six budget levels created within any one budget code:

- Job level
- Sub-job level
- Task level
- Job / expenses level
- Sub-job / expenses level
- Task / expenses level

The system defaults to a user-defined budget code. However, further budgets or forecasts can be created for any one job.

Maintenance Data

Job Management does require some maintenance data to be set up, but in general this is a limited amount, and requires little attention after the initial set-up.

The Company Profile

Before any transactions are posted, key application controls must be set up. These include entering holding accounts and active interfaces, reserving expense types to particular transactions, and finalising batch control details.

Parameter Maintenance

Most of this data simply needs a code and a description to be entered. Data within this category includes:

Managers	Status codes
Locations	Budget codes
Regions	Transaction sources
Divisions	Transaction types
Industry codes	Variation reason codes
Profit / cost centres	Analysis codes
Charge codes	Expense types
Retention types	Uplift codes
Penalty clauses	Activity codes

Job Types

The job type dictates how the system processes the job, e.g. the valid expense codes for this job type, or any default text that is to appear on invoices.

Employee Details

Each employee on a job needs to be entered on the employee file. The required data includes employee number, name, etc., plus a grade.

Employee Grades

Each employee has an associated grade code, which can be associated with a price list to calculate costs and cost to complete.

Process Codes

Certain predefined processes, invoicing for example, can be set to change a job detail's status code automatically to a user-specified value when the process takes place.

Transaction Processing

The normal day-to-day processing in Job Management consists of:

- Building quotations
- Adding and amending jobs
- Processing transactions
- Producing invoices
- Releasing job costs and income from work in progress balance sheet accounts

Building Quotations

Quotations are built up using an identical hierarchical structure to jobs, with a header, a details screen, a hierarchy where jobs, sub-jobs and tasks can be entered, and transactions which can be entered at any level.

The similarity between a quotation structure and a job structure means that when you have built your quotation you can transfer it as a budget with the headers and details also being copied across. The transfer of details is flexible, and you may decide not to copy across at all.

No postings to GL are made. The estimated costs, which can be added for any level in the hierarchy, can build up using standard functions such as part lists and user-defined transactions.

Furthermore, price overriding and global mark-up changes are available for all types of transactions.

If you have the Sales Order Processing / Advanced Order Entry (SOP / AOE) link active, then, a parts list can also be attached to a Job Management quotation. The sales order that is created will be held as suspended, if the parts list is for a quotation.

The following steps show how the quotation feature within Job Management works:

Step 1

In building a quotation, you create a quotation header with quotation details.

Step 2

From the header you may add sub-job and tasks for the quotation. This is done on the quotation work with details screen.

Step 3

Behind each quotation detail (i.e. any header, sub-job or task number), you can add a number of expense types. These expense types are added and maintained on the quotation work with expenses screen.

Step 4

Behind each expense type, you can add a number of transactions. These transactions have a reference number and a description. These transactions are added and maintained on the quotation work with transactions screen. Global

mark-ups, selling prices and margins can be adjusted in order to arrive at a final quotation price.

There are a number of copy facilities available for quotations:

- Copy a closed quotation to a live job.
- Copy a sub-job / task to another sub-job / task within a quotation.
- Copy an existing quotation to a new quotation.
- Copy a job to a quotation.

Any quotation can be revised, and the original version and earlier revisions will be kept for reference. The system will always default to the latest revision, but other revisions can still be maintained. Revisions are automatically sequenced by date and time.

Adding and Amending Jobs

To create a new job it is necessary to define a job header and a main job at the job level with optional sub-jobs and tasks. A credit check on the customer may be carried out when the job is raised (determined by job type).

Necessary information on the job level includes that which is made available across a number of jobs for analyses such as project manager, division, region etc. Other information specific to the job, such as estimated start and finish dates are also entered in the job details at job level.

Those jobs designated as fixed price offers you an opportunity to enter stage payment details. If designated as non chargeable or time & material (T&M), the stage payment screen is not revealed.

Processing Transactions

Costs attributable to those jobs which are not covered by the automatic update facilities - timesheets, other application interfaces and invoicing - are catered for by direct entry. User-defined transaction types allow for flexible analysis of items such as adjustments and cash payments.

Timesheets

Although many organisations enter timesheet data on a weekly / monthly cycle, it is possible to record the entries on a daily basis, with a periodic update of the job costs. The data recorded is used to update actual labour costs on both chargeable and non-chargeable jobs, and calculate invoice values on T&M jobs. Timesheets can be reviewed and amended before posting.

Employee Expenses

Employee expenses may be entered either from within Job Management, or from Accounts Payable. If entered from Accounts Payable, the transactions are posted to Job Management when the invoice is accepted. The employee mileage and claim amount to be charged can be automatically calculated against a job, as required.

Standing Costs

Standing costs simplify the posting of costs against a job number, where the cost could be deemed 'regular' or 'standard'. For instance, rent of machinery could be a standard cost with a fixed currency, an expense type, a transaction type and charge code. Instead of entering these details in the direct transactions option, it

is possible to choose the standing reference and enter the amount. Each standing reference has an associated set of job-numbers (i.e. job / sub-job / task). Reversible standing costs can also be set up to deal with periodic accruals.

Releasing Job Costs and Income

Costs and income can be released from GL work in progress by varying user-defined methods including:

- Automatic release without user intervention.
- User-controlled release. The release facility allows a range of control from automatically releasing all values for a job to the defined release of part of one transaction.
- Additional control can be put in place so that the use of the user-controlled release outlined above can apply only to closed jobs.

Invoicing

Invoicing can be achieved via the three methods listed below. Some of the invoicing features include:

- Multiple currencies within a job.
- Multiple delivery addresses.
- Support of statement accounts, extended terms etc.
- The facility to review invoices, change invoice values and test before final printing.
- Reprint facilities.

Invoice Methods

Stage Payments

These are automatically generated from jobs where the job type is defined as being a fixed price job. An invoice is generated whenever the date of a stage payment is reached, or passed. Stage payments can depend on either of the following:

- When a specified date is reached.
- When specified activities are complete.

Time & Materials

These are generated from jobs where the job detail is defined as 'time and materials'. An invoice is generated whenever a rechargeable cost transaction is recorded against such a job.

Direct Entry

There are facilities that enable you to enter an invoice directly. These only apply to fixed price jobs. Sundry credit notes can also be issued for fixed price jobs.

Enquiries

There are several enquiry routes within Job Management.

Company Profile Enquiry

This displays all the company profile information.

Parameter Enquiries

There are enquiries for all the application reference data entered within the system. They consist of output-only versions of the parameter maintenance function.

Main Job Enquiry

The main job enquiry requires selection of job enquiry by customer or job number - subject to status code.

The job details enquiry provides standard information on the set up details as well as values for costs, budgets, variances and invoices to date. Similar information can be provided on both sub-job and task enquiry details. Further financial details are displayed for rolled-up and this level details. This information can be reviewed in more detail, i.e.:

- Commitments (Purchase Management or Inventory Management)
- Invoices
- Price variances
- Stage payments
- Cost breakdown this level
- Indented cost breakdown
- Alternative budgets

Session Enquiries

The front-end screen requests an entry via a transaction reference number or a session number. The enquiry by session number has drill-down facilities, to view transaction, source and destination details.

Other Enquiries

The Job Enquiry By Asset Number identifies all job details that are associated with a particular asset number, where Fixed Assets is active.

The Quotation Enquiry shows the breakdown of costs against a quotation with the associated mark-up percentages needed to arrive at a proposed selling price.

The Parts List Enquiry facilitates an enquiry on the status of inventory items associated with a job together with the stock availability set against individual items (where SOP is active).

If the Production interface is being used, an enquiry will reveal all production orders associated with a job. The status of the order can be viewed and details of input (component) availability displayed.

Reports

Generally, standard reports are printed on standard listing paper with 132 print positions and 6 lines to the inch.

Parameter Reports

You can produce a report of all the existing parameter entries. If, for example, you produce a report of all the job types for a company, the report will include all valid expense types and the flags set for that job type (e.g. invoice hours text etc.).

Analysis Reports

There are three principal analysis reports that can be created. The options are Job Analysis, Sub-job Analysis and Transactions by Task.

Each report can be generated for status of open jobs, closed jobs or both. These reports detail for a range of parameters. For instance, a report can be generated that involves all job numbers, or is specific to a particular task.

Each report displays details on actual revenue to date, any commitments outstanding (i.e. from Purchase Management and Inventory Management).

- Actual costs (total and current month)
- Actual budgets (YTD and total)
- Variance value and percentages

Transaction By Task enables all your transactions that have been entered to date to be reviewed. It displays the expense type, reference numbers and value, with internal text attached.

Returned Assets Report

This report identifies the return, via the Job Management / Customer Returns interface, of any serial-controlled Inventory item that has previously been capitalised via Job Management.

Transaction Processing Report

When you enter costs through Job Management, two reports are produced automatically; a posting validation error report and a GL interface file validation report.

When you create timesheet transactions, a timesheet reconciliation report is produced. This lists employees whose timesheets have not been entered, or whose hours were less than the standard weekly hours.

When costs are posted from through Job Management to the General Ledger posting validation report and a General Ledger File validation report are produced.

For timesheet transactions, a timesheet reconciliation report is produced. This produces a report of employees whose timesheets have not been entered or whose hours have been less than their normal weekly hours as defined in the employee details maintenance.

The interface file validation report for the General Ledger contains details of sessions for Job Management cost transactions. The following details are outlined in this report.

Session Details:

- Session number.
- Company.
- User and device.
- Date & time.

Header Details:

- Transaction type.
- Document reference.
- Posting period.
- Description of job detail.
- Company.
- Source account.
- System session number.
- Currency code of transaction.
- Prime control amount.

Line Details:

- The GL account.
- Prime value control (of line).
- Base value.
- Memo quantity.
- Secondary analysis codes.
- Source system for accounts.



Warning: Warnings are also highlighted for failed postings to GL accounts via the mask.

Invoice Reports

In addition to the final invoices and credit notes, appropriate interface file validation reports are automatically produced. A report for generated invoices needing review and a report of all generated invoices are also automatically created.

Fixed Assets Report

To validate the transfer of assets, three reports are produced automatically:

- Asset creation report - a report for every asset that is being created for the first time.
- Asset postings report - a report of the costs being posted to Fixed Assets.

- Exception report - a report of any errors that would occur within creation and/or amendment of Fixed Assets if all transactions were to be processed. It also flags any transaction causing such an error so that it is not processed in the actual transfer.

A summary report is generated for any released transactions posted to General Ledger and Fixed Assets.

Prerequisites

Two applications are prerequisites for the running of Job Management:

- General Ledger
- Accounts Receivable

Implementation Guide

1. Install GL (General Ledger) and Accounts Receivable. These are prerequisite to the installing of Job Management.
2. If you are intending to make use of the Job Management - General Ledger posting interface, then you must set up the GL chart of accounts with the accounts which will be receiving Job Management postings.
3. Other applications you may wish to interface with should be installed. These are not prerequisite to the running of Job Management, and can therefore be installed after you have begun using Job Management. They may include:
 - Fixed Assets
 - Inventory Management
 - Purchase Management
 - Accounts Payable
 - SOP (Sales Order Processing) / AOE (Advanced Order Entry)
 - Production
 - Service Management
 - Customer Returns
4. Install Job Management, setting up library lists in accordance with the application installation instructions.
5. Go into the Company Profile maintenance option from the Utilities menu to set up your company. The system prevents you seeing the third details screen, because this requires the entry of parameters you have not yet had the opportunity to set up!
6. Your company now exists, and you must set up the parameters you intend to use for processing and analysis. These are found on the Parameter Maintenance menu.

If you do not want to use all of the parameter details for a job detail, you can set up your parameters so that a blank code is valid.
7. Re-enter the Company Profile maintenance option and enter the details screen which takes the parameter and holding account details. You can now enter your default parameters.

Do not change the batch number parameters within the company profile once you have started using Job Management.
8. If you are running the system on an AS/400, you must ensure that the Job Management subsystem is started before making transactions.

If the subsystem is not started, transactions will not be lost but will not be processed until the subsystem is started.
9. You can now start a job or quotation.

System Generated Data

When you set up a company profile, the system automatically creates some essential data within Job Management. The data created is as follows:

Status Codes

01 - New

99 - Closed



You will need to set up further status codes used by the system in conjunction with the process codes listed below.

Transaction Sources

IM - Inventory movement

JM - Job Management

PL - Accounts Payable

RC - Release cost

RI - Release income

RS - Release standing cost

SC - Standing cost

SR - Standing cost reversal

Process Codes

IVD - Invoiced

PDS - Partial despatch

RCV - Goods received

RTN - Return generated

SOC - Sales order created

TDS - Total despatch

Fixed Asset / General Ledger Account Types

090 - Job Management postings

Production Reschedule Policies

These codes protect the four order statuses used by Job Management. Materials Requirements Planning will be prevented from changing Job Management orders. The codes are set up in the Maintain Parameters option in the Production application, under parameter type 'WTYP'.

Planned: 41J

Confirmed: 42J

Released: 43J

Active: 44J



Note: Production orders generated via Job Management have a planning filter of J.

AOE Quotation Type

Q - Job Management quotation.

This code is set up via the Descriptions option in the Inventory Management module, under parameter type QTNO.

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Chapter 5

Service Management Overview



Introduction

Service Management is designed to operate independently or as an integrated part of other modules.

The package will support multi-branch operations within each of a number of companies.

Contract maintenance is supported by quotations, pricing and renewal to the production of maintenance invoices, including in advance, arrears or by instalment.

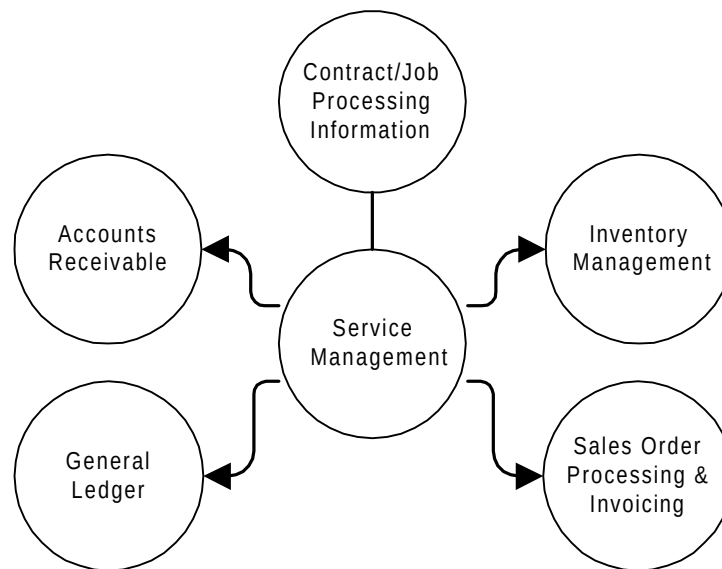
Job processing is controlled by service call logging, engineer allocation and planned maintenance. Job escalation facilities are also provided. Full enquiry features enable access to master files, transaction history, job status, and financial data.

Where System Manager is installed, security is provided by restricting access to certain functions by operator ID, or by creating tailor-made menus for each operation. This means that operators can be prevented from accessing specific functions, or information belonging to certain companies.

Relationship to Other Modules

Service Management operates under the control of System Manager. The full benefits of the system are gained by integrating it with the following prerequisite modules:

- Accounts Receivable
- General Ledger
- Inventory Management
- Sales Order Processing



You can also optionally integrate it with the following:

- Advanced Financial Integrator (AFI)
- Distribution Requirements Planning
- Customer Returns
- World Trade
- Forecasting

Module Configuration

As with all modules, Service Management can be operated for a number of companies, the characteristics of each being maintained on a control file. The module is controlled by setting up data for the company and then for each branch within the company. This would normally be done after the Accounts Receivable, Inventory Management and Sales Order Processing companies have been created.

Accounts Receivable, Inventory Management, Sales Order Processing and Service Management modules all use the same company codes. You can add branches after starting the module.

Certain data and policies are defined at the company level. These include:

- Document reference numbers: invoice, credit note, contract, job
- General Ledger accounts
- Default labour price list
- Default scheduled visit profile
- Working day start / finish times
- Hours entry only, for call reporting.

Additionally, the details of service and accounting periods are defined at company level. Up to 99 service periods per year are allowed, and are used mainly for the scheduling of planned maintenance visits. Each day of the year may be defined as a working or non-working day, to enable calculation of target response date / time for service jobs.

Up to 99 accounting periods per year are allowed and these should be consistent with those specified in the ledgers. The number of days in the accounting periods is used to enable the apportionment of deferred revenue from service contracts invoiced in advance.

Once the control parameters for a company have been defined then at least one branch will be defined. A branch is a physical or logical location that has responsibility for servicing specific customer sites.

Certain data and policies are defined at the branch level. These include:

- Branch name and address
- Printer output queue for job audit log
- Job escalation parameters: escalation time fence
- Escalation reporting interval
- Users to be alerted at each escalation level
- Default response hours.

Each user must be authorised to access at least one branch within a company, but may be authorised to several or all branches if required. This authorisation by branch will restrict the data to which a user has access.

Module Reference Data

Service Management uses a number of reference files that hold data which rarely changes, as follows:

- Standard codes and descriptions
- System parameters
- Customers
- Divisions, model groups, model sub-groups and models
- Installations
- Contract types
- Job categories
- Labour rate price lists
- Service territories, teams and field service groups
- Engineers
- Contract rates
- Scheduled visit profiles

Standard Codes and Descriptions

A descriptions, or codes / parameters file enables codes to be set up with standard descriptions. This provides for validation at the time of data entry and also the display of descriptions on both screens and reports. In some specific instances a parameter or number is also stored, as for example in the case of value added tax percentages.

Certain standard codes and parameters must be present for the successful operation of Service Management, and these are supplied with the system. Examples of code types and values that might be set up are as follows:

Code Type	Type Description	Code Value	Code Description
FLTC	Fault Code	OVH	Overheating
		ELE	Electrical Fault
EGRD	Engineer Grade	GR1	Grade 1
		GR2	Grade 2
LHTY	Labour Hours Types	BASE	Basic
		DBLE	Double Time

Setting Up Customers and Equipment

Before contract information can be recorded, or service operations entered, customer details, the location of customer sites and the equipment installed at these sites must be entered.

The set of procedures described in this section enables you to set up and maintain:

- Customer accounts, maintained through Accounts Receivable.
- Installation addresses, maintained through Accounts Receivable.
- Installed equipment details.
- Equipment models.
- Model group and sub-group maintenance.



Note that some of this information is also used by Accounts Receivable and Sales Order Processing. Care must be exercised in its maintenance to make sure that data is correctly held for all purposes.

Each set of information can be printed or used for enquiry purposes.

Before starting to set up customer and installation details in Service Management you have to use the Names And Addresses option in Accounts Receivable and the Installation Address Maintenance, in Accounts Receivable and Sales Order Processing.

Names and Addresses Maintenance

A customer account holds the customer's name and address, plus accounts-related data, such as payment terms and credit limit.

Once an account is defined, the installation site addresses for that customer can be maintained. The account address can itself be an installation site.

Additional service details are linked to account and installation addresses. Three maintenance screens are involved. The routine for maintaining these records is described in the Additional Service Details Maintenance section of Service Management.

No installation records or contracts can be entered, and no transaction processing can be performed, without customer accounts. Although some of the data held on the customer account is not used by Service Management, the following parameters are significant:

- Name and address - Displayed on many screens to allow verification of correct customer account. Printed on some reports, including invoices / credit notes.
- Geocode or postcode - Used in automatically assigning teams of engineers to jobs at the customer site, by means of the geocode / territory definitions and engineer team / field service group / territory assignment rules.

- VAT code & indicator - Used within invoice / credit note generation to establish which VAT rate to charge.
- Credit stop indicator - Used within job maintenance to provide warning that the customer is on credit stop.

Installation Address Maintenance

This option is used to maintain customer site addresses where equipment is installed.

An installation address is a customer address at which equipment is installed, which may require service. A customer account address may itself be an installation address, and there may be several additional installation addresses for the customer account.

When you set up a customer account, a three digit suffix (address code) of 000 is appended. Further installation addresses are then created through this option by assigning address codes of 001, 002, etc. to each address record for the account. The codes can be numerics or characters, as preferred.

Each installation address is identified by a unique combination of account and address code.

Additional service details are linked to account and installation addresses. Three maintenance screens are involved. The routine for maintaining these records is described in the Additional Service Details Maintenance section.

The site address must be created before the details of equipment installed at the address can be entered. No contract or transaction processing can be performed until this equipment database is present. The following data for an installation address is significant:

- Name and address - Displayed on many screens to allow verification of correct customer address. Printed on some reports, including invoices / credit notes.
- Postcode - Used in automatically assigning engineers to jobs at the customer site, by means of the postcode, or geocode / territory definitions and engineer / territory assignment rules.

Price List Options

The two price list options are both Sales Order Processing options, so turn to SOP product information for more details regarding these.

Customers

Service Management shares a common customer database with Accounts Receivable and Sales Order Processing, with maintenance routines in each module reflecting its specific requirements. For example, Accounts Receivable controls credit limits and payment terms, Sales Order Processing controls stock allocation, pricing of sold items and Sales Analysis coding, and Service Management controls pricing of service items and invoice consolidation levels.

The structure of the customer information can be manipulated to match the structure of the customer's organisation. On the one hand there may be a simple single account, on the other a statement account with many 'invoice to accounts' each with many different delivery or installation accounts.

Statement Account

This is shared with Accounts Receivable and is normally created and maintained by it. The data includes credit details, Accounts Receivable parameters and statistical codes.

Invoice Account

Invoices may be sent to an invoice only address.

Delivery or Installation Account

Each account may have many delivery or installation addresses. These are defined by Sales Order Processing and / or Service Management, by adding a three-character address code suffix to the account code.

Additional Information

Each account address, at which equipment is to be serviced, must have additional information set up, which is used solely by Service Management. The information includes the invoice destination address, invoice consolidation level (account, address code, or contract / job), service region, and price list for service parts.

Identification of a customer can be made either by entering the account and address code, or by using the powerful name and address search routines that are available in the system.

Models

Each piece of equipment defined to Service Management is identified by its model code and, normally, its serial number. Standard model codes / numbers are used to describe the different types / configurations of equipment that can be installed. Each model code is linked firstly to a model sub-group which in turn is linked to a model group.

Models may be specified as individual machines, or as peripherals to a machine.

Contract rates for different types of service contract can be held at model group, model sub-group or model code level; engineers can be assigned to perform work only on specific model groups, model sub-groups or models within a service territory.

The labour rates to be charged for service work are determined by the model on which the work is performed, since each model code holds a pointer to its appropriate labour price list.

Installations

An installation within Service Management can be defined as a customer address at which equipment or systems are installed.

The customer address is identified by an account code and address code and can have many items of equipment or systems installed. This configuration of equipment or systems at a customer location constitutes an installation record. Records should be set up for all installations irrespective of whether they are covered by service contracts.

Each piece of equipment at an installation is identified by the unique combination of its model number and serial number. If serial numbers are not known, then the model number and model quantity can be used.

The data held for each piece of equipment at an installation includes details of age, dates of delivery and installation, warranty period, location / department, complexity points and product division.

Engineers

Each engineer who will carry out service / repair work is defined on the engineer master file. Details include his name and address, radio page number, team code, grade, hourly cost rate, service region and stockroom code, together with his skills matrix.

The stockroom code provides the link to Inventory Management and must be defined therein. It represents a discrete area within a company where stock is recorded and controlled separately from other company stocks. In the case of service engineers, the stockroom could be the specific van or car of the engineer himself, or a common store or depot from which spare parts are drawn.

Service Territories And Engineer Assignment

Service Management facilitates the automatic assignment of engineer teams to service jobs, provided that certain details have been defined:

- The customer records must hold the geographic or postcode of the customer address.
- The geographic area covered by the company must be divided into service territories; each territory will be defined by a list of full or partial geocodes / postcodes.
- The engineers' teams who cover each territory must be specified.
- The field service group for the team must be defined, to show which product divisions, model groups, model sub groups and models the team will service. The field service group record contains a skills matrix to define these limits.

The system can map the customer's address to a service territory and allocate work to the team of engineers designated to operate in that territory.

The allocation of an individual engineer within the team, to a specific job, can be handled automatically or manually. Further specialisation is used in job selection according to the engineer's skill level, i.e. an engineer can be assigned to service specific model groups (and optionally sub-groups) and / or to perform work of a specific type (i.e. job category), and / or to perform work on equipment belonging to a specific division.

Contract Types

Contract types are used to define the different types of service contract offered by the company to its customers. Along with its description, each contract type record holds the normal duration of the contract, guaranteed response hours, invoice schedule (in advance, in arrears, or on completion of each scheduled maintenance visit), and whether a quotation or contract is generated at renewal time.

Billing parameters, which may be applied to the contract type, are also selected (i.e. to indicate if service charges, rental charges, or various meterage conditions will apply).

All contracts created in the system will have a user-defined contract type code. The system will default a contract type code of *NO for work on equipment not covered by a service contract.

The parameters set up for the contract type will act as defaults for the contract, but can be changed if required.

Job Categories

Job category codes are used to describe the different types of service work to be performed. A number of categories can be defined as system defaults for breakdowns, warranty and repeat calls etc, to save time at call logging.

Additionally, a table of valid job category codes for each contract type must be defined. This table is used to specify whether the customer should be charged for travel and labour hours (both fixed and hourly charges), for up to four parts groups and for miscellaneous charges, when work of a certain job category is carried out on equipment covered by the specific contract type.

Price Lists

Two types of price list are used by Service Management, for labour rates and for parts.

Labour rate price lists define the hourly labour rates to be charged to customers, where appropriate, for contract, non-contract and workshop work. The price lists include an hourly travel rate; they carry effectivity dates, to control the module of new prices, and are linked to models: the model code of the equipment being serviced / repaired determines the labour rates to be charged.

Parts price lists and discount lists define the prices and discounts to be applied when charging for spare parts fitted or used by an engineer. These lists, which also carry effectivity dates, are maintained in Sales Order Processing, but are linked to the customer additional details records in Service Management: the prices charged for service parts are determined by the customer site at which the work is done.

Contract Rates

Contract rates are defined for each contract type at model group, model sub-group, or individual model code level.

Contract rates specify the charge per scheduled maintenance visit and / or a fixed charge for each piece of equipment covered by the contract type. Contract rates carry effectivity dates to control the module of new rates, and are used by the contract pricing routines in Service Management.

Contract rates can only be used when the system has been defined for term processing; they cannot be used for monthly processing.

Scheduled Visit Profiles

Scheduled visit profiles are used to define standard schedules of planned maintenance / service visits to equipment under contract. The schedules are defined in terms of the number, type and frequency of visits.

A visit profile can be associated with a contract, model, or model group, and enables the automatic generation of the service visit schedule for any piece of equipment added to a service contract.

Contract Processing

Service Management provides flexible and comprehensive facilities for contract management, as follows:

- Maintenance
- Pricing and invoicing
- Renewal
- Quotations
- Credits

Contract Maintenance

Service contracts are identified by a combination of contract number, type and start date. Each contract relates to one or more locations of a single customer account, and may cover one or more pieces of equipment installed at these locations.

When a new contract is entered on to the system, the contract number, type, start date, customer account and address code are specified. Contract header and equipment records are created.

The contract header record includes the contract duration and end date, order reference and date, and invoice destination. The service parameters include response hours, scheduled visit profile, allow service cover while the contract is pending, and the rates to be applied. The billing parameters set the invoice term, invoicing frequency and whether charges are in advance, in arrears or per visit. The service conditions provide for setting up service, visit and rental charges at contract header level: the service and visit charges override the contract rates set by model group, model sub-group or model; the service conditions can be overridden at equipment and peripheral levels.

Once you have set up the header you must define the contract equipment. The system lists all the equipment defined for locations of that account, and you can select those items to be placed on the new contract. Equipment placed on a contract must have been previously specified as resident at a location of the account to which the contract applies.

Service conditions at equipment and peripheral levels can be added to the contract. You can create a schedule of planned service visits, by service period for the duration of the contract, for each item on a contract. This can be done automatically by the system using the service visit profile associated with the contract header, model, or model group, but you can amend the schedule if required.

Contract Pricing And Invoicing

Contracts which are invoiced in advance or in arrears are priced and invoiced as follows:

Service Pricing

- A contract is priced for a defined period of time (invoice term) which may be the same as the contract duration itself, or may be less. The charge for an invoice term can in turn be split into instalments for invoicing purposes, for example a one year term can be invoiced in quarterly instalments. Additionally, the system will pro-rata the charge for equipment added to a contract part-way through its invoice term.
- To price a contract, each piece of equipment covered is priced individually and accumulated to give a total contract price. If the piece of equipment and its peripherals have special prices held, then these are the prices to charge.
- If no special price exists, then the contract rates file is accessed (term processing only), to obtain the price per maintenance visit and any supplement fee. The total number of maintenance visits scheduled for the piece of equipment in the invoice term is then multiplied by the price per visit and the supplement fee added. The result is the gross charge for the piece of equipment.



Note: Service contracts may include fixed visit charges, to be invoiced after each visit: the contract's billing parameters may be set accordingly. Such visits are invoiced through job invoicing at call completion, not through contract invoicing.

Rental Pricing

- A rental charge may be made for equipment on contract (if rental is defined on the contract type). The charge may be applied on the contract header, in which case it will be used as the default price for each item added to the contract, or may be maintained on the contract equipment line. The charge entered will be per month if monthly processing is selected on the system parameter file, or per term if term processing has been selected.

Meterage Pricing

- The following meterage charges may be applied (if selected on the contract type). Meterage charges can only be used with 'monthly processing'.
- A minimum volume can be defined on the contract header or equipment line. The minimum volume will be multiplied by the price per copy, to reach a minimum monthly charge, which may be invoiced in single or multiple months.
- A minimum charge can be defined on the contract header or equipment line, which will be the minimum monthly charge; this may be invoiced in single or multiple months.

- Up to five copy charge bands, for each of two meters, can be defined for each contract equipment line.
- Interim pricing allows for charges to be made according to a meter estimate, or actual reading returned by the user, based on the minimum units already charged for, (or actual units used if no minimum has been entered). The frequency of interim pricing is defined on the contract header.
- Reconciliation pricing allows for charges to be made according to an actual meter reading and allows for additional meterage charges to be applied, or credits given for overcharges. The reconciliation term must be a multiple of the interim period.
- Meterage pricing can be 'pooled', i.e. the minimum volumes / values can be spread across all of the equipment on the contract, allowing the excess units on one machine to be offset against the under-usage on another machine.

Invoicing

- Contract pricing creates records on an invoices pending file. A second function, generate and print invoices, generates invoice header and line records. At this stage, invoice consolidation, the grouping of invoice pending lines into invoices, takes place as specified on the customer additional details records, i.e. by account, account / location or contract. VAT is calculated according to the parameters set on the customer account record. The invoice documents are then printed.
- Invoice generation posts the relevant transactions to the sales and general ledgers. Where advance invoicing is performed, the revenue (i.e. income) value of the invoice is deferred in the General Ledger, by generating additional postings which 'transfer' period revenue values from a deferred income account into a sales account each period.

Contract Invoice Term

When pricing a contract, the price is calculated to cover a fixed period of time known as the contract invoice term. There can be one or many invoice terms within the life (duration) of a contract.

The contract invoice term is defined as the period of time from the next term starts on date to the next term ends on date, and defines the period of price protection for the contract.

Next Term Starts On Date

The date on which the next contract invoice term to be processed will start. Contracts are selected for pricing if the next term starts on is less than or equal to the date entered on the price contracts prompt screen. This field is held and maintained on the billing parameters window.

Next Invoice Term Ends On Date

The date on which the next contract invoice term to be processed will end. This field is held and maintained on the billing parameters window.

Normal Invoice Term

The length of time in months to be added to the next term starts on date, to calculate the subsequent next term ends on date and hence define the normal

invoice term for the contract. This field is held and maintained on the billing parameters window.

Invoice Frequency

A sub-division of the normal invoice term. This designates the period of time, in months, between invoices, if the invoice term of the contract is to be invoiced in instalments. This field is held and maintained on the billing parameters window.

Contract Payable in Advance

A contract for which contract charges for any contract invoice term, or instalment of a term, are invoiced before the term or instalment.

Contract Payable in Arrears

A contract for which contract charges for any contract invoice term, or instalment of a term, are invoiced after the term or instalment.

Contract Payable Per Visit

A contract for which contract charges are invoiced as a result of a scheduled service visit being made and technically reported. The pricing of these visits is actually carried out by the Price Jobs option.

Pending Invoice Line

An invoice line, not yet assigned an invoice number or posted to the Ledgers, which is waiting to be grouped with other invoice lines to form an invoice. This grouping is dependent on the invoice consolidation level, which is set up for the customer account (i.e. the 000 record) by the Additional Service Details file maintenance option.



Note: The account code (000) sets the invoice consolidation level for all sites.

Pending Invoice Line Availability Date

The date on, or after which, the pending invoice line may be included in an invoice. For advance contracts, this will be the start date of the contract invoice term or instalment period; for arrears contracts, this will be the end date of the contract invoice term or instalment period.

Description

You can select all branches or a single branch, specify the type of billing, and enter a cut-off date relating to the contract invoice term(s). Select **F8=Submit** to submit a batch job.

Only contracts payable in advance or arrears are processed by this function; visit only contracts are excluded.

Pending invoice lines are generated for:

- All non-invoiced pieces of equipment, added to a contract during an invoice term, which have been priced previously.
- All equipment on contracts where the entered cut-off date is later than, or equal to, the next term starts on date of the contract.

Pricing Routine

Contract pricing is governed by the conditions set up in contract maintenance, at header or equipment levels, or at both.

Default Prices from the Contract Rates File



Note: Default pricing can only be used for fixed service, rental and visit charges. It will not apply to copy / vend contracts, where special contract conditions must be set up.

If the service conditions, at contract header or contract equipment levels for the fixed service, rental or visit charges, are not overwritten, their values default from the contract rates file. Contract pricing picks up the fixed service, rental and visit charges from the contract rates file, for the appropriate effective date, equipment group, contract type, term and currency.

For a particular currency, the contract rates record used will be that for the contract type and term, effective on the 'pricing date'. The 'pricing date' is derived as follows: if the 'use start date rates' field on the contract header is set to '1=Yes', or if there is a 'price held until date' which is later than the 'next term starts on date', the pricing date is the contract start date; otherwise, the pricing date is the 'next term starts on date'.

The equipment model group, model sub-group and model are compared to the same fields on the contract rates record; the record with the best match will be selected. If no record is found, the contract could be left with no charge: special service conditions could be set up by the user for the contract. The recommendation is that records should be created on the contract rates file for as many combinations of equipment and contract types as are necessary.

The price is calculated, by multiplying the visit charge by the number of scheduled maintenance visits for this piece of equipment which occur in the 'contract invoice term': between the 'next term starts on' and the 'next term ends on' dates.

The fixed service charge, from the contract rates record, is added; the charge is deemed to apply to the 'normal invoice term', and is adjusted to allow for the 'contract invoice term': the number of days in the 'contract invoice term' is divided by the number of days in the 'normal invoice term', and this factor is applied to the fixed service charge.

The rental for the contract term is included, but separately from the service and visit charges.

Contract Conditions Prices at Header / Equipment Levels

If a price is keyed into the Contract Conditions window at header or equipment levels, this price is deemed to apply to the 'normal invoice term' specified on the contract. The 'normal invoice term' is converted to days, by adding the 'normal invoice term' in months to the 'next term starts on' date, to get the 'normal invoice term ends on date', and to work out the number of days included between these two dates. The number of days between the 'next term starts on' date and the 'next term ends on' date (from the contract header) is also calculated. The ratio, of the latter number of days to the former, is calculated and the same ratio is applied to the contract conditions price, to get the actual price for the contract, or for the specific piece of equipment, for the contract invoice term.



Note that for complete instalment periods, the invoices are calculated to be of equal value.

Pro-Rata Pricing

The price for equipment added late or removed early from a contract will be calculated on a pro-rata basis, as follows:

The number of days in the 'normal invoice term' of the contract is calculated, as described above.

For equipment added late, the contract invoice term is the number of days between the date of addition of the piece of equipment to the contract, and the 'next term ends on' date, or the equipment removal date, whichever comes first.

For equipment removed early, the contract invoice term is the number of days between the date of addition of the piece of equipment, or the 'next term starts on' date, whichever comes last, and the 'next term ends on' date, or equipment removal date, whichever comes first.

The ratio of the two sets of days (i.e. contract invoice term and normal invoice term) to each other is calculated. This ratio is applied to the calculated price for the normal invoice term, to get the actual price for the contract, or for the specific piece of equipment, for the number of days that it has been on the contract.



Note: Equipment actually removed, or scheduled for removal, within an invoice term already processed, will not have an automatic credit generated; for this, use the Contract Credit option.

Contract Renewal

This function can be run at any time and prompts the user for a date representing the expiry date limit of contracts for renewal. The system reads through all active contracts and determines which ones to renew, by comparing the contract end date with the entered date. If the contract end date is less than or equal to the entered date, renewal is performed.

In order to renew a contract, the equipment on the contract must still be eligible for the contract type, i.e. the contract charge rates must be in effect.

A new set of contract records will be generated with either the same contract number or a new, system-assigned contract number depending on an indicator on the company profile. The contract start date for the new contract will be the day after the existing contract end date. Other contract details are copied to the new contract.

The status of new contracts will be set as either 'pending start date' or 'quotation', depending on the 'quotation required at contract renewal' flag on the contract type record. The user can, if required, amend the contract details generated (through contract maintenance procedures) before quotations or invoices are generated.

Contract Quotations

Quotations are created in the same way as contracts, but have a special status of 'quotation'. They may only be created for contracts with service and visit charges; rental and meterage contracts cannot have system-generated quotations.

Contract quotations can be produced on pre-printed stationery, to show account / site details, contract number, type, start date, duration, customer order number, equipment covered, number of scheduled maintenance visits and price. The quotation price is calculated in the same way as the contract invoice price.

Quotations can be converted to contracts, by using the contract maintenance facility to change the status from 'quotation' to 'pending contract'. A facility is also provided for generating quotations from existing contracts. This can be used, for example, if a customer requires a quotation for a different type of contract, covering the same equipment as his existing contract.

Contract Credits

Credits can be processed at any time, to refund part, or all, of a contract invoice, and to cancel any pending invoice records.

The contract, or the invoice, or the pending invoice period from / to is selected. The selected record(s) may be amended to affect the operative date and value(s) involved. The selected record may be discarded, a full or partial credit may be given, or the original charge may be credited and a different amount re-invoiced.

The resultant invoices and credit notes are produced when the option to Generate And Print (Contract) Invoices is next taken. Any cancelled pending invoice records are removed from the work file at the same time.

Job Processing

An important function of any service organisation is to complete, in an efficient and timely manner, both ad hoc customer service requests and planned maintenance work according to the contractual terms agreed.

On completion of service jobs, any chargeable work should be invoiced promptly in order to optimise cash flow.

Service Management provides the following facilities to enable these objectives to be met:

- Planned maintenance job generation
- Call logging
- Engineer work assignment and diary
- Engineer technical reporting
- Job pricing & invoicing
- Job escalation
- Remote communications links to and from engineers

Planned Maintenance Job Generation

This is a process that is run as required. Load PM Jobs creates a service job for each installation that has a planned maintenance visit scheduled for periods falling within a user-specified range. The function allows the schedule to be created for up to six periods in advance of the current period. The service schedule details, held for equipment covered by contract, provide the base data for processing.

The process will attempt to assign an engineer team to each service job, by comparing the customer geographic code with the territory definitions, and the model type requiring service with the field service group that covers the division, model group, model sub-group and model. If a specific engineer was entered against the equipment on the installation record, this engineer will always be assigned (if this method of selection is defined on the system parameter file, on screen 5).

If there is more than one piece of equipment at an installation address, on the same contract, requiring a service visit in the same period, then this procedure will normally generate only one job covering all the equipment. However, an option can be set on the company profile to generate an individual job for each piece of equipment.

Call Logging

This facility enables unscheduled / breakdown calls from customers to be recorded, and corresponding service jobs generated, for subsequent allocation to engineers. It also enables the details of existing jobs outstanding to be amended.

Addition of New Job

The logging of a new job requires entry of either the customer account code (or the account can be found using alpha search data), the service contract number, the serial number, or the model number / serial number. Any of these enables access to the customer installation record which holds the details of the equipment at the customer site. A new job can only be logged if an installation record exists for the customer address requesting service. A facility to create installation details for non-account customers is provided at this point.

If the customer is on credit hold / stop, a warning message is displayed and the call status is set to 'credit hold', but the job may still be entered.

If a contract number has been entered, all the equipment on the contract is displayed. If an account code and address code have been entered, all the equipment at the installation is displayed. Each item to be included on the new job can then be selected from the displayed list. If a specific model and serial number have been entered, the job details for the item are prompted for immediately.

If the customer has a number of items of equipment on site and cannot identify which of them requires the service, the job can be logged against any available machine line. When the engineer completes the job and reports on it, the true model / serial number must then be identified to the system.

The system will check for previous calls (for qualifying job categories), to determine if this is considered to be a repeat call: this compares the number of days and / or meter units which have elapsed since the previous call, with the values defined on the volume segment file for the model. The system allocates a job number; the account name and address is displayed and may be amended just for this job. Job line details are displayed, showing the contract / warranty cover; response hours are calculated by reference to the special serial numbers file, the contract line, the contract header, or the contract type (possibly reduced by the 3-D Matrix percentages). A target start date and time is set by adding the retrieved hours to the date and time the call is logged. The date and time the call is logged may be altered while the call is in 'Addition' mode, to allow for calls taken out of hours (e.g. by overnight answering service).

A history enquiry, showing details of previous jobs on the piece of equipment, is displayed in a window if the call is a repeat, or can be accessed by a function key. A customer order number may be entered; it is essential if requested on the customer additional details record and any elements of the call are chargeable.

The job category displayed for the call defaults to the value determined by the user in job category maintenance and may be amended. Customer contact name and fault code / description, for the item of equipment to be serviced / repaired under the new job number, are entered. The response hours can be altered if required, which will re-calculate the target response date / time for the job, which can then be monitored by the escalation procedure. Alternatively, if the customer requires an engineer to visit at a specific date / time, an appointment date and time can be entered. At this stage, an engineer can be assigned to the job, by displaying a selection window of the engineers in the team and their skills.

Amendment to an Existing Job

Access to existing service job records can be achieved through various routes – by entry of customer account code / address code, or serial number, or model / serial number, or contract number / type (if the customer has a service contract), or customer order number, or job number. Alternatively, the customer account can be accessed by specifying up to three search parameters (relating to pre-defined alpha search data fields from the account / address record).

A list of all current jobs on the system for the installation is displayed; the list includes both scheduled service and unscheduled / breakdown jobs. On selection of a job number, a list of all the equipment included on the job is displayed (a job may cover one, or more than one piece of equipment at an installation depending on the setting on the parameter file), along with job 'header' information, i.e. the job number, creation date and time, customer order number, customer contact, service contract number and type. On selection of an individual piece of equipment the details of job category, engineer assigned, customer contact, fault code, are displayed and can be amended as required.

Engineer Work Allocation

Engineer work allocation may be carried out by running the automatic call allocation program, or manually.

If automatic call allocation is required, the program is initiated by taking an option on the service operations menu. In addition, the Service Management sub-system must be running.



Note: The sub-system library list must be tailored to take account of:

- the use of consolidated files libraries in place of IN, OE, SL and G files libraries.*
- the version of Business Manager in use, for adjusting the four IPG libraries.*



If you are running System 21 Version 3 on an AS/400, then use the CHGJOB command and **F4**, enter OSLSSP/OSLS2P3, then select **F10** and page down to the library list.

Automatic call allocation is a sleeper job, that will automatically schedule engineers within a team, to the jobs which have been assigned to the team in job logging. Assignment takes place on the basis of finding an available engineer, based on his current workload and skill profile, and who can achieve an estimated time of arrival that is equal to or less than the target date and time for the call. If the selection criteria cannot be met, an error message is displayed for the call, and a non-assignment reason code is entered on the call detail screen.

Calls can be scheduled manually, in which case the same criteria and checks for engineer selection are applied.

The function allows outstanding jobs to be despatched to service engineers, who phone or call in, to request work. It also enables entry of status updates from engineers on their current jobs, to indicate for example, that parts are awaited or the work is complete.

A list of jobs assigned to a specific engineer or team of engineers, or a list of all jobs for the branch can be displayed. Further selection parameters enable either planned jobs or unscheduled (call out) jobs, and jobs of certain status, to be extracted. The list of jobs displayed will be sequenced in order of target date / time, the most urgent jobs first.

Jobs for a team can be displayed graphically: colour blocks represent each job and the colour denotes the urgency of the job. Unscheduled jobs for the team are displayed in the lower part of the screen; scheduled jobs are shown under each engineer in the upper part of the screen. Jobs can be moved with the cursor and a function key. Job line details can be displayed, for team and engineer work queues, by positioning the cursor on any job and pressing 'Enter'.

The valid status codes for a piece of equipment on a job are:

Open;

Assigned;

Scheduled;
Despatched;
Work in Progress;
Requires Telephone Call;
Awaiting Parts;
Complete: Not Documented.
Status codes can be amended as required.

Engineer Technical Reporting

This is the facility used to enter the details of completed work from the engineer's reports, and is accessed by selecting the completed job line with 99.

A technical report is made up of four sections:

Header Details

Details of the engineer number, job number, visit date, report reference, time on site, customer and engineer travel time, are entered for each new technical report. Distance travelled can also be recorded and charged

Labour Hours

Labour hours spent on each piece of equipment are entered, qualified by fault code, job category, machine section and sub-section repaired, and corrective action. Hours can be booked at up to four different labour rates, i.e. basic plus three overtime rates. The hours are classified as non-chargeable (0%), or chargeable (100%), according to the charge matrix for the contract type / job category combination defined; the charge percentage can be changed to any intermediate value between 0% and 100%.

Parts Used

Spares parts used in carrying out the service or repair of each piece of equipment are entered, analysed by job category. For each part number, the quantity used is entered. The parts are classified as non-chargeable (0%), or chargeable (100%), according to the charge matrix for the contract type / job category combination defined. The charge percentage can be changed to any intermediate value between 0% and 100%.

Parts Ordering

The engineer uses this function to order spare parts to complete an existing job or a return job.

This reminds the engineer that parts have been shipped for the job when in the Call Reporting Parts Used Screen.

The engineer can order and use items that are not defined as being stocked in the relevant stockroom. The system creates the stockroom/item link automatically.

Service Management keeps track of the order for the job number. When a technical report is next entered, the parts are automatically displayed on the parts used screen, to prompt the keying in of their usage. Where parts are shipped but not used on the job, the available stock in the engineer's stockroom is updated.

The service stockroom, supply stockroom and central warehouse must be set up in a DRP network if you want to run DRP net change processing. If no network is set up, the net change triggers are not set.

The sequence of operations to set up a DRP network covering service, supply and central stockrooms is:

- Set up the calendar for the current year in the Calendar Maintenance option in DRP.
- Set up periods for the central and supply stockrooms using the Period Maintenance option in DRP.
- Set up the central and supply stockrooms using the Stockroom Maintenance option in DRP.
- Set up customers for the central, supply and service stockrooms using the Customer Maintenance option in DRP. The internal customer has to be set up in Accounts Receivable, using the Names And Addresses option.
- Create the network using the Network Maintenance option in DRP, giving the service network a code linked to the central stockroom.
- Set up the planners using the Planner Maintenance option in DRP. Use the central stockroom with user-defined detail for the planner, and set up a planner linked to the supply stockroom.
- Set up items using the Distribution Item Master Maintenance option in DRP.
 - For the central stockroom the source of supply should be set to 1, the order policy should be A, and the local stockroom should be the central stockroom.
 - For the supply stockroom the source of supply should be set to 2, the order policy should be A, and the local stockroom should be the supply stockroom. Use the central stockroom for the centre and the central stockroom.

The sequence of actions for a DRP order for Service Management parts is:

- Amend the delivery name and address, if necessary; **Enter** to fill in details of order lines, and **F8=Update** to complete the DRP order.
- Manually allocate the DRP order, using the Order Amend (Transcriptional) option in SOP.
- Print one pick note at a time, using the Picking Notes option in SOP. Ensure that the 'from' and 'to' order numbers are the same.
- Confirm despatch from the DRP order lines, despatch date and despatch method only, using the Allocation By Order option in SOP. No charges data is relevant.
- Confirm the shipment receipt using the Record Transfer Out option in DRP. Then revert to Service Management and technical reporting.

Miscellaneous Costs

Miscellaneous costs incurred in carrying out the service or repair of each piece of equipment are entered, analysed by job category. For each cost, a cost / charge type code (with its file, or user overwritten description), the cost value and charge value are entered / displayed, or the charge is derived by entering a percentage uplift.

A service job may require more than one engineer visit to complete the work, in which case several technical reports can be entered. The prime report is given the suffix-00, with each subsequent report suffix being incremented by the

system. After the final report is entered, the user indicates that the job is finished and fully documented. At this point, the job becomes eligible for pricing and invoicing.

Job Pricing And Invoicing

Service jobs, which are not fully covered under the terms of a service contract, need to be charged to the customer. The pricing and invoicing of jobs is performed after they have been completed through the technical reporting function.

The decision as to which jobs, or elements of jobs, are chargeable is made by referencing the user-defined matrix of contract types and job categories. For the particular combination, the matrix indicates whether labour and / or travel distance and / or travel time and / or miscellaneous costs and / or four types of parts are chargeable to the customer.

Six types of charge can potentially appear on a job invoice:

Contract Visit Fee

Service contracts can include fixed visit charges, to be invoiced after each visit: the contract's billing parameters being set accordingly. Such visits are invoiced through job invoicing at call completion.

The fixed visit charge is obtained from the contract conditions at equipment or header levels, or from the contract rates file.

Labour

The labour charge is calculated from the number of hours booked on the engineer report, multiplied by the hourly rate on the labour price list held for the model. The hourly rate used will be either a contract, or a non-contract, rate depending on whether the piece of equipment was covered by contract when the job was logged. The hourly rate will be uplifted by a premium factor for work performed at overtime rates.

A fixed labour charge may be applied, if required by the job's charge matrix set up in Contract Type / Job Category Maintenance.

Travel Labour

The travel time is calculated from the number of customer travel hours and minutes booked on the engineer's technical report, multiplied by the hourly travel rate from the labour price list held for the model.

A fixed travel labour charge may be applied, if required by the job's charge matrix set up in Contract Type / Job Category Maintenance.

Parts

The charge for parts used by the engineer is calculated according to the price list and discount list defined for the customer site. The price and discount list details are obtained from Sales Order Processing.

Miscellaneous Costs

The charges are based on the values entered through the technical reporting function.

Chargeable Travel

The charges may be based on a standard distance value, the actual distance driven, or zones. Data is set up on the Additional Service Details record for each site.

All jobs, even non-chargeable ones, should be processed through the job pricing and invoicing functions, since the cost values of labour, engineer travel time, miscellaneous costs and parts, for all jobs, are calculated by these routines.

Pending Invoice Line

An invoice line not yet assigned an invoice number, which is waiting to be grouped with other invoice lines to form an invoice.

Description

A batch job is submitted by this menu option to price jobs for the current branch.

Only job lines which have been fully technically reported are processed by this function. This is when **F10** has been taken against any technical report line relating to the job line.

Pending invoice lines are generated for all job lines which have been fully technically reported but not yet invoiced. If a job bill to account exists, this is used to generate the pending invoice. If there is no job bill to account but there is an ownership account, this is used instead. If there is no ownership account, the site account is used.

The price charged for each job line is built up from the following five individual charges:

Labour Charge

Each labour hours entry from each technical report for the job line is priced individually.

A labour charge is only made, if the contract type (of the contract which covered the piece of equipment when the job was created) and job category (entered against the labour hours on the technical report) combination is defined as chargeable for labour, on the cover type / job category file.

Fixed charges: these are applied, prior to any variable labour charges, if defined on the cover type / job category file.

Variable charges: these are applied on the time remaining, after any fixed charges. The labour hours, on each technical report entry, are multiplied by the premium factor (from the codes file LHTY record) for the labour hours type (indicates basic, time and a half, double time etc.).

The result is multiplied by an hourly labour rate from a labour rate price list for the appropriate currency, which is derived as follows:

The labour rate price list is taken from the customer additional details, then the model file. The price list contains four, hourly labour rates: contract, non-contract, workshop and travel. If the engineer is a workshop engineer, that rate is used; otherwise, for work, depending on whether the work has been carried out under a valid contract or not, the contract or non-contract rate is used; and if travel time is involved, on contract or not, the travel rate is used.

In identifying the actual price list record to be used, the pricing date is taken into account, i.e. the price list record effective at the pricing date will be used. The pricing date is determined as follows:

If the equipment is not covered by contract:

- Pricing date=Visit date (from technical report)

If the equipment is covered by contract:

If the Use Start Date Rates field on the contract which covers the equipment is set to on:

- Pricing date=Contract start date

If the Use Start Date Rates field on the contract, which covers the equipment, is set to off, and the job is a planned (scheduled) job, and the visit date is before the price held until date on the contract which covers the equipment, then:

- Pricing date=Contract start date

If not, or if price held until date is zero:

- Pricing date=Visit date

If the job is a 'callout' (breakdown) job and the call logged date is before the price held until date on the contract which covers the equipment, then:

- Pricing date=Contract start date

If not, or if price held until date is zero:

- Pricing date=Call logged date

The labour cost value is always calculated, whether or not the labour is chargeable. The cost value, in base currency, is the number of labour hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer who performed the work). No premium factor is applied to the cost rate.

Parts Charge

A charge for a part is only made if the contract type (of the contract which covered the piece of equipment when the job was created) and job category (entered against the part number on the technical report) combination is defined as chargeable for parts on the cover type / job category file. When reporting parts used on a job, the charge % may be reset from 0% to any value up to 100%.

Parts are priced by retrieving the parts price, discount list, and currency codes held on the Additional Service Details. Using the price and discount list records, defined for these codes in Sales Order Processing (and effective for the pricing date, as derived for the labour pricing calculations), the prices and discounts are calculated in the appropriate prime currency.

The cost value of parts used is always calculated, whether or not the parts are chargeable. Parts costs are always set up in Inventory Item / Stockroom Details in base currency. The cost value is calculated in base currency, not within the Price Jobs function, but when Inventory Management is updated with the detail of the parts used; this takes place when **F9=Complete Technical Report Line**, or **F10=Complete Job Line Report**, is taken within technical reporting.

Miscellaneous Charge

A miscellaneous charge is only made if the contract type (of the contract which covered the piece of equipment when the job was created) and job category

(entered against the miscellaneous charge on the technical report) combination is defined as chargeable for miscellaneous charges on the cover type / job category file.

The charge value of each miscellaneous charge line is the charge value entered on the line itself in the specified currency.

The cost value of each miscellaneous charge is always calculated, whether or not the item is chargeable. The cost value is the cost value entered on the line itself in the specified currency. If the specified currency is not the base currency, the system automatically converts the prime currency value to base.

Travel Charge

Fixed charges: these are applied prior to any variable travel charges, if defined on the cover type / job category file.

Variable charges: these are applied on the time remaining after any fixed charges.

The prime currency travel charge is calculated as the billable customer travel time entered on the technical report, multiplied by the travel rate obtained from the labour rates price list. No premium factor is applied to this rate.

The base currency travel cost value is always calculated for engineer travel time. The cost value is the number of travel hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer on the technical report).

Contract Visit Charge

If the job line being invoiced is part of a planned maintenance job (i.e. has been generated via the Load Planned Maintenance Jobs option), and the contract under which the job is performed is currently defined as being invoiced per visit (i.e. Adv / Visit / Arr=1 on the contract's billing parameters), a contract visit charge is generated (unless calculated as zero).

The contract visit charge is calculated as described in the previous section Price Contracts: the number of visits covered by the job is multiplied by the visit fee. The visit fee may default from the contract rates file, or be specified as a contract condition at header and / or equipment line levels.

Engineer Diary

Service Management provides an engineer diary, which enables the planning and scheduling of non-machine-related jobs and other commitments for each engineer. Job appointments are automatically entered in the diary for the engineer, at the time they are made in call logging or in work allocation.

The next five working days (as defined in the daily calendar file) are displayed, starting from the system (current) date. Each day in the diary is free format; bookings can be entered for any date, time and duration, or for a date and time 'from' with a date and time 'to'.

Diary entries

Diary entries are made as required and are used to reserve the times for events such as holidays, training courses, sickness, dental appointments, etc. As entries for engineers are scheduled for a date and time, they will appear both in the engineer's work queue and in the graphical display of his jobs.

The diary is accessible from the Engineer Work Allocation routine.

Job Escalation

Service Management can alert you to any service jobs that are approaching their response time and are still outstanding. Job escalation runs so long as the Service Management sub-system is active.



Note: The sub-system library list must be tailored to take account of:

- the use of consolidated files' libraries in place of IN, OE, SL and GL files' libraries.*
- the version of System Manager in use, for adjusting the four IPG libraries.*

All service jobs created in the system will be stamped with a target response date and time, which is calculated as follows:

For call out jobs (i.e. those entered to the system using the Job Logging / Maintenance function), the target response date and time is calculated as:

Date / time call logged + response hours

... where response hours is derived from the special serial numbers file, or the contract which covers the piece of equipment if a contract exists. If the equipment is not covered by contract, the response hours default to the value for the contract type *NO. Manual overrides of the response hours may be made, if special circumstances arise.

In calculating target response date / time, the system takes into account working / non-working days, and the standard service 'window' for the company, i.e. the times of the day between which service is provided. These are all user-defined.

For planned jobs (i.e. those generated from the service schedules defined for contracts), the target response date and time is calculated as:

Target date=Date of last working day of the service period in which the job is scheduled

Target time=Start time of the working day

You define the parameters which determine the escalation time fence, the interval between each escalation step, and the users who are to be alerted at each step. The system monitors each outstanding job, comparing its target date / time with the current system date / time. Once the job enters the defined escalation time fence, it will start to escalate and will have its escalation step recalculated at frequent intervals: the escalation step moving from high to low value as the job approaches its target date / time. Escalation step zero denotes that the job has reached (or gone beyond) its target time.



Note: The system assumes a 7-day working week. If less than this is actually worked, the system still escalates what would be Monday's work on a Sunday, not on the preceding Friday.

As a job moves from one escalation step to the next, the system can send a message to pre-defined users to alert them to the fact that the job has been escalated to the next level, thereby allowing them to take appropriate action.

Jobs will be removed from the escalation procedure once they reach a certain status. This is user-defined within the job line status code record; for example, escalation may be required to stop, once the job is allocated to an engineer, or once the engineer has arrived and started work. A screen enquiry allows the escalation status of jobs to be viewed easily, using various selection parameters.

Example of Call Escalation

Escalation Step	Reporting Times - (1 Hour Intervals)	
0	16:00	Target Time
1	15:00	
2	14:00	
3	13:00	
4	12:00	Escalation Time Fence (4 hours) starts

In the above example, the escalation time fence is set to 4 hours and the reporting interval to 1 hour.

A job with target time of 16:00 will start to escalate at 12:00.

Between 12:00 and 12:59, the escalation step will be calculated as 4, because the job is within 4 hours of target.

Between 13:00 and 13:59, the escalation step will be calculated as 3, because the job is within 3 hours of target, etc.

At 16:00, or later, the escalation step will be calculated as 0, because the job has reached or passed its target time.

Remote Communications Links to and from Engineers

Outgoing and Incoming job processing records can be held on files interfaced to a communications gateway. Records are validated and may be corrected centrally, or returned to the appropriate engineers for correction.

A variety of carriers and mobile terminals can be integrated with Service Management. Users are able to select their own preferred route for the gateway, the communications link and the remote devices.

Whether or not the remote communications link is active, normal job processing routines continue without any apparent change.

Time Zones

This function enables you to operate across different time zones. The zones are specified as a number of hours difference (called the 'offset time') from the system time. You can specify zones at branch level and at customer additional details level.

Once you have specified the offset time, when service calls are received the customer's date and time is recorded with them, and are used to calculate target times.

The time zone offset times are set up in Inventory, taking the Descriptions option from the Maintenance menu and adding times under the major type TIMZ.

Enquiries

Service Management provides a comprehensive range of enquiries for:

Master Files

These generally show the same details as their counterpart maintenance routines.

Transaction History

A job history enquiry is available for viewing details of historical service work performed against equipment. An invoice enquiry is available for viewing details of a service invoice, whether it be a contract, job, or sundry invoice.

Status

An escalation enquiry provides up-to-date information on the escalation status of service jobs.

Loading Service Management Data

The following data must be loaded for each service company which is to be processed within the live module.

The data is divided into two groups:

- Basic data - data used mainly for reference purposes, which is low in volume and not subject to ongoing change, e.g. codes.
- Major data - data which constitutes the bulk of the module and is subject to regular change, e.g. customers, contracts, etc.

Some of the data listed below is marked optional. This means that, in system terms, the data is not essential to running the module; but note that, in practice, the data may well be essential, e.g. contracts.

Basic (Parameter) Data

- Company profile
- System parameters file
- Branch records
 - one for each service branch
- User / branch authority records
 - one for each user and branch (to which the user is authorised) combination
- Codes / parameter file records.
 - System-required parameters are identified by * in the first position of the parameter type description: these must be set up, or copied from the demonstration data supplied. In addition, set up appropriate code values for each of the following types:
 - ABSC (Engineer absent)
 - CCRC (Contract cancellation reasons)
 - CHGT (Miscellaneous costs / charges)
 - CORA (Corrective actions)
 - DAYT (Day types)
 - EGRD (Engineer grades)
 - FLTC (Faults)
 - LHTY (Labour hours / overtime types)
 - REGN (Service regions)
 - RRES (Credit release reason)
 - STAT (Equipment status)
 - TXTO (Text destinations)
- Calendar control record
- Daily calendar file (& rebuild period end dates)
 - for all years over which current service contracts are spread
- Division records
 - one for each service division
- Model group / sub-group records
 - one for each model group / sub-group combination
- Geocode / territory records (optional)

- one for each service territory, with a list of geocodes (postal codes) within the territory
- Field service group
 - one for each group or equipment (division / model family / model to be serviced)
- Team
 - one for each engineer team
- District
 - one for each work controller, to authorise work on teams
- Engineer records
 - one for each service engineer
- Labour rates price list records
 - one for each set of labour rates to be linked with models (and for different currencies where World Trade and Multi-Currency are active)
- Zone charge (optional)
 - one for each zone (and currency where World Trade and Multi-Currency are active)
- Model records
 - one for each model code
- Volume segment
 - minimum one, optionally more for each model, or group of models
- Job category records
 - one for each job category code
- Scheduled visit profile records (optional)
 - one for each standard schedule of planned maintenance visits
- Contract type records
 - one for each contract type code
- Contract type / job category records
 - one for each valid combination of contract type, job category, (and currency where World Trade and multi-currency are active) with invoicing indicators
- Contract rates records (optional)
- Escalation parameters (optional)
- Tax codes (for World Trade and multi-currency only.)
 - for each country's contract type / job category combinations
- Inventory descriptions file

- DEPT (Departments)
SS - Service workshop
S3 - Service
- MANT (Mantle groupings)
A - 'Attack' customers
C - Central Govt. Depts.
L - Local Govt. Depts.
M - Multi-nationals
S - 'Special' customers
- MFID (Manufacturer's ID)
* - Mfr not specified
BRIT - Briton Ferry Steel Co Ltd
EAGL - Eaglesbush Nuts& Bolts
FALC - Falcon Electrical
GRU - Herbert Grudgings (Spares)
JONE - Jones Britannia Foundry Ltd
ROKI - Rock Island Arsenal (NJ) Ltd
STEP - Steptoe & Son Precision Engrs
TAY - Taylor Engineers (Rothley)
WRAY - Wray Optical Co Ltd
- OTHR (Customer search parameters)
END1 - Cust account no. 9
END2 - Phone no.332
START1 - Cust account no. 2
START2 - Phone no. 313
- PTYP (Item type)
4 codes are essential to categorise parts:
C - Consumables
J - Major spares
N - Minor spares
S - Supply items
2 codes assist analysis of machine sales:
M - Machine sale via SOP
P - Peripheral sale via SOP
- SPED (Urgency indicator)
1 - Special courier: ETA TBA
2 - By noon tomorrow: promised
3 - By noon tomorrow: guaranteed
4 - 24 hour delivery: guaranteed
5 - By 17:30 today
6 - By 22:00 Monday
- STKU (Stockroom usage code)
B - Service spares stockroom
W - Service workshop: company stock
X - Service workshop: customer stock
- TIMZ (Time zones)
E01 - East + 1 hour
E02 - East + 2 hours
E03 - East + 3 hours

W01 - West - 1 hour
W02 - West - 2 hours
W03 - West - 3 hours

Major Data

- Customer account records
 - one for each customer account.
- Installation address records
 - one for each site address, within a customer account.
- Customer additional details (Service Management) records
 - one for each customer site.
- Installation records
 - for each customer site, details of each piece of equipment at the site.
- Contract records (optional)
 - one for each active service contract, with details of each piece of equipment covered and service schedules.

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Chapter 6

Transport Planning Overview



Introduction

For many companies, transport related costs form a significant portion of the total distribution budget. If you increase the efficiency and reduce costs due to effective planning and management of the transport department, they can have a real effect on a company's profits.

To optimise the utilisation of transport resources and load fill, you need to make accurate information available to the transport department as early as possible in the delivery cycle. This allows efficient scheduling of deliveries in the short term, and assists strategic transport planning in the longer term.

Transport Planning provides a wide spectrum of facilities for planning and day-to-day management of transportation. Together with the appropriate Distribution applications, it provides the comprehensive solution for companies managing their own fleet and those using third party carriers or a combination of both.

General Concepts

You can make deliveries in four basic ways, using:

- Own transport facility
- Third party transport
- Parcel service
- Collection by customer.

Whichever method, or combination of methods you use, you take the following operational steps to plan the transport:

- Build loads
- Assign transport resource
- Manage movements
- Consign loads

Building Loads

You have to consider a number of factors when you are building a load. These factors fall into three categories: customer; product; and vehicle.

Customer Factors

- Geographical position
- Limits on vehicles and container types accepted
- Carriers accepted
- Days of the week
- Times of the day
- Authorisation required for delivery
- Exclusive deliveries.

Product Factors

- Temperature control
- Security.

Vehicle Factors

- Maximum capacity (weight, volume, number of containers)
- Driver restrictions (HGV required.)
- Availability of vehicle and driver.

You need to continuously view the forward delivery load. The forward view should present the requirements in the appropriate units (weight, volume, number of containers) by Transport centre, date, time, route, and/or vehicle type restriction.

Typically, the day before despatch (or same day) you finalise the load, with the vehicle and carrier assigned. You then issue the picking and preparation documents.

Assign Transport Resource

The procedure for assigning the transport resource differs for companies using their own transport from those using third party.

Using third party, you must notify the carrier of:

- Vehicle requirements
- Loading date, time and place
- Drops
- Required delivery dates and times
- Weights/volume/number of containers per load
- Product details.

You can then agree the rates as part of the confirmation. Typically, this is a specific confirmation of a long standing agreement, based on rate tables supplied by the carrier. This allows you to accurately calculate and accrue transport costs, and match costs to carrier invoices.

If you are using your own transport facilities, determine the requirements and assign a vehicle and driver. You should continuously review the despatch requirements to maximise utilisation and minimise the number of part loads sent out.

You often need to assign more than one carrier. For example if you use a master load to send a complete load to a destination, where it is split into sub-loads. Then use local carriers to deliver these sub-loads on smaller vehicles.

Manage Movements

This includes all of the tasks necessary to get the goods on the right vehicle at the right time and in the most efficient manner. You can use Transport Planning to produce the following paper work:

- Picking notes
- Despatch notes
- Load sheets
- Bills of lading (consignment notes)
- Invoices

The important fact is that you pick the load as a consignment. You can pick by order or by carrier or route or individual load and then marshall the load.

There can be different requirements for confirmation of the different movements, but normally document confirmation with exception is used. In the case of despatch notes, you can print a clean document before consigning the load to the carrier.

Consign Loads

Once loaded, you confirm the following transport details before despatch:

- Carrier used.
- Weight or volume of load and number of containers.
- Date and time of despatch.
- Vehicle and driver details - such as registrations or employee number.
- Driver instructions.
- Security details - including tag number, type and colour assigned to the vehicle.
- Thermostat settings (for chilled and frozen compartments).

The last three are particularly relevant for internal transport.

The driver normally sign a copy of the bill of lading to accept the load, that is the load has been consigned to the carrier.

Relationship To Other Modules

Transport Planning operates under the control of System Manager, and is an advanced module which extends the functionality of base modules.

To use Transport Planning, you must have:

- Inventory Management
- Sales Order Processing

You can use Transport Planning with:

- Advanced Order Entry
- Distribution Requirements Planning (DRP) - You use the DRP interface to plan transport for depot replenishment as well as for retail deliveries.
- EDI Application Interface - You use the EDI interface to electronically receive sales orders and make them available for transport planning.
- Telesales
- Warehouse Management - You can produce a consolidated pick per load (or route or carrier).
- Accounts Payable - Once authorised, you can automatically transfer carrier invoices to the Accounts Payable invoice log.
- Advanced Financial Integrator (AFI) - Advanced Financial Integrator can post carrier charges and payment variances into the General Ledger, based on user defined rules.

Transport Planning Configuration

You can operate Transport Planning for a number of companies.

Within each transport company, you set up transport centres by assigning a number of Inventory stockrooms. You can then manage transport on behalf of a number of physical depots.

The transport planner has a view of the transport requirements for each defined stockroom, but can build loads on behalf of each or amalgamate them onto combined loads. The former is for shipping direct from the depots and the latter for transshipping prior to despatching from a single delivery depot.

You can associate the transport centre with multiple sales companies, so the planner can manage transport across commercial companies or divisions within an organisation.

The AFI interface can post the calculated carrier freight charges and actual payment variances to multiple General Ledger companies.

Maintenance Of Set Up Data

As this is an advanced application module, you set up much of the Transport Planning data as part of the base modules. However, you must enter additional data for the maximum benefit.

Transport centres

A transport centre is a delivery depot, where you plan, pick and despatch transport loads. You maintain a number of control flags that affect the processing of loads:

- **Add Orders Immediately To Load** - If set to yes and Transport Planning finds a load with the same route, date, transport centre, and vehicle type, with sufficient capacity, it automatically adds the order to this load. If set to no and Transport Planning finds a suitable load, it suggests this to the planner. If Transport Planning does not find a suitable load, you must create a suitable load and add the order to this load, using the load build screen.
- **Release Loads Before Goods Ready** - Sales Order Processing maintains a total of orders per load and number of orders confirmed as despatched. If you set this flag to no, then Transport Planning will not release loads and print documentation unless these totals tally.
- **Print Documentation Immediately After Load Released** - If set to yes, then immediately after releasing a load, Transport Planning prints the load sheet, and bill of lading, if required. If set to no, the load prints in the next batch print run.
- **Bypass Load Despatch** - If set to no, you must confirm each load as despatched individually. This date and time stamps the load as it leaves the depot.
- **Load Numbers** - you can maintain sequential load numbers per transport centre with a user-defined prefix, or maintain a range of load numbers per transport company.

Transport planners

For security reasons you must define each Transport Planning user as a transport planner. You can authorise each planner to one or more transport centres. You allocate each planner a default centre. The centre description is shown at the top of all screens. A planner can change to any other transport centre to which you have authorised them.

You maintain the following control flags per planner:

- **Release Load** - determines if a planner can release loads for printing of documentation and processing.
- **Override Capacity Errors** - determines if the planner can build loads over the capacities defined to the vehicle type.
- **Authorise Payments** - determines if the planner can transfer carrier invoices to Accounts Payable.

- Authorise Adjustments - determines if the planner can transfer carrier invoices to Accounts Payable if there is a difference between the calculated charge and the payment amount.

Vehicle types

If you require capacity checking or analysis by vehicle type, you must maintain vehicle types.

For each vehicle type, you enter a code, description, and load capacity, by either maximum weight, or maximum volume.

If you assign a vehicle type to a load, Transport Planning outputs an error if you exceed any specified capacity.

Routes and Nodes

A route represents a journey between two geographical locations undertaken by at least one carrier (internal or external). You define the route by a code, description and, optionally, an origin and destination, which can be the same for circular routes. You can also specify, per route, if you require a bill of lading (BoL), its format and number of copies.

If you specify the origin and destination, you must define them as nodes, that is geographical start and end points for the route.

You can enter a default carrier and ship time and vehicle type for the route. When you create a load for this route, Transport Planning adds these defaults to the load.



Note: You associate the default route and drop sequence with delivery points using the Delivery Profile Maintenance option.

Carriers

You must enter for each carrier, internal or external, a code, name, and address.

You can also enter parameters to rate the load, that is calculate transport charges. If you have Accounts Payable installed, specify a supplier account to transfer a matched carrier invoice from Transport Planning onto the Accounts Payable invoice log.

If you want to check carrier route combinations, then define the routes each carrier covers, together with any route specific rating parameters.

Carrier rates

If you want Transport Planning to rate loads, that is calculate carrier freight charges, you must enter sets of rates and associate them with carriers. Each set of rates has a code and expiry date. You can set up future rates to use automatically when current rates expire.

There are four rating methods available:

1. A table of charges based on the load gross capacity, with a fixed rate per drop and a maximum and/or minimum drop charge. You can specify a surcharge value or percentage to accommodate unforeseen.
2. The same as for 1, except that the charge is based on the gross capacity of each drop.

3. A charge based on the gross capacity dropped into each area. For each area you enter the fixed charge per unit of capacity, the Surcharge value or percentage, the drop minimum and maximum charge and the area minimum charge
4. As for 3, except that you maintain the charge per area in a table based on gross capacity.



Note: The capacity can be gross weight, load volume, or number of containers.

Areas (zones)

You can define areas, by a code and description so that you can group load drops into areas (or zones). You assign the area code to the appropriate delivery points, then you can produce documentation and rate loads split by area.

Item transport details

You must maintain additional item details so Transport Planning can calculate the capacity requirements of each item. These are:

- Weight of the goods, expressed in a standard unit of weight, per selling unit
- Volume of the goods per selling unit.
- The number of issue units which fill a default container (pallet, case, and so on).
- The tare weight of the default container.
- The volume of the default container.

You must specify at least one of the capacities.

Delivery profiles

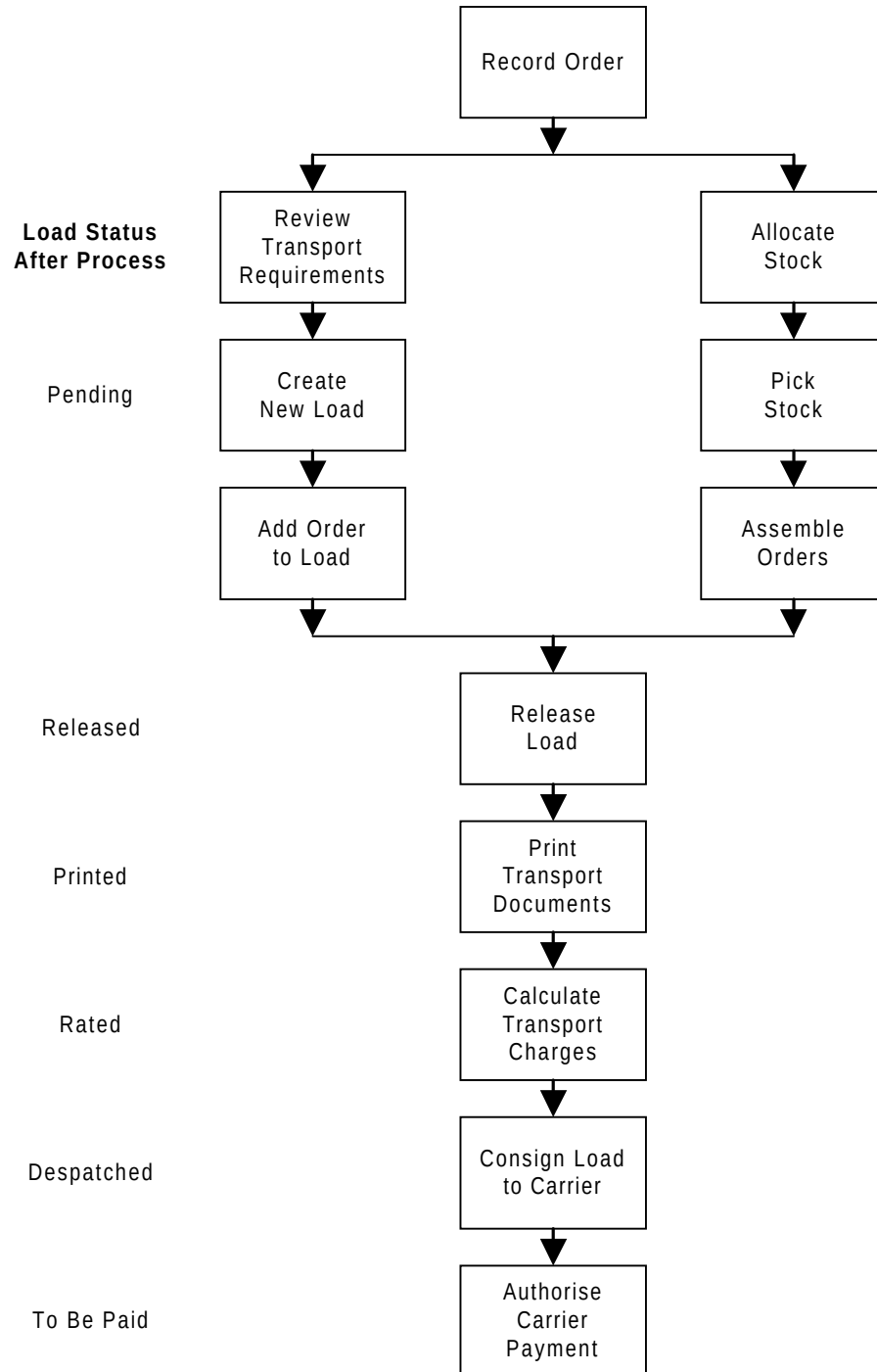
You must set up a delivery profile for each customer delivery point. The details needed are:

- Default Route, Drop and Carrier
- The area or zone used to split the drops for documentation and rating.
- If this load is exclusive to the customer, that is it cannot contain orders for other customers.
- If you must confirm the delivery at this delivery point requires.
- The days and times the customer accepts deliveries. (For memo purposes only).

You also maintain, for each transport company, when Transport Planning first creates transport requirements from Sales Order Processing. This can be either after you take the orders, when you allocate stock, when you create a picking note, or when you confirm the despatch.

Processing

The following diagram represents the business processes:



You can transfer the transport requirements of orders from Sales Order Processing immediately after taking the order, on allocation of stock, creation of pick notes or on confirmation of despatch. Any under or over pick or despatch confirmation is automatically reflected in Transport Planning.

You can automate the building of loads for specific transport centres. The transport planner only needs to create the pending load headers in advance and Transport Planning then adds the appropriate orders (up to any specified maximum capacities defined for the vehicle type.)

You can consign loads to the carrier manually, via the confirm load despatch screen, or automatically after printing and rating, where appropriate, the load.

You only need to match and authorise carrier invoices for rated loads.

Finally you close and copy the load to history, for future analysis.

Review Transport Requirements

In Transport Planning, the load build screen displays all the information necessary to assemble orders onto loads. The details displayed include:

- Gross weight or Volume or Number of containers.
- Sourcing stockroom.
- Default route and drop sequence.
- Default carrier and vehicle type.
- Ship date and time.
- Customer's name and account
- Suggested load number (if any).

You can use the following functions to display more details:

- Pending loads prompt (that is loads not yet released).
- Load details enquiry.
- Order enquiry.
- Delivery profile enquiry.

Orders with fixed or special instructions are highlighted with a flashing asterisk.

Also, an asterisk is displayed against the customer if there are any delivery restrictions.

Create New Load

You can create new load headers individually on line or altogether via a batch function.

Transport Planning can automatically generate load numbers or you can control them manually.

You must enter, as a minimum, a route and ship date.

You can set up skeleton load headers to represent a standard journey and then copy this each time you create a load.

If you have not set up a load header in advance, you can create it when you build a load.

Add Orders To Load

In the load build screen, you can select a number of orders to add to new or existing loads.

If you specify a vehicle type, then Transport Planning checks that its capacity is not exceeded.

You can move the order to alternative routes or resequence the drops or both.

When you add orders to loads, Transport Planning automatically removes them as requirements from the load build screen.

Release Load

Before you release a load, you can make any final amendments to drops, assign any driver, vehicle registration or security seal details via the load maintenance screen.

When you release a load, you can print the necessary transport documentation either immediately or next time you select a print run.

Once you release a load, it is no longer available for planning. You must return the status back to pending before you can amend any details.

Print Transport Documents

You can print load sheets, and optionally bills of lading, for each released load. These are described in detail in the Documents section of this overview.

Details of the route, drop and load print on each pick and despatch note, in Sales Order Processing. You can generate the notes for a specific carrier, route or load.

If Warehouse Management is installed, you can produce consolidated pick lists to efficiently pick for loading of vehicles.

Calculate Transport Charges

Transport Planning can calculate the expected carrier charge for selected loads.

The calculations are based on:

- Charge per capacity unit (gross weight, volume, containers).
- Penalty charge (If below specified minimum).
- Charge per drop.
- Minimum drop charge.
- Maximum drop charge.
- Surcharge value/percentage.
- Minimum charge per area.
- Maximum charge per area.
- Minimum charge per load.

You can maintain the charge rates (per unit or per drop) as a fixed amount or in a table based on capacity, that is weight, volume or number of containers.

There are four methods of automatic calculation:

- A charge based on the capacity, maintained in a table;
- A load charge based on the sum of drop charges which in turn are based on drop capacity, maintained in a table;
- A charge based on a fixed unit rate multiplied by the capacity dropped into each area (zone) contained in the load;
- As above except that the charge per area is based on a table of capacity ranges, rather than a fixed rate.

If Advanced Financial Integrator is installed, then you can post the calculated charges (prior to payment) into the General Ledger, so selected P & L and accrual accounts are debited and credited respectively.



Note: You can post to the General Ledger journals based on user defined attributes of the load, customer, product and carrier. The value may be apportioned over the products contained in a load, based on gross weight, volume or number of containers.

Consign Load To Carrier

You can confirm that each load is despatched. This makes sure that the load is date and time stamped when it leaves the depot.

You can bypass this stage, by allowing Transport Planning to automatically consign the load once the load sheet is printed.

Authorise Carrier Payment

When you receive a carrier invoice, you can match it against the calculated charge for one or more loads.

If there is a discrepancy, only authorised users can authorise the payment. Once authorised, Transport Planning automatically transfers the invoice and load details to the invoice log in Accounts Payable (if installed).

If Advanced Financial Integrator is installed then you can update the General Ledger with any variances between the rated and paid values.

Enquiries

You can group enquiries into three categories:

- Planning - What is to be done?
- Static information - What are the parameters?
- Status - Where are things?

Planning Enquiries

The Outstanding Loads Enquiry displays orders not included on loads.

If you select a range of dates, this shows the future outstanding requirements requiring planning of transport.

Static Information

You can view details for the following:

- Carriers
- Carrier route restrictions
- Delivery profiles
- Delivery time slots.

Status Enquiries

You can view the following load details:

- Pending loads (loads not yet released).
- Load details (load header information).
- Load drops (all drop details for selected load).
- Load orders (all orders for selected load).
- Load consignee (consignee details for selected load).
- Load totals (quantity/value totals for selected load).

You can use the Loads by Order enquiry to view the current status of an order, this includes which loads, if any, the order is assigned to. Select one of the loads to directly access the load enquiries.

Reports

Before you run a report, you can enter selection and sequencing parameters. All Transport Planning reports are defined to Report Generator so you can easily amend them.

Transport Planning prints the standard reports on standard listing paper with 132 print positions and 6 lines to the inch. The procedure or program number prints on the top left and the page number and date on the top right of the report.

Planning/Status Reports

The Load Status report provides a summary, per date or route, of the total capacity requirements of outstanding orders with a view of any current loads. Use this report to establish which routes are under or over committed before building loads.

The Outstanding Orders report provides details of all orders not included on loads, sequenced by route or drop within date. Use this report to view the detailed capacity requirements. This is particularly useful when you identify an over commitment on the Load Status report.

The Load Details report details all the drops and orders for selected loads.

Documents

You can print, and reprint:

- Load Sheets - print these for all released loads (including master loads). They list all drops for each load selected (in ascending or descending drop sequence). It serves three major purposes:
 - As a marshalling document for all despatch notes, especially if they are from more than one stockroom.
 - As an aid to loading the vehicle in the best sequence for dropping.
 - For the driver to use as an action and confirmation list.
- Bills Of Lading (BoL) - a more formal document than the load sheet, normally produced for selected external carriers. This can be in any one of four formats:
 - One BoL per load; One totals line; List of despatch note numbers.
 - One BoL per drop or delivery point; One totals line; List of despatch note numbers.
 - One BoL per load; One totals line per drop; List of despatch note numbers per drop.
 - One BoL per load; One totals line per area; List of all despatch note numbers on the load.

Financial Integrator Interfaces

The following functions relate to the data extraction via the AFI:

1. *Rating* - Manual or automatic rating creates charge details for loads that AFI can then extract for accrual accounting of transport costs.
2. *Payment* - Payment authorisation (manual or with Accounts Payable attached) updates charge details with payment information and transfers the invoice details to Accounts Payable. You can post any variances between rating and payment values to GL via AFI.
3. *Load Cancellation* - If you cancel a load that was rated but not extracted by the AFI, it creates a rating posting and a subsequent rating cancellation posting when you extract using AFI. Where you have extracted the rating posting by AFI, it only creates a rating cancellation posting. For loads you have not rated, no AFI extraction takes place.

Prerequisites

Applications

For Transport Planning to work correctly you must have the following installed:

- System Manager
- Inventory Management
- Sales Order Processing.

Transport Planning also interfaces with the following order capture modules:

- Advanced Order Entry
- Distribution Requirements Planning (DRP)
- EDI Application Interface
- Telesales
- Warehouse Management.

Transport Planning interfaces with the following financial modules:

- Accounts Payable
- General Ledger
- Advanced Financial Integrator.

Data

Items

You must define all items processed by Transport Planning to an associated sales or inventory stock holding company.

Customers

You must define all customer delivery points processed by Transport Planning to an associated sales company.

Transport Planning Configuration

You need to consider the following when you set up the data:

- Transport descriptions in Inventory
- Company profile
- Transport centres
- Planners
- Areas
- Routes and nodes
- Vehicle types
- Carriers
- Items
- Delivery profiles.

Transport Descriptions File Entries

You must define some entries to the Inventory descriptions file.

The following Description Identity Codes must be present for the Inventory company you use as the basis for the Transport Company:

TPAR

Adjustment reason.

TPBL

This is the bill of lading format (code length=2).

- 01 - One document per load.
- 02 - Document per drop/delivery point.
- 03 - One document with drop details.
- 04 - One document details by area.

TPCC

Commodity code.

TPCT

Carrier type.

TPCY

County.

TPDC

This is the capacity (code length=1).

- 1 - Weight.
- 2 - Volume.
- 3 - Containers.

TPIC

This when you can create transport requirements (code length=1).

- 1 - Order.
- 2 - Allocation.
- 3 - Pick.
- 4 - Commit (not used).
- 5 - Confirm pick (not used).
- 6 - Confirm despatch.

TPLS

This is the load status (code length=2).

- 10 - Pending (on hold, being planned).

- 20 - Aprinting (awaiting printing).
- 30 - Allocated (not used).
- 40 - On pick (not used).
- 50 - Picked (not used).
- 60 - Arating (awaiting rating).
- 70 - ACOD (awaiting confirmation of load despatch).
- 80 - Aauth.pay (awaiting payment authorisation).
- 90 - Aledg.post (awaiting posting of payment to the General Ledger).
- 95 - Ahist (awaiting drop to history).
- 99 - Cancelled.

TPPM

Freight payment method.

TPRM

This is the rating method (code length=2).

- 01 - Load and rate table.
- 02 - Drop and rate table.
- 03 - Area.
- 04 - Area and rate table.

TPSE

This is the serious errors (code length=2).

- 01 - Sales order not found.
- 02 - Pick note header not found.
- 03 - Item record not found.
- 04 - Stockroom balance not found.
- 05 - Route specified not found.
- 06 - Time out for requirements update.
- 07 - Details changed after AFI posting.
- 08 - Details changed after override.
- 09 - Global route change warning.
- 10 - Possible capacity problems.
- 11 - Item transport profile not found.

RSNC

Reason code.

TTYP

This is the text type (code length=1).

K - Carrier.

C - Customer.

O - Order.

T - Transport load.

USGC

This is the text usage (code length=1).

I - Internal.

P - Planner.

E - External.



Note: The Text Type and Usage Combinations are:

KI - Internal carrier text

CP - Planner customer text

OE - External order text

TE - External transport load text.

WAIT

This is the wait type for temporary record locking (code length=6).

BATCH - Batch lock request. Define BATCH entry with a limit of 5 for the record lock attempts and a rate of 20 for the wait seconds.

INTER - Interactive lock request. Define INTER entry with a limit of 5 for the record lock attempts and a rate of 2 for the wait seconds.



Note: The WAIT Description Identity controls how long a job waits for the release of a temporary record lock and how many times to retry to achieve a lock for updates.

Company Profile

The Transport company profile defines a number of primary defaults:

- When should the transport centre receive requirements from Sales Order Processing?
- Should transport load references be created at company or transport centre level?
- What are the significant decimal places for capacity types? (Weight, volume and number of containers.)
- What level do you plan orders, header or line detail? This determines if Transport Planning uses the stockroom code at header or line level to determine which the transport centre to place the order line requirements into.

Transport Centres

Transport centres define a delivery depot that process order requirements and loads. Consider the following:

- How many transport centres do you want to define for the company?
- How many inventory stockrooms do you want to associate with each centre?
- Should these stockrooms span sales companies, do you allow multiple company planning by a single centre?
- What is the default capacity type for the centre; weight, volume or number of containers?
- Should requirements be assigned to loads on initial creation, therefore bypassing manual load build and allowing Transport Planning to automatically add requirements to suitable loads?
- Should loads be available for release prior to confirming the despatch of all the orders assigned to the load?
- Should transport documentation be printed online when you release the load?
- Do you require confirm of load despatch?

Planners

You must define a planner profile for all users of Transport Planning. Consider the following issues:

- Can the planner plan loads across a number of transport centres?
- What level of authorisation should you assign to this planner. Can the planner:
 - Release loads?
 - Override capacity errors?
 - Authorise payments and adjustments?

Areas

You must define areas or zones so you can rate loads and produce appropriate documentation.

You should consider how these areas relate to a transport centre, that is, is it a geographical location or a radial pattern.

Routes And Nodes

You must define all routes together with optional start and end points. Consider the following issues:

- Are internal or external carriers associated with routes?
- Are vehicle types a factor, do certain vehicle types run on particular routes?
- Are origins and destinations applicable, is this between two locations rather than a circular trip back to the transport centre?
- Is formal documentation required for each route, do you need a bill of lading, or is a load sheet sufficient?

Vehicle Types

You only need to define vehicle types if you require analysis of transport requirements against vehicle capacities. Consider the following issues:

- Internal or external types or both (as defined by carriers).
- Are discrete weight and volume capacities important (these may be set as infinite requiring manual review/control of load capacities)?

Carrier Types

You must define all carriers. Consider the following issues:

- Is this internal transport or external carrier services.
- Is rating applicable?
- Are there any carrier route or rating restrictions?

Carrier Rates

You only need to define carrier rates if you want to calculate transport charges. This is generally only a feature of external transport of goods and rates are generally communicated on a contract basis via the carrier. Consider the following issues:

- Which of the rating methods is appropriate?
- How many rate codes and effectivity dates do you want to define?



Note: You may need to enter a large amount of data.

Items

You should review item details to assess capacity requirements for their transportation. Consider the following issues:

- Net weight and volumes per item issue unit.
- Tare weights (per issue unit or percentage of net weight) for any item wrappings.
- Quantity per default container.
- Container tare weight and volume.

Delivery Profiles

You must define all customer and distribution centre delivery points. Consider the following issues:

- Route and drop sequence.
- Default carrier.
- Does the customer require exclusive loads?
- Do you need to confirm delivery?
- Does the customer have specific times you can deliver?

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Chapter 7

Vendor Scheduling Overview



Introduction

Companies frequently need to purchase goods on a repetitive basis, and the accepted method of raising discrete purchase orders and receiving goods can be burdensome.

In addition, there is often a requirement to source supply from different vendors with whom a contract may or may not exist. The Vendor Scheduling module allows you to set up contracts and then source the periodic supply from one or more vendors, using MRP, MPS or DRP.

If any one item appears on more than one contract, the Vendor Splitting Rules will be defined. These rules show how suggested requirements are allocated across multiple vendors and / or across multiple contracts for the same vendor.

Only items that have the Vendor Scheduling indicator set on the Item Stockroom file are processed by Vendor Scheduling.

Features of Vendor Scheduling

The features of the Vendor Scheduling Module are:

- Full purchase contract control
- Optional contract restrictions (e.g. maximum value)
- Support for automated multi sourcing with splitting rules
- On-line review and maintenance of schedules
- Schedule send via fax or EDI.
- Supports re-ordering of consumable items (including automatic option)
- Fully integrated with other modules

Contract Maintenance

A contract may be maintained for purchase of one or more items from a vendor with details of effective dates, currency, total quantity, and value. More than one contract may be entered into for the same vendor and items. Items may be attached to the contract with appropriate price structures, sourcing rules and contract maximums for quality and value. Appropriate standard enquiries may be called to assist in this task.

Contract Control

When a buyer has identified a source of supply and has completed all contractual negotiations, a vendor contract will be established to formalise the agreement. Each contract in Vendor Scheduling is for one vendor, although there may be many contracts for each vendor. The contract is divided into two main components;

- The contract header details which relate to the contract as a whole
- The contract item details which are specific to an individual item

Individual companies can authorise different users to have access to Contract Maintenance and can further restrict those users who have the authority to authorise a contract. Once established and authorised the contract is printed and sent to the vendor for confirmation.

Each contract can optionally have further restrictions defined on the header and at item level. Traditionally many companies released blanket purchase orders that were open forever and had no quantity limits. The potential problem with this scenario is, if an employee ordered 5,000,000 items instead of 5,000 (due to a keying error) the liability is enormous. Using contract restrictions available within Vendor Scheduling can minimise these liabilities, particularly as the planning and re-ordering process becomes more automated which leads to reducing the non-value adding activities of an organisation.

The restrictions available are:

- Effective from and to dates
- Maximum values (in whichever currency)
- Maximum quantity

Each of the above can be applied to header and / or item detail level. As these restrictions are firm, i.e. the users cannot order more than the restriction allows, Vendor Scheduling includes a report to show which contracts are due to expire within X weeks, more than Y% complete and so on. The manual and automatic release options also produce exception reports showing contracts where dates or quantities have expired.

There is a lot more data that can be defined against a contract. For example, users can define up to three authorisation periods to override those set at company level (typically Fabrication, Material, and Forecast) as well as stipulating whether the schedule will be printed, and how often, transmitted by fax or EDI and whether history will be kept. The Vendor Scheduling module will also automatically track and update cumulative receipts, cumulative quantity ordered, and cumulative quantity returned for each item and each contract.

Multi Sourcing Rules

Many organisations have a policy of ordering certain high volume items from more than one vendor. This is supported in Vendor Scheduling. For each item a sourcing table may be set up which will be used to calculate vendor and quantity for any supply requirements. The sourcing rules, at the highest level, determine the effective dates of the sourcing rules and whether to source by price, vendor percentage or overall percentages.

At item level the rules determine the proportions that are placed on each vendor or more particularly against each contract and any batching rules that apply. The individual vendor's rules may or may not be attached to a contract.

The choice of three algorithms to be used to split the requirements for an item across multiple vendors are:

- Least cost
- Percentage of individual requirements
- Percentage of overall requirements

Least Cost

This rule allocates the particular requirement to the supplier which has the lowest price for the quantity required. There are two places this information is held, either for a particular contract, or from the item / supplier profile. The process is as follows:

For the requirement, find the lowest price for the particular quantity, firstly looking at the contract price breaks, then the item / supplier profile price breaks. If only one supplier has the lowest price, then allocate the entire requirement to this supplier. The prices are converted to base currency and compared at base currency level before deciding the vendor to chose.

If two suppliers have the same price for the same quantity and one of them is the preferred supplier on the item details (in inventory), then this supplier will be used for the requirement. If both are defined, then the primary one will be used first, then the secondary etc. If neither supplier is defined then the system will use the supplier with the highest year to date value (from the P/L period balances files), else the system will allocate in alphabetical order.

Percentage of Individual Requirements

This rule allows the allocation of one individual requirement to more than one vendor.

The multi-sourcing details file defines the criteria which will determine the allocation of the requirement. A vendors supply values will only be included in the overall picture in circumstances where a contract is entered if the effective dates on the contract (or contract item) make the item current. Each criteria will be considered in turn.

The apportionment value indicates the proportion of the requirement to be allocated to the specified vendor, if no other restrictions to the supply quantities are applied. This is the first criteria considered in the processing to determine the initial amount to be allocated to each vendor. All the proportion values for the vendors are to be accumulated (where a contract is entered, the value is only to be included if the contract item is current, which is determined by checking the dates on the contract and contract items file). Each individual vendors value is then divided by this total and the result multiplied by overall item requirement to give a basic requirement per vendor. This value will not be rounded until the final figure is derived. If proportion values are not entered, then progress down the list of vendors in sequence attempting to allocate all to the first vendor if possible.

Each vendor will be considered in turn for a complete set of rules, before progressing to the next vendor in the sequence.

After determining the apportioned value, the contract cumulative quantities to date must be checked (from the contract item details file), if the item is purchased against a contract, to ensure that the contract maximum has not been exceeded. If the contract quantity has been exceeded, the system will allocate to this supplier up to the contract limit, and attempt to allocate the remainder to other suppliers. If this is not possible, the system will retrieve the flag from the company profile which indicates if contract quantities can be exceeded, if this is 1, then the outstanding amount will be allocated to the original supplier, else the supply cannot be satisfied. The contract values can only be exceeded once, so if on performing this last potential reallocation, the contract 'ordered to date' values are already greater than the 'maximum quantity' of the contract item, then no further requirements can be added to the contract.

If a contract item maximum quantity is entered, this value must not be exceeded. If the derived value above exceeds this amount, then the maximum is to be ordered and the remainder allocated to the next vendor in the sequence, if this exceeds his maximum, the same is to apply to all vendors on the list. If all requirement has not yet been satisfied, then the remainder is to be added to the preferred vendor defined at the multi-sourcing rule header.

Minimum Quantity

This is the minimum quantity the vendor will accept on any individual requirement. If the amount is allocated to this vendor does not reach this amount, then the system will ignore this supplier and move to the next in the sequence if the round to minimum indicator is not set.

Round to Minimum

If this flag is set to 1, and the quantity allocated does not reach the minimum quantity, then the system will round up the quantity to the minimum amount. E.g. if the amount to be allocated is 300, and the minimum quantity is 5000, the system will allocate 5000 to this supplier.

Round to Multiple

If this field is entered, then the system will round the amount allocated up to a whole number multiple of this amount.

If no rounding has occurred to the amount originally allocated, the system will round the quantity up to the next multiple of the purchase unit of measure.

After allocating an amount to the first vendor, the system reapportions the amount of the original requirement left to the remaining vendors and allocates as with vendor 1. This is repeated down the list in sequential order until the original requirement is satisfied. If the end of the list is reached and there is still some demand unsatisfied, then this is added to the preferred vendor defined on the multi-sourcing rule header file.



Note: If rounding (or reallocating any 'left over'), causes the maximum quantity to be exceeded then there is no recalculation, or any attempt at correction for this particular circumstance.

Percentage of Overall Requirements

This is very similar to the proportion of individual requirements, but the amount is allocated in accordance with the overall allocation amount to date for the particular rule. These values are maintained at multi-sourcing rule item level by schedule release.

This rule allows each individual requirement to be allocated to a particular vendor as a whole, but the vendor may vary.

The system retrieves the quantities to date for each item as recorded on the multi-sourcing rule detail file. It calculates the total for all vendors and divides each vendors quantity by the total to determine the proportion allocated to each vendor to date. Next the system will calculate the required proportion in the same manner as for the individual requirements (dividing the individual proportions on the multi-sourcing rules by the total for all proportions). Next, the actual to-date proportions are subtracted from the multi-sourcing rule proportions and the vendor with the greatest difference is determined. The entire requirement is allocated to this vendor.

The maximum rule does not apply.

Round to minimum / round to multiple apply as for rule 2 above.

If the allocated amount falls below the vendors minimum, and the flag to round up to minimum is 0, use the vendor who is second in the list of furthest away from the annual percentage, and so on. If the requirement falls below the minimum fall to all suppliers, and the rounding flags are 0, then the whole amount is allocated to the preferred vendor.

Create Schedules From MRP / MPS or DRP

With the base data established, the supplier scheduler reviews the results of the latest MRP / MPS / DRP planning processes and then runs the Vendor Scheduling function. The scheduler then reviews the results of the process with the Schedule Maintenance facility. This will show all schedules that have a source of supply problem that should be corrected before releasing the schedule.

Automatic Schedule Release and Schedule Maintenance

Auto Release

Vendor Scheduling offers the capability to fully automate the reordering process. Following an MRP planning run, an automatic update of supplier schedules can be initiated which will action all the MRP purchase suggestions for items controlled by Vendor Scheduling and purchase order lines are created. The automatic update function also provides a simulation facility which prepares a pre-list report of all schedules without actually confirming the process. The report lists contracts that have insufficient quantity remaining to fulfil the requirements of this planning cycle.

Processing

The schedules will be read in the sequence of the vendor / item / stockroom / contract. For each schedule only the current schedule will be processed (since all past schedules will have the history flag set to 1).

If the selection is set to only include schedules due for release then only process the schedule if the expected date of this schedule, on the schedule header is less than or equal to the assumed release date which has been entered.

Historic schedules that have not been issued or have not updated purchase order files will be purged in a separate function.

The processing of a schedule involves the update of the purchase orders. Once a schedule has been processed its status will be set to 'update / to issue' so that it is picked up by the print run. No updates will occur if the pre-list run type is taken.

Purchase Order Creation and Update

For each schedule, the date of the end of the FAB authority will be obtained from an ascending hierarchy of contract line and company profile. The schedule details will be read up to this date and converted into a purchase order.

If there is one purchase order for the contract then the purchase order will be obtained from the contract header. If there is one purchase order for all items in this run the purchase order will be the next number from P/O company profile. If there is one purchase order for each item in this run a new purchase order will be retrieved for each item.

Purchase orders will be created in the currency from a hierarchy of contract and vendor and in a unit of measure from a hierarchy of contract details and item purchase unit.

Landed costs will be attached if they exist at item / supplier level.

Schedule Maintenance

If schedules are generated that contain errors; e.g. insufficient quantity available on the contract, then these schedules will be reviewed using the Schedule Maintenance facility. All schedules available for release are able to be viewed and amended using this facility.

Schedules are printed and can be sent to suppliers using any combination of three different ways: post, by direct fax link or by EDI. The choice of which method is

defined on the contract for each vendor / item combination. The schedules are printed in a 'one item pre page' vertical format on A4 size layout. Vendor Scheduling also offers the capability to send schedule data directly by fax or EDI.

Method of Use

Supply suggestions are converted into Vendor Schedules by comparing the pricing and sourcing parameters, from one or more vendors, to determine the preferred vendor(s) at any one time. The pricing and sourcing data may be contracted or may be for items not under contract. Similarly the same item may be under two different contracts for the same vendor.

In addition to MRP, MPS or DRP, Inventory re-ordering will look at the sourcing rules for items that are multi-sourced and create a schedule albeit for a single delivery, when the stock levels reach the re-order point. Items not defined as vendor scheduled on the item stockroom file will not be processed by vendor sourcing.



Warning: It is critical that all MRP purchased items that are sourced through Vendor Schedules have MRP filters associated to not allow reschedule in, reschedule out, increase quantity, or decrease quantity. The only allowable values are for suggest delete. This will allow the Sourcing Rules and Contracts within Vendor Scheduling to action the demand requirements correctly.

The MRP runs, including net change, may be run any number of times between sourcing runs, since vendor sourcing will always pick up suggestions from the last run.

Once the source of supply has been determined you may review the schedules for all vendors / contracts and the individual vendor's schedule with the ability to override the system generated quantities and dates. Thus there will be no conflict between the number of times vendor sourcing is run and schedule review / release.



Warning: It is critical that the schedule print program is run after release and before the next vendor sourcing run.

After reviewing the schedule the user may release it, in which case purchase order lines will be created or amended and the new schedule will be ready to be sent to the customer.

Schedules are released and printed and sent to the supplier by post, fax or EDI.

Facilities are provided to automate the schedule sourcing, release and print / send functions so as confidence in the system grows the purchase officer(s) spends less time managing the system and more time on exceptions and other value adding activities.

The Vendor Sourcing Process

A batch run processes the source data from MRP, MPS, DRP, and inventory re-ordering.

Whenever run is selected, then the last run date and time for the user will be displayed.

MRP and MPS

This is a batch run which performs the multi-sourcing. The source data files for MRP and MPS are identical.

The function reads the supply file by company and active model code (retrieved from the MRP company profile field), retrieving the company, stockroom, item, planner, quantity, and due date / time (demand / supply date). Only those records which satisfy any selection criteria entered, have a record status not equal to processed, and are purchase items with the vendor schedules indicator set on the Item Stockroom profile will be selected for sourcing.

DRP

This functions in a similar manner to the MRP / MPS run, however, included in the selection will be the code or DRP centre entered on the selection screen.

The selections to be included are exactly the same as for the MRP / MPS items, but also select only those items within the entered distribution centre.

Inventory Reordering

The process to run Inventory ROP calculations to create vendor sourcing records as opposed to requisitions, is to allow items which are not controlled by MRP etc. to be purchased against a contract, or with multi-sourcing rules applied.

Any items which satisfy the criteria for inclusion into the schedule files are processed by vendor sourcing.

Item Stockroom

Only item records from the item stockroom file that have a vendor sourcing indicator set to 1 will be processed by vendor sourcing. This means that for some items in a stockroom their purchase is controlled by Requisitions and some items may be controlled by Vendor Scheduling.

Running Multi Sourcing

Each module source record is passed through the multi-sourcing algorithms to ensure the correct vendor is selected. Even if the item has no multi-sourcing algorithms the requirements are still to appear on the schedule file. This will allow the requirements to be reviewed by the schedule maintenance function.

If an item has a contract, but no multi-sourcing rules, then only the contract maximums are to be checked.

Output

The two output files are the schedule header and schedule detail files.

After the allocated amounts have been decided for each vendor, the schedule records are created, with the appropriate priority if required. The most important

priority has the highest number. If a particular schedule has more than one priority applicable, then the highest numbered priority is to apply.

The priority is user defined on the purchase management parameters file, type PRTY. The number to be updated on the schedule header file as the priority is the value field from the parameters file.

The following priorities are defined on the parameters file in descending order:

95=Invalid Vendor Code

90=Exceeded Contract / Not Ordered

80=Contract Unauthorised

70= Exceeded Contract / Ordered

60=Exceeded Contract / Ordered on Alternative Vendor

50=Significant Change Period 1 - for future use.

40=Significant Change Period 2 - for future use.

30=Significant Change Period 3 - for future use.

20=Significant Change Period 4 - for future use.

10=Significant Change Period 5 - for future use.

05=Supply placed on Preferred Vendor

00 (Blank)=Normal

Priorities 95, 90, 80, 70, and 60 have already been decided when running the multi-sourcing rules. Priority 05 is to apply if the multi-sourcing rules have 'failed' and supply have been placed on the preferred vendor, without any problems with contract cumulatives. The priority codes and associated descriptions are held on the Inventory Descriptions file type PRTY.

Enquiries

There are four main enquiries which will allow you to review the schedule information on the system.

Contract Enquiry

This allows you to view the contract details. The enquiry is solely to provide static information concerning the contracts.

Contract Item Enquiry

This enquiry allows you to view the Contract Item Details.

Sourcing Rules Enquiry

This enquiry allows you to enquire on multi sourcing rules set up for particular items.

Schedule Enquiry

This allows you to view details relating to a schedule that has been previously released and sent.

Reports

There are a number of reports which you can use to keep track of your schedules.

Contract Report

This allows you to list the contract details. The report is solely to provide static information concerning the contracts.

Contract Exception Report

This report allows you to determine those contracts which are about to expire, reach the limit etc. By careful selection of criteria those contracts which are reaching the end of their life, or are close to the maximum values defined can be easily identified.

Multi Sourcing Report

This report allows you to report on multi sourcing rules set up for particular stockroom and / or items.

Schedule Exception Report

This option allows you to list, by exception, the schedules which have been created. This will enable you to easily identify those items which may need manual intervention prior to the scheduling being issued.

Schedule Summary Report

This report allows you to list the created schedules in a summary format selected by particular criteria to limit the report.

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Chapter 8

Warehousing Overview



Introduction

Warehousing provides different levels of support for different organisations, or different warehouses within the same organisation. You can use Warehousing to provide a tracking system for a limited range of key items. You can expand this to encompass all aspects of warehouse control and management.

The map of the warehouse can include proximity algorithms. You use these algorithms as one way of deciding where you locate items.

You can base picking and marshalling by order, journey, customer, and due date. Warehousing takes account of expiry dates, lots or batches, pallet numbers and warehouse organisation when allocating stock to picking instructions.

Warehousing is designed to fit into the operational environment of a warehouse, you only need to enter the minimum of data to operate it. All you normally need to do is confirm the lists of tasks. The database easily interfaces to technical advances in warehousing such as hand-held terminals.

Relationships to Other Modules

The only module which is a pre-requisite for the operation of Warehousing is Inventory Management.

Warehousing is designed to be used in conjunction with, and enhance the following modules:

- Sales Order Processing
- Purchase Management
- Production Definition Management
- Production Order Control
- Production Schedule Control.

Once you have installed Warehousing and fully activated a stockroom as a warehouse, links are switched on between the operational modules and the warehouse to provide additional location management and control.

You can automatically print pallet labels for items received against a purchase order. The labels can include put-away instructions. You record despatches for sales orders in Sales Order Processing and you can pick, marshall, and despatch, issues for production orders and schedules. You can make receipts from production orders and schedules into a warehouse and start put-away routines.

Warehousing Configuration

You can operate Warehousing for a number of companies, keeping the characteristics of each on control files. Warehousing uses the control data stored for Inventory Management. The precise control, however, is at warehouse level because you could have different policies in different parts of the organisation.

The policies you control at each warehouse are:

- How you give instruction and record actions.
- Check digits used to confirm actions.
- Stock rotation.
- The sequence in which to produce action lists.

All actions resulting in an Inventory receipt automatically locate the item in the receiving location. Similarly, when you require picking, Warehousing recommends that you move stock for items from certain locations to the marshalling location before issue.

You can define any locations set up as the marshalling and receiving location. However, we recommend that you set up warehouse areas so that each contains a single location with a Location Code which is the normal name for receiving and marshalling in the warehouse. You can set up these areas to satisfy the requirements of each individual warehouse.

Accuracy and discipline are the fundamentals of the warehousing operation. If warehousing detects an error, it directs a message instantly to a nominated user.

You must set up for each item managed by Warehousing an item/warehouse profile that defines its rules and characteristics. You can set up defaults for the warehouse to minimise the amount of data entry required when you maintain these profiles.

Since a user of Warehousing is normally at a particular site, you can define a default warehouse. This makes the day-to-day running easier particularly for recording transactions and making enquiries.

Maintenance of Reference Data

You set up much of the data for Warehousing when you install the base modules. You must, however, enter additional information to get the maximum benefit from Warehousing.

You can maintain the following information:

Warehouse Profiles

Use the warehouse profile to:

- Set up the warehouse policies, such as confirmation check digits.
- Set up the special areas for receiving and marshalling.
- Define who receives the error messages.
- Define defaults used in the item/warehouse profiles.
- Activate or de-activate the warehouse.

Area Profiles And Dimensions

These define the shape and structure of the warehouse. You define a series of areas, each of which is divided into a number of locations. For example, you could define three areas, Floor used for the receiving and marshalling locations, Main Store to represent most of the aisles, rows and bays in the warehouse and Picking to represent the pick locations.

You use up to four dimensions to define the area, such as aisle, bay, row. The dimensions are used to build up the code that identifies individual locations. Each code can be up to six characters in length, but the final location code (the concatenation of all the dimension codes) cannot exceed twelve characters.

Item/Warehouse Profiles

You must associate each item kept in a warehouse with a location. Warehousing makes recommendations for where to put items, when and from where replenishment should take place, and from where items should be picked. You must define a profile for each item in each warehouse. The profile contains five type of information:

- Packaging - This relates the item in inventory stockroom issue units of measure to the packaging types used in the warehouse. It also defines the basis of stock rotation.
- Fixed Locations.
- Put-Away - This defines the steps Warehousing uses when making recommendations as to where an item should be put.
- Commit - This defines the steps Warehousing uses when making recommendations of where an item should be picked from.
- Pick Location - You can set up information including replenishment trigger and the replenishment quantity. You can also define the maximum, in a single pick, to take from this location before the pick is made direct from bulk.

Packaging Types

You move many items in a warehouse using pallets or containers of some kind. You must define these to Warehousing and then create the relationship between the pack and item. You must set up a default pack type for each item.

Location Types

You chose the size, shape, and use of locations to meet particular storage requirements, including the number and distribution of pack types. Warehousing uses this, together with the number of packs per location, to decide whether to put away a receipt in a location. You must then define the distribution of location types throughout the warehouse areas. Also you can remove locations from use by tagging them as void, and limit their use to single-item.

Warehouse Map

The warehouse map assigns location types to locations to determine the characteristics of the locations. For example, the locations within the Main Store area can have a location type of Standard Pallet Racking which limits the locations to storing only one pallet at a time.

Action Rules

You should only move an item in a warehouse as a result of an instruction. You must decide the best location for the item depending upon factors such as:

1. The item itself.
2. How it is packed.
3. Does it require special conditions?
4. Is the item picked from a picking location?
5. Can mixed lots be put in the same location?
6. Must the receiving area be cleared quickly?

You can create Warehousing rules to mimic these decisions. You can set up the following to control the recommendations for put-away, pick and replenish:

1. Use the pick location.
2. Use the fixed locations.
3. Use an item rule. This reproduces a line-of-sight check. To create the rule you must define a series of locations to test in relation to the pick location.
4. Use an area rule. This type of rule can make use of the proximity data set up for an area or the randomising algorithm. This rule reviews a large number of locations to answer such questions as:
 - Find the first not-full location in this part of the warehouse, closest to the receiving bay
 - Select randomly a not-full location.
5. Use stock rotation dates, such as expiry and arrival date.

Resources

Resources are warehouse equipment, such as trolleys and fork lift trucks. You can define a resource then relate it to pack types within a location type. For example, you can relate trolley to a pack type of pallet in a location type of floor.

Check Digits

You use check digits or return codes to make sure that the warehouse staff have visited the defined location. You confirm the action by entering the check digit. For an intra-warehouse movement, you must enter both check digits. To use check digits:

1. You set up the warehouse to use check digits.
2. Set up a set of default check digits for the warehouse. You can override check digits for each or all areas.
3. If you use check digits, but do not set up a default, Warehousing uses modulus 23 to generate digits.

Processing

You can divide most of the work in a warehouse into three types of tasks:

- Putting away items.
- Moving items to optimise storage, picking and despatching.
- Picking and despatching.

Warehousing makes recommendations for each of the three tasks, and produces instruction lists on paper and on screen. You can confirm these tasks have been completed by entering 1 against the action, or, if you use check digits, entering the appropriate digits. If you do not use check digits, you can set Warehousing to confirm actions by default. You can override those actions.

You can change actions to show you used different locations when carrying out the instructions.

You can carry out some of the processing in a background subsystem or in batches. Running jobs in batches can be especially useful for printing, where special stationery for work planning is required.

Receiving and Putting Away

When you receive and put away an item you must record the specific actions and process the recommendations made by Warehousing.

The steps to receive an item and put it away are:

Inventory Receipt

Record the inventory receipt in one of the operational modules, Purchase Management, Inventory Management, Production or Sales Order Processing. This increases the inventory balance at the stockroom. You must therefore increase stock in the warehouse. The sub-system moves the item into the Receiving location. This balances the warehouse and inventory without an explicit additional action in the operational module. However, as Warehousing can operate with multiple packaging types, you can confirm or change the packaging type when you receive the item into a warehouse.

The receiving sub-system:

- Moves the item into the receiving location.
- Determines the number of packs.
- Identifies the rules for putting the item away.
- Makes enough recommendations to put the item away into the warehouse.
- Produces one recommendation for each unique combination of item, lot, packaging type and location.

Warehousing plans the item out of receiving and into the locations.

Printing Documents

Before you confirm an action, Warehousing must generate an instruction. If you requested pallet labels, Warehousing prints them, otherwise it a list. The labels and list only print in the warehouse when requested.

Confirmation of the Action

You can confirm put-away by entering a single character or check digit found at the location. You can use the sub-system to update of the database so the update is normally complete within seconds of finishing the entry.

Intra Warehouse Move

You must be able to record that you need to move stock within a warehouse and that you have moved stock. You may need to move stock in the warehouse because you are moving bulk to a picking area or refurbishing part of the warehouse. If the need for action is immediate, you can report the move to Warehousing after the event.

Warehousing produces a list of instructions which you can see on a screen and confirm or change.

Warehousing distinguishes between movements generated for replenishment and those manually requested. You can produce these on separate lists so you can assign the proper priority for the different actions. When you confirm the actions there are different overrides are appropriate for the different moves. For example, if this is a replenishment move, you cannot change the item or the move to location.

Picking and Despatching

You initially allocate stock to sales orders in Sales Order Processing or Production Control and Costing. This allocate against the available stock figure in the nominated stockroom but does not identify where to find the stock. You use Warehousing to find the stock.

Warehousing decides the stock location by reviewing such factors as:

- The quantity to despatch.
- Stock rotation.
- Physical proximity of location.
- The practicality of picking a number of order lines together.

The picking and despatching cycle has the following events:

Request Picking Lists

Warehousing considers requests to despatch generated from Sales Order Processing or requests to issue generated from Production Control and Costing. Warehousing performs a series of tasks to print of the documents necessary to pick and despatch the items. These are:

- Select a range or group of orders.
- Sequence and combine them.
- Decide from which locations to pick the stock.
- Print a pick list organised to suit the warehouse
- Print appropriate marshalling lists which cross-reference picking lists and despatch or issue notes.
- Print the despatch or issue notes.



Note: Kit parent and non-stock items are not committed and do not print on the warehouse pick list.

Confirm Pick

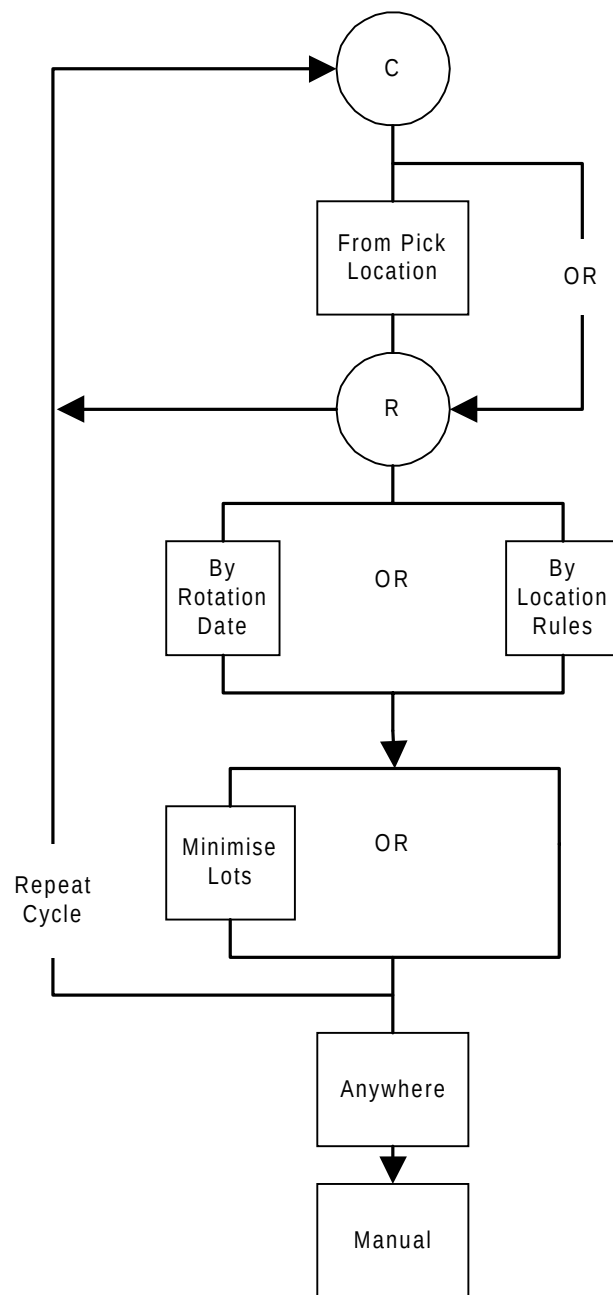
You can confirm the pick by entering either the check digit or return code. You can override the module instruction by changing both the location and the quantity. This will result in Warehousing recording an error. When you confirm pick, Warehousing transfers the stock to the marshalling location.

Confirm Despatch / Issue

You can confirm despatch from the marshalling location at order level using the standard picking note, despatch note or shop documentation produced by Sales Order Processing and Production control and costing.

You can produce reports to help marshall complex loads. The reports link picking lists and despatch or issue notes. If the despatch has taken place as intended, you can use a single function to confirm it. For despatches you can add delivery and freight charges, and reprint the despatch note with actual despatch quantities.

Commit and Replenish Cycle



Warehousing performs this process for each commit action. A commit action is at the level of item, lot, stock rotation date, pack type or pack number.

Enquiries

You can group enquiries into one of three categories:

Status

You can use status enquiries to display what is in a location, or where you store a location. You can also sequence the enquiries by location, area or item rule.

You can display:

- Item details
- Location contents
- Location details.

Action Lists

There are three types of action lists:

- Put away
- Move
- Pick.

You can display the unfinished lists. You can also display details of despatch instructions sent but not yet actioned and then access to the full order enquiry within Sales Order Processing.

Movements

You can display the warehouse movements and the reported error conditions. If an action is requested but not then confirmed as instructed, an error record is written. You can use multiple selection criteria for both movements and errors enquiries.

Reports

You can use selection and sequencing parameters to define a report before running a report. Standard reports print on standard listing paper with 132 print positions and 6 lines to the inch.

All reports print the procedure or program number in the top left, and the page number, date and time to the top right.

Implementation

Before implementing Warehousing you must consider:

- How Warehousing and other Modules should be used.
- How many warehouses should be set up.

Procedure

There are two possible implementation procedures:

- Warehousing with Inventory already installed
- Warehousing with Inventory newly installed

Use the appropriate procedure as a guide.

We recommend that whichever procedure you use, you follow a project plan. This is so you can capture and record the necessary data before entering the stock count.

The basic steps for a successful implementation are:

- Master file data collection and input.
- Work load organisation.
- Suspend processing of stock.
- Initial date take-on and reconciliation.
- Resume processing of stock.

Warehousing with Inventory Already Installed

This assumes that you are familiar with Inventory Management and you have been trained to use Warehousing.

Master File Data

Define the data required to make Warehousing functional. The amount of effort required depends upon the number of items, locations, pack and location type details which you need to define.

Work Load Organisation

It is essential that during the stock count exercise and the early stages of processing with Warehousing that you control the movement of stock. Try to minimise the activities for receiving and issuing stock.

Receipts

Review any receipts which have been booked into stock but have not been put away. Then:

- Put the items away, reducing the items at the receiving area to those not yet booked into stock.
- If the receipts were made through Purchase Management transfer them back to either goods inwards or inspection. Then receive the items into stock after activating the warehouse.
- Leave the items at the receiving location. Record the stock at Data Take-on and then move manually after activating the Warehouse.

Issues

Process all requests for issuing stock. Pick Items to despatch and move them to the Marshalling location. Where possible despatch the items and confirm the actions. If you leave items at Marshalling, you must enter the full details required for Confirming Despatch.

Suspend Processing of Stock

Suspend stock movement into, around and out of the warehouse whilst you perform and reconcile the stock count for Data Take-on. Complete any outstanding transactions.



Note: From this point until you activate the warehouse, suspend the processing of Inventory, Purchasing, Production and Sales Order Processing.

Initial Data Take On/Reconciliation

Do a full stock count including every item and every location in the Warehouse. Enter the details using Data Take-On programs.

Reconcile the count to the Inventory lot and stockroom balances, discrepancies are shown on the report. Verify any variances. Once the count is cleared as correct,

update the location details using the Reconciliation with Update option. This also makes appropriate adjustments to the Inventory balances.

Resume Processing of Stock

Activate the warehouse.

Resume transaction processing in all areas of the warehouse. You can record transactions in the modules previously suspended.

Receipts and issues are now subject to the control of Warehousing.

Warehousing with Inventory Newly Installed

You must undertake training so you can understand both Inventory Management and Warehousing.

Master File Data Collection and Input

Enter the data required to make both Inventory and Warehousing functional. Make sure that you enter dependencies in the appropriate order. The amount of effort required depends upon the number of items, location, pack and location type details that you need to define.

Work Load Organisation

During the stock count exercise and the early stages of processing with Warehousing, you control the movement of stock.

Review items that are in the receiving area. Where possible do the following:

- Put the items away before the stock count.
- Record, at data take-on, the items in the receiving area. You must enter manual moves after activating the warehouse.
- Leave the items at the receiving location, do not record the stock during data take-on. You receive the items after activating the warehouse.

Process items to despatch through to the marshalling area. It is better to start with as few incomplete despatches as possible.

Suspend Processing of Stock

Suspend the movement of stock whilst performing the stock count for Data Take-on.

Movement of stock can resume after activating the warehouse.

Initial Data Take On / Reconciliation

Do a full stock count of every item and location in the warehouse. Enter the details either through the Data Take-On programs or by recording Physical Stock Adjustments in Inventory.

If you use Data Take-On, use Reconciliation with Update to set the Inventory balances.

Resume Processing of Stock

Activate the warehouse.

You can now resume transaction processing in the warehouse.

Receipts and issues are now subject to the control of Warehousing.

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Chapter 9

Distribution Requirements Planning Overview



Introduction

Distribution Requirements Planning provides the necessary data for matching customer demand with the supply of products at various stages in the physical distribution system.

DRP also helps management to anticipate future requirements in the field, more effectively deploy inventory to meet customer-service requirements, and more rapidly adjust to the vagaries of the market-place.

DRP evaluates the demand at the Distribution Centres (DCs) and generates supply plans for centres at a product level. These time-phased plans, or schedules, provide the information planners need to match supply and demand.

The planner can then confirm suggested supply orders, check the feasibility of suggested changes, monitor data integrity and assess the impact of new situations.

Relationship To Other Modules

DRP is one of the Advanced Application Modules which extend the functionality of Base Application Modules.

The applications which are pre-requisite for the operation of DRP are:

- Inventory Management
- Sales Order Processing
- Purchase Management
- Requisitioning
- General Ledger
- Accounts Receivable
- Accounts Payable

DRP also interfaces with, and enhances, Warehousing, Production, World Trade, and Forecasting.

At DCs, when you confirm shipment receipt, you can enter warehousing details to include the items in the warehousing put-away processing.

At the Supply Centres, you can pick items by Distribution Order, within Warehousing. A Distribution Order is an internal order for transferring goods from Centre to DC, DC to DC or DC to Centre. It is generated from a DRP suggestion or entered manually, and is handled as an internal order within Sales Order Processing.

If the Supply Centre is a Production site, the Distribution Orders are considered as demand within the Production systems MPS/MRP processing runs. (To achieve this define items at the Centre to Production).

If World Trade is attached, you can add or amend Distribution Orders in both Distribution Requirements Planning and World Trade (only DRP files are updated). When you transfer of goods, which cross an EC Border (via order processing/warehousing confirmation of despatch), movement records are written for processing in World Trade and are used to produce Intrastat and ESL reports, and to post the acquisition VAT due on receipt of the goods.

You can transfer sales forecasts from Forecasting into any DRP Centre or DC, at item or minor item group level.

DRP Configuration

You can operate Distribution Requirements Planning for a number of companies. You maintain the characteristics of each company on control files.

DRP uses the control data stored for Inventory Management. However, you need to enter further information at company level specifically for DRP.

You can control the following DRP policies at each company:

- Method of forecast consumption.
- Maximum number of lines per distribution order - that is a supply order from Centre to DC. You can change this when you create orders from DRP suggestions.
- Default calendar parameters.
- Percentages of quotations to include DRP runs (only if Advanced Order Entry is installed).
- Default absolute forecast error tolerance percentage.
- Calculation of safety stocks.
- Period forecasting.

Consumption of Forecast

You can assess demand by comparing the sales forecasts with actual sales orders by period and to plan for the greatest.

Also you can assess demand using the Consumption of Forecast. This compares cumulative forecasts with cumulative demand. This is because if a period demand exceeds the period forecast it may be because demand forecast for a following or prior period had either arrived early or late. That is the quantity forecast is accurate but the period due is not. Consider the following demand:

<u>Period Start</u>	<u>1 Mar</u>	<u>8 Mar</u>	<u>15 Mar</u>	<u>22 Mar</u>	<u>Total</u>
Forecast	100	100	100	100	400
Actual Sales	150	90	30	110	380
Demand A	150	90	60	100	400
Demand B	150	100	100	110	460

Where Demand A uses forecast consumption, and B does not.

There are three modes of Forecast Consumption:

- Consumption throughout a DRP run without reference to period ends.
- Consumption of forecast over individual periods.
- Forecast consumption over individual weeks.

These generate different demands for re-supply which mean different opening and closing stock positions, affecting stock holding costs and customer service.

The mechanics of consumption of forecast are:

1. Sum the forecast to date.
2. Sum the actual sales to date.
3. Set the demand as the greater of forecast and actual sales.
4. Sum the consumed demand to date, excluding the current period.
5. Calculate the consumed demand for the current period as Demand to date (from 3) - Consumed Demand to date (from 4).

Reconsidering the previous example, then:

<u>Period Start</u>	<u>1 Mar</u>	<u>8 Mar</u>	<u>15 Mar</u>	<u>22 Mar</u>	<u>Total</u>
Forecast	100	100	100	100	400
Actual Sales	150	90	30	110	380
Forecast to date	100	200	300	400	
Actual Sales to date	150	240	270	380	
Demand to date	150	240	300	400	
Consumed Demand	150	90	60	100	400

Calendar Parameters

You can enter a default calendar code for the company. DRP planning runs use this for any Centre or DC which does not have a calendar assigned to it.

You can enter the following period details:

- Number of periods per year.
- Period Names.
- Number of days per period.

The DRP Calendar Maintenance function uses these when generating a calendar.

You can also maintain a standard week template. This specifies the normal working week. DRP Calendar Maintenance displays and uses this when generating calendars.



Note: The DRP planning runs suggest actions only on working days.

Maintenance of Reference Data

Before you use Distribution Requirements Planning you must plan and prepare carefully.

Consider the supply chain network and flow of products between them, particularly the relationship between distribution centres and stockrooms. For planning purposes, you can associate multiple stockrooms with a distribution centre yet for any item only define one stockroom as the primary receiving and issuing store.

As this is an advanced application module, you set up much of the primary data as part of the base modules. However, you must enter additional information to get the maximum benefit from DRP.

Reference Information Required

You need to maintain Information for the following:

Company

This contains control and parameter information. You must set up the company in Inventory Management. It defines the default calendar, maximum number of lines on the Distribution Order, whether you require consumption of forecast, quotation percentage (if Advanced Order Entry is installed) and forecast error tolerance.

Calendars

This defines, by year, alternative calendars used for planning. Different Centres or DCs and seasonal profiles can use different calendars. You can enter a number of calendars per company. For each calendar, you can maintain up to five week templates specifying the working days in a normal week, and the dates to which it applies.

Periods

This defines for a Centre, or DC, the forecast periods used for planning. You maintain the period intervals for the planning runs per Centre or DC. You can specify the intervals by calendar days. You can specify up to 46 periods. The DRP planning runs assess the supply and demand in each period, beginning at the start date for the run.

Stockrooms

This defines the stockrooms associated with a site and controls which stock balances, demands, and supply orders DRP considers during a planning run. You can define any number of Inventory Stockrooms to a single Centre or DC. You must have defined the stockroom to Inventory Management and specified periods for the Centre / DC. You must assign each stockroom an SOP dummy internal customer number, to reflect distribution orders raised for that stockroom.

If there is only one stockroom per site, you can also assign the stockroom code to the Centre or DC. You can only assign a stockroom to one Centre or DC.

For multiple stockrooms, the DRP planning runs extract stock, supply and demand information for all stockrooms at the selected DC or Centre.

Customers

This defines internal customer codes associated with stockrooms for distribution centres to generate order details and despatch documentation that support stock transfers between Centres / DCs. You must enter customer numbers for each DRP defined stockroom. These are valid customers from within Sales Order Processing with a delivery sequence code of 001. These customers hold the internal distribution orders raised for supply to that particular stockroom.

Supply Networks

This defines the supply chain planning relationship. You must specify the relationship of Centres or DCs within a network. You must have defined periods and stockrooms for each Centre / DC.

You can define multiple networks for a company and include sites in more than one network, normally as a Centre in one level and a DC in another. You must set up a supply network for each level of supply, normally, the Centre of one network is a DC, supplied by an alternative Centre, within the other network. You can define a site to any number of networks, using the same DC or Centre.

Simply give a network a code, description, Centre, and all DCs supplied by this centre.

Planner Codes

This defines the person responsible for planning an item or range of items. You must maintain at each centre and DC each planner who will use DRP. You enter personal details and a cross reference to the purchase officer number in Purchase Management.

You assign this planner code to items, for which the planner is responsible, in the item master details.

Item Group Profiles and Forecasts

For each item group for any centre or DC you can define the percentage sales of each item and maintain sales and stock forecasts by value or quantity. You enter an annual forecast spread over the weeks in the specified year.

Item Sales Forecast and Stock Levels

You can generate Item Sales Forecasts from the item group forecasts and profile. You can maintain the forecast manually.

You can generate Item Stock Levels from the item group profile and target stock quantity and range of dates. You can maintain these manually.

Net Change Reason Codes

This controls which operational transactions set net change processing flags and affect the efficiency and run times of DRP processing in net change mode.

Supplier Schedule Profile

This controls the format and scope of vendor supply schedule reports.

Seasonal Profiles

you can define period, weekly and daily indices which you can use to spread sales forecasts and stock build requirements interactively and automatically.

Processing Profiles

To update inventory when you despatch and receive distribution orders, you must set up the DRP Processing Profile record in Inventory Management. Refer to the Utilities section of the Inventory Management Product Information for further details.

Item Master Details

This defines by item and Centre / DC the re-supply rules used for planning. You must define items to the local stockroom within Inventory Management. You must define the Centre or DC to at least one network.

You do not need to define an order policy at the centre for a centrally supplied item unless the item is included in the DRP run for the Centre.

For DRP to control how its supply is administered you must enter the following:

- How much is ordered when a net requirement has been calculated?
- Where is the main source of supply?

To satisfy the first requirement specify an order policy and the appropriate quantities. The choice is as follows:

Order Policy	Description	Mandatory Quantities
A	Discrete. Order the quantity required.	None.
B	Discrete above minimum. Order quantity required if greater than minimum. Otherwise use minimum quantity.	Minimum order quantity.
D	Fixed quantity. Generate multiple orders of fixed quantity to satisfy the required quantity.	Fixed order quantity.
G	Number of days supply. Accumulate the requirements over specified number of days and suggest an order to satisfy the demand.	Number of days supply.
H	Fixed quantity above minimum. Order quantity must be greater than minimum and divisible by the multiple quantity.	Minimum order qty. Multiple order qty.

Satisfy the second requirement by specifying whether you purchase the item locally or centrally or manufacture it centrally. If you supply the item centrally, you must enter a centre and stockroom defined to the Centre. DRP then creates a distribution order line for this central stockroom.

You must enter item master details for the item at each DC it is sold from. If you centrally supply the item, you must also enter details at the supply centre.



Note: If the supply centre is a production site, define these items to the Production System and plan them via MPS (after evaluating the DC demand and passing demand to the centre as distribution orders).

Otherwise, define the items at the centre to DRP and the central supply plan created within it.

Starting to use DRP

When you have set up the basic data you can start using DRP.

The system can access data maintained through other modules; for example, from Inventory Management - stock; from Sales Order Processing - outstanding sales orders quotations (if AOE installed); from Purchase Management - supply from external sources; from Forecasting - Item and Item Group sales forecasts.

If there are outstanding transfer orders between distribution centres, you must enter the details manually using the ad hoc Distribution Order Maintenance functions. If the number of orders are significant and on an existing computer system, it may be more cost effective to transfer data using a conversion routine commissioned for the purpose.

Implementation Procedure

After installation, the recommended implementation steps for DRP are:

1. Within Sales Order Processing, create customer records for each stockroom, with a delivery address number of 001.
2. Make sure the necessary processing profile records are in Inventory Management. Refer to Inventory Management Product Information (Utilities section) for further details.
3. Create the DRP company profile record, using the same company code as the one defined to Inventory Management.
4. Create the DRP calendars for this year, and any subsequent years.
5. Design the DRP networks, and how / when they are processed.
6. Create the DRP period records for each centre and DC.
7. Create the DRP stockroom records for each centre and DC.
8. Assign the internal customer number to the stockroom within DRP Customer Maintenance.
9. Create the DRP network and planner records.
10. Define the Item to all stockrooms at centres and DCs via Inventory Management.
11. Create the distribution item master records for every centre or DC which stocks or sells the item.
12. Create item group profiles and forecasts.
13. Create seasonal profiles.
14. Generate item sales forecasts.
15. Maintain the generated sales forecasts, where appropriate.
16. If you require stock building, enter Item group stock level parameters and create item stock levels.
17. Maintain the generated item stock level forecast, where appropriate.
18. If you are using net change DRP, set up net change reason codes.
19. If you require the supplier schedule report, define supplier schedule profiles.



Note: If you only create sales forecasts in Forecasting, you do not need to do steps 12 to 14. However, if you maintain seasonal profiles then Forecasting applies them when you transfer into DRP forecasts.



Note: You can run DRP based on actual demand only, that is excluding forecasts

Only steps 1 to 11 are mandatory.

Forecast Manipulation

You can enter Sales Forecasts at item group (minor) level and then spread, on a time phased basis, over defined time bands down to period, weekly and daily levels in stages either by:

- Interactively selecting definable seasonal profiles.
- Invoking automatic spreading.
- Direct entry and manipulation of forecasts and spread indices.

You can maintain the percentage spread of sales amongst items within an item group. This is known as the item group profile.

From the item group profile and item group sales forecasts, you can automatically generate and then maintain time phased item level sales forecasts.

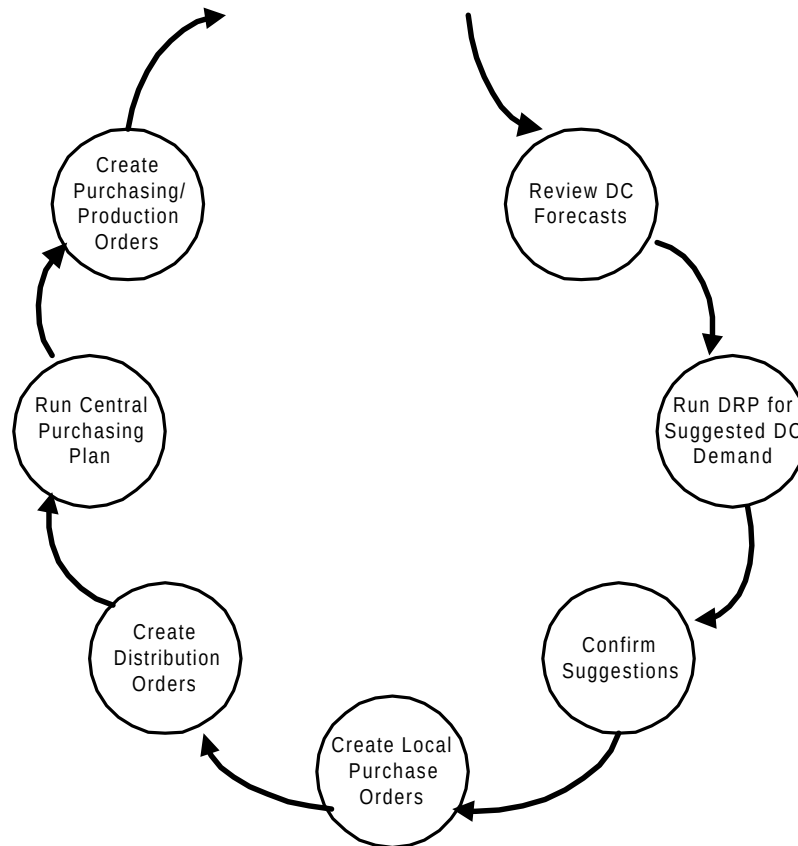
You can also build item level forecasts directly, even if the items do not belong to a defined item group or are not included in the group level forecast.

You can transfer sales forecasts into DRP from Forecasting. This avoids entering manual forecasts and takes into account the last three years' sales demand.

You can also enter stock forecasts to specify stock building requirements. Creating demand, in addition to sales, to build stocks covering future periods when supply would be constrained and insufficient. Typically, this is where there is a highly seasonal demand and a belief that suppliers will not have the capacity to meet the requirements unless they are smoothed and pulled forward.

Processing

The DRP Cycle



- Review DC Forecasts - Use DRP Forecast Maintenance options to review and update the stock and sales forecasts for relevant items.
- Evaluate DC Demand - DRP, via DRP processing run, assesses the supply and demand figures for all DCs in a specified network, suggests new supply orders and amendments to existing supply orders, and deletes all previous supply suggestions.
- Confirm Suggestions - Review and accept the suggestions, then use Requisitioning to convert suggestions to requisitions. You can amend these within Requisitioning before creating purchase or distribution orders.



Note: All outstanding requisitions sourced from DRP for the DC stockrooms are discarded and new ones added from the latest DRP suggested orders for the DC.

- Create Local Purchase Orders - Within Requisitioning, convert the purchase requisitions created from DRP into purchase orders on the relevant external suppliers. Transfer the purchase into Purchase Management.
- Create Distribution Orders - Within Requisitioning, convert the distribution requisitions, for supply from a centre, into distribution orders. Transfer the

distribution orders into Sales Order Processing. The order lines are placed against the centre stockroom defined to the item in DRP. You can also enter distribution orders directly into Sales Order Processing via distribution order entry or amend. Specifying the from and to stockrooms, to automatically create the link between the two stockrooms. This creates an internal sales order for the customer defined for the stockroom which receives the supply.



Note: The distribution order is an internal sales order, so no invoice details are raised. The customer is the one defined to the stockroom which will receive the supply.

When you confirm despatch it adds the distribution order lines to the DRP shipments in-transit file. If World Trade is attached and the transfer is across an EC border, it creates a movement record for World Trade to use to produce Intrastat and ESL reports, and to post the acquisition VAT due on receipt of the goods.

When you receive distribution order lines at the stockroom, confirm the goods as received via DRP processing.

- Run Central Supply Plan - Assesses the supply and demand figures for the selected supply centre, suggests new purchase orders from external suppliers and recommends amendments to purchase orders.



Note: If the centre is a production site, you must define the items to production not DRP, and the centre run should be an MPS run not a DRP one. Otherwise, this centre run is self-contained within DRP.

- Create Purchasing/Production Orders - If the centre run was within DRP, create the requisitions from DRP suggestions and subsequently make into purchase orders using Requisitioning. If the centre run was within Production, create the purchase requisitions and then convert into purchase orders, using Requisitioning. Also, you can create production orders from MPS suggestions within Production Master Production Scheduling.

Run Processing Options

In Distribution Requirements Planning you can select the items for re-planning in the DRP run processing options (DC, Centre and Network planning runs) in a number of ways.

There are three distinct run types as follows:

1 - Full regeneration

Re-plans all DRP items for the selected network or centre.

2 - Net Change

Selects a subset of items for re-planning on the basis of changes in the supply, demand, stock and/or replenishment parameters since they were last planned. This is achieved using net change triggers.

The Distribution Management applications set triggers using a set of defined reason codes.

You can define a net change reason as a type of business transaction which potentially affects the supply, demand, stock and/or replenishment parameters of a particular item.

The reason codes currently set up are:

Code	Description	Applications Where Triggers Set
01	Forecast vs. Actual Comparison	DRP
02	Forecast Generation	Forecasting
03	Forecast Maintenance	DRP
04	Distribution item Master	DRP
05	Unplanned Inventory Transaction	Inventory Management
06	Requisition Maintenance	Requisitioning
07	Unplanned Issue or Receipt	Inventory Management/Warehousing
08	Purchase Order Maintenance	Purchase Order Management
09	Distribution Order Maintenance	Sales Order Processing / Requisitioning

Example

You raise a purchase order in Purchase Management for replenishment of item I1 at stockroom S1.

If stockroom S1 is attached to a centre or DC in DRP, and item I1 is defined as a DRP item at that centre or DC, then a net change trigger is written for that particular location / item / reason code combination (provided that trigger has not already been set).

When you review net change triggers for a centre or DC before submitting a net change planning run, the number of triggers set for each reason code is displayed. Use this information to select whether you process some or all of the items which have set triggers under any of the defined reason codes.



Note: When a particular item is re-planned at a particular DRP location (centre or DC), all net change triggers are deleted for that location / item combination. This is irrespective of the DRP run type chosen for processing.

3 - Selective Processing

Selects a subset of items for re-planning on the basis of the selection of ranges of DRP items, DRP Planners, Inventory minor item groups, Inventory cost / usage elements and / or preferred suppliers.

You can make these selections before submitting any of the DRP Planning Runs (DC, Centre or Network runs).

An item will only be re-planned at a particular centre or DC under the following conditions:

- It is within the range of items selected, *and*
- The DRP planner assigned to the item is within the range of DRP planners selected, *and*
- The Inventory minor item group to which the item belongs is within the range of Inventory minor item groups selected, *and*
- The Inventory cost / usage element calculated for the item is within the range of cost / usage elements selected, *and*
- The preferred supplier defined on the Inventory item / stockroom details is within the range of preferred suppliers selected.

DRP Cycle Summary

This represents a one-level DRP cycle run. To accommodate multi-level, you can repeat or partially repeat this cycle, for each required network. You can define a supply centre in one network as a DC in another network.

Confirm Shipment Receipt

This is the day-to-day functional part of DRP, as distinct from the planning runs.

When you confirm distribution order lines as despatched from supply stockrooms, they are added to a shipments in-transit file. At this point they are considered to be in-transit between the two stockrooms.

On receiving the shipment, you confirm receipt via the relevant DRP processing option. This allows you to receive all order lines or selectively receive lines, with or without amendments.

You must confirm the quantity of items recorded as despatched from the From stockroom as received. You can do this across any number of shipment receipts until you confirm the full quantity. If there is a discrepancy, you must receive the full despatched quantity and do an Inventory stock adjustment, with an appropriate reason for the difference.

When you confirm receipt, the details of each shipment received is output to a report.

You can enter warehousing details if the receiving stockroom is defined as a warehouse to Warehousing.

The Inventory stock figures are updated, and movement records created. This transaction is regarded as a transfer in to the receiving stockroom. Similarly, distribution order lines confirmed as despatched at the supply stockroom are treated as transfers out (within history and movement records).

On receiving an item, the costs are updated, based on the cost held on the in-transit record in the Inventory Management application.

Enquiries

You can group enquiries into one of three categories:

- Planning - What is to be done?
- Static Information - What are the parameters?
- Status - Where are things?

Planning

You can view all supply and demand information generated by any of the DRP planning runs via the schedule enquiries.

There are separate, almost identical, DC and Centre schedule enquiries. This allows for access restriction, if required.

Static Information

You can enquire on the ordering policies and parameters for each item, at any selected Centre or DC.

Status

You can view the current stock situation for any item / stockroom combination, via the Inventory Management Enquiries.

Use the Shipments in-transit enquiry to view the items in transit between stockrooms. This gives all distribution order lines despatched by the supplying stockroom but not yet confirmed as received at the receiving stockroom.

Reports

You can enter selection and sequencing parameters before running a report.

Standard reports print on standard listing paper with 132 print positions and 6 lines to the inch. The reports print the procedure or program number on the top left and the page number and date of run on the top right of the report.



Note: The schedule and valuation reports both have 198 print positions.

As with the enquiries, you can group the reports into Planning, Static Information and Status.

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Chapter 10

Workshop Management Overview



Introduction

Workshop Management is a comprehensive control tool which greatly simplifies the administration of activities concerned with the repair and overhaul of equipment at the service provider's premises.

Workshop Management is a complementary product to Service Management, both of which can run concurrently using the same data structure, and both are fully integrated with System 21 Financial and Inventory modules.

It supports multi-branch operations within each of a number of companies. It also supports contract maintenance, from quotations, pricing and renewal to producing maintenance invoices, including in advance, arrears or by instalment.

Job processing is controlled by call logging, work control and planned maintenance. You can also escalate jobs. Full enquiry features mean you can access master files, transaction history, job status, and financial data.

Where System Manager is installed, you can restrict access to certain functions by operator ID, or create tailor-made menus for each operation. This means you can prevent operators from accessing specific functions, or information belonging to certain companies.

Relationship to Other Modules

Workshop Management operates under the control of System Manager. The full benefits of the system are gained by integrating it with the following prerequisite modules:

- Accounts Receivable
- General Ledger
- Inventory Management
- Sales Order Processing.

You can also integrate it with the following:

- Advanced Financial Integrator (AFI)
- Distribution Requirements Planning
- World Trade
- Forecasting.

Workshop Management Configuration

Company Level Information

You can operate Workshop Management for a number of companies, maintaining the characteristics of each on a control file. You control Workshop by setting up data for the company and then for each branch within the company. You normally do this after creating the Accounts Receivable, Inventory Management and Sales Order Processing companies.

You define certain data and policies at the company level. These include:

- Document reference numbers: invoice, credit note, contract, job
- General Ledger accounts
- Default labour price list
- Default scheduled visit profile
- Working day start / finish times
- Hours entry only, for Job Reporting.

You also define service and accounting periods at company level. You can enter up to 99 service periods per year. They are mainly used to schedule planned maintenance visits. You can define each day of the year as a working or non-working day, so you can calculate target response date / time for jobs.

You can enter up to 99 accounting periods per year. These should be consistent with those specified in the ledgers. The number of days in the accounting periods is used to apportion deferred revenue from contracts invoiced in advance.

Branch Level Information

Once you have defined the control parameters for a company, you must define at least one branch. A branch is a physical or logical location that has responsibility for servicing specific customer sites.

You can add workshop branches after starting Workshop Management.

You define Certain data and policies at the branch level. These include:

- Branch name and address
- Printer output queue for job audit log
- Job escalation parameters: escalation time fence
- Escalation reporting interval
- Users to alert at each escalation level
- Default response hours.

You must authorise each user to access at least one branch within a company, but you can authorise the user to several or all branches. This authorisation by branch restricts the data to which a user has access.

Reference Data

Workshop Management uses a number of reference files that hold data which rarely changes:

- Standard codes and descriptions
- System parameters
- Customers
- Divisions, model groups, model sub-groups and models
- Installations
- Contract types
- Job categories
- Labour rate price lists
- Service territories, teams and field service groups
- Engineers
- Contract rates
- Scheduled visit profiles.

Standard Codes and Descriptions

You use the descriptions, or codes / parameters file to set up codes with standard descriptions. At data entry this validates the information and displays descriptions on both screens and reports. In some specific instances, you can store a parameter or number, for example value added tax percentages.

For Workshop Management to work correctly, certain standard codes and parameters must be present and these are supplied with the system.

Setting Up Customers and Equipment

Before you can record contract information or enter service operations, you must enter customer details, location and the equipment installed at the sites.

You must set up:

- Customer accounts, maintained through Accounts Receivable.
- Installation addresses, maintained through Accounts Receivable.
- Installed equipment details.
- Equipment models.
- Model group and sub-group maintenance.



Note: Some of this information is also used by Accounts Receivable and Sales Order Processing. Make sure that data is correct for all purposes.

You can print and enquire on all the information.

Before you can set up customer and installation details in Workshop Management you must enter the customer details using the Names and Addresses option in Accounts Receivable and the Installation Address Maintenance, in Accounts Receivable and Sales Order Processing.

Customer Name and Address Details

You enter additional customer details for Workshop Management using the Customer Additional Details Maintenance option.

You cannot enter installation records or contracts, or process transaction, without customer accounts. Although Workshop Management does not use some of the customer account data, the following parameters are significant:

- Name and address - Displayed on many screens and printed on some reports, including invoices / credit notes.
- Geocode or postcode - Used to automatically assign teams of engineers to jobs, by means of the geocode / territory definitions and engineer team / field service group / territory assignment rules.
- VAT code and indicator - Used within invoice / credit note generation to establish which VAT rate to charge.
- Credit stop indicator - Used within job maintenance to provide warning that the customer is on credit stop.

Installation Address Details

An installation address is a customer address at which equipment is installed. A customer account address can be an installation address, and there can be several additional installation addresses for the customer account.

When you set up a customer account, an address code of 000 is added. You can then create further installation addresses by assigning address code suffixes of 001, 002, and so on to each address record for the customer.

Each installation address is identified by a unique combination of account code and address code.

You link additional customer details required by Workshop Management to account and installation addresses.

You must create the site address before entering the details of equipment installed at the address. You cannot perform any contract or transaction processing until this equipment database is present. The following data for an installation address is significant:

- Name and address - Displayed on many screens and printed on some reports, including invoices / credit notes.
- Postcode - Used to automatically assign engineers to jobs at the customer site, by means of the postcode, or geocode / territory definitions and engineer / territory assignment rules.

Price List Options

The two price list options are both Sales Order Processing options, see SOP product information for more details.

Customers

Workshop Management shares a common customer database with Accounts Receivable and Sales Order Processing. Accounts Receivable controls credit limits and payment terms, Sales Order Processing controls stock allocation, pricing of sold items and Sales Analysis coding, and Workshop Management controls pricing of service items and invoice consolidation levels.

You can manipulate the customer information to match the structure of the customer's organisation. You can have a simple single account, or a statement account with many Invoice To accounts, each with many different delivery or installation accounts.

Statement Account

This is shared with Accounts Receivable and is normally created and maintained by Accounts Receivable. The data includes credit details, Accounts Receivable parameters and statistical codes.

Invoice Account

You can send invoices to an invoice only address.

Delivery or Installation Account

Each account can have many delivery or installation addresses. Define these in Sales Order Processing or Workshop Management or both, by adding a three-character address code suffix to the account code.

Additional Information

You must set up additional information, for each account address, where you service equipment. Workshop Management uses this information and it includes the invoice destination address, invoice consolidation level (account, address code, or contract / job), service region, and price list for service parts.

You can identify a customer either by entering the account and address, or by using the available name and address search routines.

Models

You identify each piece of equipment defined to Workshop Management by its model code and, normally, its serial number. You can use standard model codes and numbers to describe the different types or configurations of equipment. Link each model to a model sub-group and link the sub-group to a model group.

You can specify models as individual machines, or as peripherals to a machine.

You can hold contract rates for different types of service contract at model group, model sub-group or model code level. You can assign engineers to only work on specific model groups, model sub-groups or models.

You can determine the labour rates to charge by the model on which the work is performed, since each model code points to the appropriate labour price list.

Installations

You define an installation within Workshop Management as a customer address at which equipment or systems are installed.

You identify the customer address by an account code and an address code. The site can have many items of equipment or systems installed. You should set up installation records for all installations irrespective of whether they are covered by service contracts.

You identify each piece of equipment at an installation by the unique combination of model number and serial number. If you do not know serial numbers, then use the model number and quantity.

The data held for each piece of equipment includes details of age, delivery and installation dates, warranty period, location, department, complexity points and product division.

Engineers

You must define each engineer who services or repairs equipment on the engineer master file. Details include name and address, team code, grade, hourly cost rate, service region, stockroom code, and skills matrix.

The stockroom code is the link to Inventory and must be defined in Inventory Management. It represents a discrete area within a company where you record and control stock separately from other company stocks. This stockroom could be a van or the workshop stores.

Territories And Engineer Assignment

Workshop Management can automatically assign engineer teams to service jobs, if you have:

- Defined the customer's geographic or postcode on the customer record.
- Divided the geographic area covered by the company into territories; defined by a list of full or partial geocodes or postcodes.
- Specified the engineers' teams who cover each territory.
- Defined the field service group for the team, to show which product divisions, model groups, model sub groups and models the team services. The field service group record contains a skills matrix to define these limits.

Workshop Management can map the customer's address to a service territory and allocate work to the team of engineers designated to operate in that territory.

You can automatically or manually allocate an individual engineer within the team to a specific job. In job selection, you can further specialise according to the engineer's skill level, that is you can assign an engineer to service specific model groups (and optionally sub-groups) or to perform work of a specific type (job category), or to perform work on equipment belonging to a specific division or any combination of any of the above.

Contract Types

You can use contract types to define the different types of service contract offered by the company to its customers. Each contract type record holds the normal duration of the contract, guaranteed response hours, invoice schedule (in advance, in arrears, or on completion of each scheduled maintenance visit), and whether you generate a quotation or contract at renewal.

You can apply billing parameters to the contract type. These indicate if service charges, rental charges, or various meterage conditions apply.

All contracts have a user-defined contract type code. Workshop Management defaults a contract type of *NO for work on equipment not covered by a service contract.

The parameters set up for the contract type are defaults for the contract, but you can change them, if required.

Job Categories

You use job categories to describe the different types of work required. You can define a number of categories as defaults for breakdowns, warranty and repeat calls to save time when you log a call.

You must also define a table of valid job categories for each contract type. This table specifies whether you charge the customer for: travel and labour hours (fixed or hourly charges), up to four parts groups and miscellaneous charges, when you do work of a certain job category on equipment covered by that contract type.

Price Lists

Workshop Management use two types of price list; for labour rates and for parts.

Labour rate price lists define the hourly labour rates charged for contract, non-contract and workshop work. The price lists include an hourly travel rate and effectivity dates, and are linked to models. The model code of the equipment determines the labour rates charged.

Parts price lists and discount lists define the prices and discounts charged for spare parts fitted or used by an engineer. You maintain these lists, which also have effectivity dates, in Sales Order Processing. You link them to the customer additional details records in Workshop Management: the prices charged are determined by the customer site at which the work is done.

Contract Rates

You define contract rates for each contract type at model group, model sub-group, or individual model code level.

Contract rates specify the charge per scheduled maintenance visit and a fixed charge for each piece of equipment covered by the contract type. Contract rates have effectivity dates to control new rates.

Contract pricing routines use contract rates, but only when the system has been defined for term processing; not for monthly processing.

Scheduled Visit Profiles

You use scheduled visit profiles to define standard schedules for planned maintenance and service visits to equipment under contract. You define the schedules in terms of the number, type and frequency of visits.

You can associate a visit profile with a contract, model, or model group. You can use them to automatically generate the service visit schedule for any equipment added to a service contract.

Contract Processing

Workshop Management provides flexible and comprehensive facilities for contract management, as follows:

- Maintenance
- Pricing and invoicing
- Renewal
- Credits.

Contract Maintenance

You identify service contracts by a contract number, type and start date. Each contract relates to one or more locations of a single customer, and can cover one or more pieces of equipment installed at these locations.

When you enter a new contract, you create contract header and equipment records.

The contract header record includes the contract duration and end date, order reference and date, and invoice destination. The service parameters include response hours, scheduled visit profile, if there is cover while the contract is pending, and the rates applied. The billing parameters set the invoice term, frequency and whether charges are in advance, in arrears or per visit. The service conditions provide for setting up service, visit and rental charges at contract header level: the service and visit charges override the contract rates set by model group, model sub-group or model; the service conditions can be overridden at equipment and peripheral levels.

You set up the contract equipment records once the header is validated and saved. You select items, from a list of all the equipment defined for the customer account code, to place. You can only select equipment previously specified as resident at a location to which the contract applies. You can add service conditions at equipment and peripheral levels. You can create for each item on a contract, a schedule of planned service visits, by service period for the duration of the contract. You can create this automatically using the service visit profile associated with the contract header, model, or model group, and then amend the schedule as required.

Contract Pricing And Invoicing

Workshop Management prices and invoices contracts invoiced in advance or in arrears as follows:

Service Pricing

- A contract is priced for a defined period of time (invoice term) which can be the same or shorter than the contract duration. The charge can be split into instalments for invoicing; for example you can invoice a one year term in quarterly instalments. Workshop Management pro-ratas the charge for equipment added to a contract part-way through its invoice term.
- To price a contract, each piece of equipment is priced individually and accumulated to give a total contract price. If equipment and its peripherals have special prices, then these are the prices charged.
- If no special price exists, then Workshop Management uses the contract rates file (term processing only), to get the price per maintenance visit and any supplement fee. The total number of scheduled maintenance visits in the invoice term is multiplied by the price per visit and the supplement fee added to get the gross charge for the piece of equipment.



Note: Service contracts can include fixed visit charges, invoiced after each visit. You invoice these visits through job invoicing, not contract invoicing.

Rental Pricing

- The rental charge can be on the contract header, this is the default price for each item added to the contract, or can be maintained on the contract equipment line. The charge entered will be per month if you selected monthly processing on the system parameter file, or per term if you selected term processing.

Meterage Pricing

- You can only use meterage charges with monthly processing.
- You can define a minimum volume on the contract header or equipment line. The minimum monthly charge is the minimum volume multiplied by the price per copy, which you can invoice in single or multiple months.
- You can define a minimum charge on the contract header or equipment line, this may be invoiced in single or multiple months.
- You can define up to five copy charge bands, for each of two meters, for each contract equipment line.
- Use interim pricing to charge for a meter estimate, or actual reading returned by the user, based on the minimum units already charged for, (or actual units used). Define the frequency of interim pricing on the contract header.

- Use reconciliation pricing to charge according to an actual meter reading. You can apply additional meterage charges or give credits for overcharges. The reconciliation term must be a multiple of the interim period.
- You can pool meterage pricing, that is you can spread the minimum volumes or values across all equipment on the contract, so the excess units on one machine can be offset against the under-usage on another machine.

Invoicing

- Contract pricing creates records on an invoices pending file. The generate and print invoices option generates invoice header and line records. Invoice consolidation, which groups invoice pending lines into invoices, takes place as specified on the customer additional details records, that is by account, by account and location or by contract. VAT is calculated using the parameters set on the customer account record. The invoices are then printed.
- Invoice generation posts the transactions to the sales and general ledgers. If you invoice in advance, the revenue value of the invoice is deferred in the General Ledger, by generating additional postings to transfer period revenue values from a deferred income account into a sales account each period.

Contract Invoice Term

When pricing a contract, the price is calculated to cover a fixed period of time known as the Contract invoice term. There can be one or many invoice terms within the duration of a contract.

The Contract invoice term is the period of time from the Next Term Starts On Date to the Next Term Ends On Date, and defines the period of price protection for the contract.

Next Term Starts On Date

The date on which the next Contract invoice term starts. Contracts are selected for pricing if this date is less than or equal to the date entered on the Price contracts prompt screen. Maintain this field on the billing parameters window.

Next Invoice Term Ends On Date

The date on which the next contract invoice term ends. Maintain this field on the billing parameters window.

Normal Invoice Term

The normal invoice term for the contract. Maintain this field on the billing parameters window.

Invoice Frequency

A sub-division of the normal invoice term. This is the period of time, in months, between invoices, if the contract is invoiced in instalments. Maintain this field on the billing parameters window.

Contract Payable in Advance

A contract for which contract charges are invoiced before the term or instalment.

Contract Payable in Arrears

A contract for which contract charges are invoiced after the term or instalment.

Contract Payable Per Visit

A contract for which contract charges are invoiced as a result of a scheduled service visit. These visits are priced by the Price Jobs option.

Pending Invoice Line

An invoice line, not yet assigned an invoice number or posted to the Ledgers, waiting to be grouped with other invoice lines to form an invoice. The grouping depends on the invoice consolidation level, set up for the customer account (000 record) using the Additional Service Details option.

Pending Invoice Line Availability Date

The date on, or after which, the pending invoice line may be included in an invoice. For advance contracts, this is the start date of the contract invoice term or instalment period; for arrears contracts, this is the end date of the contract invoice term or instalment period.

Description

You can select all branches or a single branch, specify the type of billing, and enter a cut-off date relating to the contract invoice terms. This only processes contracts payable in advance or arrears.

Pending invoice lines are generated for:

- All non-invoiced pieces of equipment, added to a contract during an invoice term, which have been priced.
- All equipment on contracts where the entered cut-off date is later than, or equal to, the Next Term Starts On Date of the contract.

Pricing Routine

Contract pricing is governed by the conditions set up in contract maintenance, at header or equipment levels, or at both.

Default Prices from the Contract Rates File



Note: You can only use default pricing for fixed service, rental and visit charges. It does not apply to copy / vend contracts, where you must set up special contract conditions.

If you do not overwrite the Service Conditions at contract header or contract equipment levels for the Fixed Service, Rental or Visit Charges, they default from the Contract Rates file. Contract pricing selects the fixed service, rental and visit charges from the Contract Rates file, for the appropriate effective date, equipment group, contract type, term and currency.

For a particular currency, the contract rates record used is for the contract type and term, effective on the Pricing date. The Pricing date is derived as follows: if the Use start date rates field on the contract header is set to 1=Yes, or if there is a Price held until date which is later than the Next term starts on date, the pricing date is the contract start date; otherwise, the pricing date is the Next term starts on date.

The equipment model group, model sub-group and model are compared to the same fields on the contract rates record; the record with the best match will be selected. If no record is found, the contract could be left with no charge: special

service conditions could be set up by the user for the contract. The recommendation is that records should be created on the Contract Rates file for as many combinations of equipment and contract types as are necessary.

The price is calculated, by multiplying the visit charge by the number of scheduled maintenance visits in the Contract invoice term, for this piece of equipment.

The fixed service charge, from the contract rates record, is added; the charge applies to the Normal invoice term, and is adjusted to allow for the Contract invoice term: the number of days in the Contract invoice term is divided by the number of days in the Normal invoice term, and this factor is applied to the fixed service charge.

The rental for the contract term is included, but separately from the service and visit charges.

Contract Conditions Prices at Header / Equipment Levels

If you enter a price on the Contract Conditions Window at header or equipment levels, this price applies to the Normal invoice term specified on the contract. The Normal invoice term is converted to days, the Term Ends On date and the number of days between these two dates is calculated. The number of days in the term is also calculated. The ratio of the latter number of days to the former is calculated and applied to the contract conditions price, to get the actual price for the contract, or for the specific piece of equipment, for the contract invoice term.



Note: For complete instalment periods, the invoices are of equal value.

Pro-Rata Pricing

The price for equipment added or removed from a contract is calculated on a pro-rata basis.

The number of days in the Normal invoice term of the contract is calculated, as described above.

For equipment added late, the contract invoice term is the number of days between the date the equipment was added and the term ends date, or equipment removal date, whichever comes first.

For equipment removed early, the contract invoice term is the number of days between the date the equipment was added or the term start date, whichever comes last, and the term end date, or equipment removal date, whichever comes first.

The ratio of the two sets of days (that is contract invoice term and normal invoice term) is calculated and applied to the calculated price for the normal invoice term, to get the actual price for the contract, or for the specific piece of equipment, for the number of days that it has been on the contract.



Note: Equipment removed, or scheduled for removal, within a processed invoice term will not have an automatic credit generated; for this, use the Contract Credit option.

Contract Renewal

You can run this function at any time. It prompts for a date representing the expiry date limit of contracts for renewal. The system reads through all active contracts and determines which ones to renew, by comparing the contract end date with the entered date. If the contract end date is less than or equal to the entered date, the contract is renewed.

In order to renew a contract, the equipment on the contract must still be eligible for the contract type, that is the contract charge rates must be in effect.

A new set of contract records will be generated with either the same contract number or a new, system-assigned contract number. The contract start date will be the day after the existing contract end date. Other contract details are copied to the new contract.

The status of new contracts will be set as either pending start date or quotation, depending on the Quotation Required At Contract Renewal flag on the contract type record. You can amend the contract details generated (through contract maintenance procedures) before generating quotations or invoices.

Contract Credits

You can process credits at any time, to refund part, or all, of a contract invoice, and to cancel any pending invoice records. You can select the record to amend by contract, invoice, or period.

You can amend the contract and invoice record to affect the operative date and values involved. You can discard the selected record, or give a full or partial credit, or credit the original charge and re-invoice a different amount.

To produce the resultant invoices and credit notes, select the Generate and Print (Contract) Invoices option. This procedure also removes any cancelled pending invoice records from the work file.

Job Processing

An important function of any service organisation is to complete, in an efficient and timely manner, both customer service requests and planned maintenance work according to the contractual terms agreed.

When you have completed a service jobs, you want to invoice any chargeable work promptly to optimise cash flow.

Workshop Management provides the following facilities to help meet these objectives:

- Planned Maintenance Job Generation.
- Call Logging.
- Work Control.
- Engineer Diary.
- Technical Reporting.
- Job Pricing and Invoicing.
- Job Escalation.

Planned Maintenance Job Generation

Run this process as required. Load PM Jobs creates a service job for each installation that has a planned maintenance visit scheduled for periods within a specified range. You can create schedules for up to six periods in advance of the current period. The service schedule details, held for equipment covered by contract, provide the base data for processing.

The process attempts to assign an engineer team to each job, by comparing the customer geographic code with the territory definitions, and the model type requiring service with the field service group that covers the division, model group, model sub-group and model. If you entered a specific engineer against the equipment on the installation record, this engineer will always be assigned (if this method of selection is defined on the system parameter file).

If more than one piece of equipment at an installation address, on the same contract, requires a service visit in the same period, then this procedure normally generates only one job covering all the equipment. However, you can set an option on the company profile to generate an individual job for each piece of equipment.

Call Logging

Use this facility to record unscheduled or breakdown calls and generate service jobs. You can also amend the details of outstanding jobs.

Addition of New Job

To log a new job you must enter customer account details, the service contract number, the serial number, or the model number and serial number. You can only log a new job if an installation record exists for the customer address requesting service. You can create installation details for non-account customers.

If the customer is on credit hold or stop, a warning message displays and the call status set to Credit Hold, but you can still enter the job.

You must select each item to include on the job from the displayed list of equipment. If the customer has a number of items of equipment on site and cannot identify which of them requires the service, You can log the job against any available machine line. When the engineer completes the job you must enter the true model and serial number.

Workshop Management checks for previous jobs (for qualifying job categories), to determine if this is a repeat call. This compares the number of days or meter units or both elapsed since the previous call, with the values defined on the volume segment file for the model. Workshop Management allocates a job number. The response hours are calculated by reference to the special serial numbers file, the contract line, the contract header, or the contract type (possibly reduced by the 3-D Matrix percentages). A target start date and time is set by adding the retrieved hours to the date and time the call is logged.

The job category defaults to the value entered in job category maintenance. Enter the customer contact name and fault code and description. You can change the response hours, and therefore the target response times. Alternatively, you can enter an appointment date and time. You can also assign an engineer to the job.

Amendment to an Existing Job

You can select the job you want to amend by customer, serial and model number, and contract type or number.

You can amend the job category, engineer assigned, customer contact and fault code for each piece of equipment on the job.

Work Control

You can either use the Work Control option to manually assign engineers, or get Workshop Management to automatically assign engineers using the automatic call allocation program.

If you require automatic call allocation you must start the job and make sure the Workshop Management sub-system is running.



Note: The sub-system library list must be tailored to take account of: the use of consolidated files libraries in place of IN, OE, SL and G files libraries and the version of Business Manager in use, for adjusting the four IPG libraries.



Tip: If you are running System 21 Version 3 on an AS/400, then use the CHGJOB command and **F4**, enter OSLSSP/OSLS2P3, then select **F10** and page down to the library list.

Automatic call allocation is a sleeper job, that automatically schedules engineers within a team, to the jobs assigned to the team in job logging. Assignment is on the basis of available engineer, based on current workload and skill profile, and who can achieve an estimated time of arrival equal to or less than the target date and time for the call. If the selection criteria cannot be met, an error message is displayed for the call, and a non-assignment reason is entered on the job detail screen.

You can schedule jobs manually, in which case the same criteria and checks for engineer selection are applied.

Use the Work Control function to despatch outstanding jobs to engineers and enter status updates on engineers' current jobs.

You can display a list of jobs assigned to a specific engineer or team of engineers, or for the branch. You can extract planned or unscheduled jobs, and jobs of certain status. The list of jobs display in order of target date and time, with the most urgent jobs first.

You can display jobs for a team can graphically: colour blocks represent each job and the colour denotes the urgency of the job. Unscheduled jobs for the team display in the lower part of the screen; scheduled jobs display under each engineer in the upper part of the screen. You can move jobs, and display job line details for team and engineer work queues.

Engineer Technical Reporting

Use this facility to enter the details of completed work. You access it by selecting the completed job line with 99.

A technical report is made up of four sections:

- Header Details - engineer number, job number, visit date, report reference, time on site, customer and engineer travel time.
- Labour Hours - hours spent on each piece of equipment, qualified by fault code, job category, machine section and sub-section repaired, and corrective action. You can book hours to a basic rate and three overtime rates.
- Parts Used - Spares parts and quantities used, by job category.
- Miscellaneous Costs - analysed by job category, you enter a cost or charge type, the cost value and charge value or enter a percentage uplift to derive the charge.



Note: Both parts and labour can be chargeable (100%) or non chargeable (0%) or any percentage in between.

If a job requires more than one engineer to complete the work, you can enter several technical reports. The prime report has the suffix-00, with each subsequent report suffix incremented by the system. After you enter the final report, indicate the job is finished and fully documented. At this point, you can price and invoice the job.

Job Pricing And Invoicing

Workshop jobs, not fully covered under the terms of a service contract, must be charged to the customer. You price and invoice the jobs after you complete them through the technical reporting function.

The decision as to which jobs, or elements of jobs, are chargeable is made by referencing the user-defined matrix of contract types and job categories. For the particular combination, the matrix indicates whether labour or travel distance or travel time or miscellaneous costs or four types of parts or any combination of the above, are chargeable to the customer.

Six types of charge can appear on a job invoice:

- **Contract Visit Fee** - Contracts can include fixed visit charges to invoice after each visit: the contract's billing parameters being set accordingly. You invoice these visits through job invoicing at job completion. The fixed visit charge is obtained from the contract conditions at equipment or header levels, or from the contract rates file.
- **Labour** - The labour charge is calculated from the number of hours booked on the engineer report, multiplied by the hourly rate on the labour price list for the model. The hourly rate used will be either a contract, or a non-contract, rate depending on whether the piece of equipment was covered by contract when the job was logged. The hourly rate will be uplifted by a factor for overtime rates. You can apply a fixed labour charge, if required by the job's charge matrix set up in Cover Type / Job Category Maintenance.
- **Travel Labour** - The travel time is calculated from the number of customer travel hours and minutes booked on the engineer's technical report, multiplied by the hourly travel rate from the labour price list held for the model. A fixed travel labour charge may be applied, if required by the job's charge matrix set up in Contract Type / Job Category Maintenance.
- **Parts** - The charge for parts used by the engineer is calculated according to the price list and discount list defined for the customer site. The price and discount list details are obtained from Sales Order Processing.
- **Miscellaneous Costs** - The charges are based on the values entered through the technical reporting function.
- **Chargeable Travel** - The charges may be based on a standard distance value, the actual distance driven, or zones. Data is set up on the Additional Service Details record for each site. All jobs, even non-chargeable ones, should be processed through the job pricing and invoicing functions, since the cost values of labour, engineer travel time, miscellaneous costs and parts, for all jobs, are calculated by these routines.

A batch job is submitted by this menu option to price jobs for the current branch. Only job lines which have been fully technically reported are processed by this function.

Pending invoice lines are generated for all job lines which have been fully technically reported but not yet invoiced. If a job bill to account exists, this is used to generate the pending invoice. If there is no job bill to account but there is an

ownership account, this is used instead. If there is no ownership account, the site account is used.

The price charged for each job line is built up from the following five individual charges:

Labour Charge

Each labour hours entry from each technical report for the job line is priced individually.

A labour charge is only made, if the contract type and job category combination is defined as chargeable for labour, on the cover type / job category file.

If you have defined fixed charges, they are applied before any variable labour charges. Variable charges are applied on the time remaining, after any fixed charges. The labour hours, on each technical report entry, are multiplied by the premium factor (from the codes file LHTY record) for the labour hours type. The result is multiplied by an hourly labour rate from a labour rate price list for the appropriate currency, which is derived as follows:

The labour rate price list is taken from the customer additional details, then the model file. The price list contains four, hourly labour rates: contract, non-contract, workshop and travel. If the engineer is a workshop engineer, that rate is used; otherwise, the contract or non-contract rate is used; and if travel time is involved, the travel rate is used.

In identifying the actual price list record to be used, the pricing date is taken into account, that is the price list record effective at the pricing date will be used. The pricing date is determined as follows:

- If the equipment is not covered by contract: Pricing date=Visit date (from technical report)
- If the Use start date rates field on the contract which covers the equipment is set to 1=Yes: Pricing date=Contract start date
- If the Use start date rates field on the contract, which covers the equipment, is set to 0 (No), and the job is a planned (scheduled) job, and the visit date is before the Price held until date on the contract which covers the equipment, then: Pricing date=Contract start date
- If not, or if Price held until date is zero: Pricing date=Visit date
- If the job is a callout (breakdown) job and the call logged date is before the price held until date on the contract which covers the equipment, then: Pricing date=Contract start date
- If not, or if Price held until date is zero: Pricing date=Call logged date

The labour cost value is always calculated, whether or not the labour is chargeable. The cost value, in base currency, is the number of labour hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer who performed the work). No premium factor is applied to the cost rate.

Parts Charge

A charge for a part is only made if the contract type and job category combination is defined as chargeable for parts on the cover type / job category

file. When reporting parts used on a job, the charge % may be reset from 0% to any value up to 100%.

Parts are priced by retrieving the parts price, discount list, and currency codes held on the Additional Service Details record. Using the price and discount list records, defined in the Sales Order Processing application (and effective for the pricing date, as derived for the labour pricing calculations), the prices and discounts are calculated in the appropriate prime currency.

The cost value of parts used is always calculated, whether or not the parts are chargeable. Parts costs are always set up in Inventory Item / Stockroom Details in base currency. The cost value is calculated in base currency, not within the Price Jobs function, but when the Inventory Management application is updated with the detail of the parts used.

Miscellaneous Charge

A miscellaneous charge is only made if the contract type and job category combination is defined as chargeable for miscellaneous charges on the cover type / job category file.

The charge value of each miscellaneous charge line is the charge value entered on the line itself in the specified currency.

The cost value of each miscellaneous charge is always calculated, whether or not the item is chargeable. The cost value is the cost value entered on the line itself in the specified currency. If the specified currency is not the base currency, the system automatically converts the prime currency value to base.

Travel Charge

Fixed charges: these are applied prior to any variable travel charges, if defined on the cover type / job category file.

Variable charges are applied on the time remaining after any fixed charges.

The prime currency travel charge is calculated as the billable customer travel time entered on the technical report, multiplied by the travel rate obtained from the labour rates price list. No premium factor is applied to this rate.

The base currency travel cost value is always calculated for engineer travel time. The cost value is the number of travel hours multiplied by the engineer cost rate (taken from the engineer master record of the engineer on the technical report).

Contract Visit Charge

If the job line being invoiced is part of a planned maintenance job and the contract under which the job is performed is currently defined as being invoiced per visit a contract visit charge is generated (unless calculated as zero).

The contract visit charge is calculated as described in the previous section Price Contracts: the number of visits covered by the job is multiplied by the visit fee. The visit fee may default from the contract rates file, or be specified as a contract condition at header and / or equipment line levels.

Engineer Diary

Workshop Management provides an engineer diary, which enables the planning and scheduling of non-machine-related jobs and other commitments for each engineer. Job appointments are automatically entered in the diary for the engineer when you log a call or in work allocation.

The next five working days are displayed, starting from the system date. Each day in the diary is free format; bookings can be entered for any date, time and duration, or for a date and time from with a date and time to.

Diary entries are made as required and are used to reserve the times for events such as holidays, training courses, sickness, dental appointments, etc. As entries for engineers are scheduled for a date and time, they will appear both in the engineer's work queue and in the graphical display of his jobs.

The diary is accessible from the Work Control option.

Job Escalation

Workshop Management can alert you to any jobs approaching their designated response time and still outstanding. Job escalation runs so long as the Workshop Management sub-system is active.



Note: The sub-system library list must be tailored to take account of the use of consolidated files' libraries in place of IN, OE, SL and GL files' libraries and the version of System Manager used for adjusting the four IPG libraries.

All service jobs created are stamped with a target response date and time, which is calculated as follows:

Call-out Jobs

For jobs entered using the Job Logging / Maintenance function the target response date and time is:

Date / time call logged + response hours

Where response hours is derived from the special serial numbers file, or the contract if a contract exists. If the equipment is not covered by a contract, the response hours default to the value for the contract type *NO. You can make manual overrides of the response hours if special circumstances arise.

When calculating target response date and time, Workshop Management takes into account working and non-working days, and the standard service window for the company, that is hours service is provided. These are all user-defined.

Planned Jobs

For jobs generated from the service schedules defined for contracts, the target response date and time is calculated as:

Target date = Date of last working day of the period in which the job is scheduled

Target time = Start time of the working day

You can define the escalation time fence, the interval between each escalation step, and the users alerted at each step. Workshop Management monitors each outstanding job, comparing its target date and time with the current system date and time. Once the job enters the defined escalation time fence, it starts to escalate and with the escalation step recalculated at frequent intervals: the escalation step moving from high to low value as the job approaches its target date and time. Escalation step zero shows that the job has reached, or gone beyond, its target time.



Note: Workshop Management assumes a 7-day working week. If you work less than this, Workshop Management escalates Monday's work on a Sunday, not on the preceding Friday.

As a job moves from one escalation step to the next, Workshop Management can send a message to pre-defined users to alert them so they can take appropriate action.

Workshop Management removes jobs from the escalation procedure once they reach a certain status. You define these within the Job Line Status Code Record. You can view the escalation status of jobs using various selection parameters.

Example of Call Escalation

Escalation Step	Reporting Times - (1 Hour Intervals)	
0	16:00	Target Time
1	15:00	
2	14:00	
3	13:00	
4	12:00	Escalation Time Fence (4 hours) starts

The escalation time fence is set to 4 hours and the reporting interval to 1 hour.

A job with target time of 16:00 starts to escalate at 12:00.

Between 12:00 and 12:59, the escalation step is 4, because the job is within 4 hours of target. Between 13:00 and 13:59, the escalation step is 3, because the job is within 3 hours of target, and so on.

At 16:00, or later, the escalation step is 0, because the job has reached or passed its target time.

Warranty Claims

This is a financial module attached to Workshop Management that you can use to create credit notes for dealers because of warranty claims against the supplier. This module uses all the customer, parts and models information you entered in Workshop Management.

If a customer can make warranty claims, you must set the Warranty Claim flag on the customer additional details to either parts, labour or both parts and labour. You also set the labour rate list to use for the warranty claim in the customer additional details.

You can use this function to:

- Enter the warranty claim data, such as model, fault, purchase date, customer details, dealer information.
- Record any parts used during the repair.
- Validate the data entered.
- Accept the claim and automatically create a Sales Order Processing credit note for labour or parts or both.
- Reject the claim with the appropriate reason and produce a letter stating the reason for the rejection.
- Maintain the credit status of the customer making the claim.

The validation process checks all the claim details and recommends if the claim should be accepted, rejected or whether you need further information before you can process the claim further.

This function also provides facilities to analyse the warranty claim data. You can:

- Search for other faults and repairs on this model and serial number.
- Check if parts have been claimed for before using the same invoice number.
- Display and audit all stages that the claim has passed through.

Once you have accepted or rejected a claim, Workshop Management archives the claim.

Enquiries

Workshop Management provides a comprehensive range of enquiries for:

- Master Files - These generally show the same details as their counterpart maintenance routines.
- Transaction History - The job history enquiry shows details of historical service work performed against equipment. The invoice enquiry shows details of a workshop invoice, whether it be a contract, job, or sundry invoice.
- Status - An escalation enquiry shows up-to-date information on the escalation status of workshop jobs.

Loading Workshop Management Data

You must load the following data for each workshop company you want to process within the live module.

The data is divided into two groups:

- Basic data - data used mainly for reference purposes, which is low in volume and not subject to ongoing change, such as codes.
- Major data - data which constitutes the bulk of the module and is subject to regular change, such as customers and contracts.

Some of the data listed is marked Optional. This means that, in system terms, the data is not essential to running the module; but in practice, the data may well be essential, for example, contracts.

Basic (Parameter) Data

- Branch Records - one for each service branch
- Calendar Control Record
- Codes / Parameter File Records - System-required parameters have a * at the beginning of the Parameter Type description: these must be set up, or copied from the demonstration data supplied. In addition, set up appropriate code values for each of the following types:
 - ABSC (Engineer absent)
 - CCRC (Contract cancellation reasons)
 - CHGT (Miscellaneous costs / charges)
 - CORA (Corrective actions)
 - DAYT (Day types)
 - EGRD (Engineer job grades)
 - FLTC (Faults)
 - LHTY (Labour hours or overtime types)
 - RRES (Credit release reason)
 - STAT (Equipment status)
 - TXTO (Text destinations)
- Company Profile
- Contract Type Records - one for each contract type code
- Contract Rates Records (optional)
- Cover Type / Job Category Records - one for each combination of cover type, job category, (and Currency where World Trade and Multi-Currency are active) with invoicing indicators
- Daily Calendar File (& Rebuild Period End Dates) - for all years over which current service contracts are spread
- District - one for each work controller, to authorise work on teams
- Division Records - one for each service division
- Engineer Records - one for each workshop engineer
- Escalation Parameters (optional)
- Field Service Group - one for each group or equipment (division / model family / model to be serviced)
- Job Category Records - one for each job category code
- Labour Rates Price List Records - one for each set of labour rates (and for different currencies where World Trade and Multi-Currency are active)

- Model Group / Sub-Group Records - one for each model group / sub-group combination
- Model Records - one for each model code
- Scheduled Visit Profile Records (optional) - one for each standard schedule of planned maintenance visits
- System Parameters File
- Team - one for each engineer team
- Tax Codes (for World Trade and Multi-Currency only.) - for each country's contract type / job category combinations
- User / Branch Authority Records - one for each user and branch combination
- Volume Segment - minimum one, optionally more for each model, or group of models
- Zone Charge (optional) - one for each zone (and currency where World Trade and Multi-Currency are active)
- Inventory descriptions file
 - DEPT (Departments)
 - MANT (Mantle groupings)
 - MFID (Manufacturer's ID)
 - * - Mfr not specified
 - OTHR (Customer search parameters)
 - PTYP (Item type)
 - 4 codes are essential to categorise parts:
 - C - Consumables
 - J - Major spares
 - N - Minor spares
 - S - Supply items
 - 2 codes assist analysis of machine sales:
 - M - Machine sale via SOP
 - P - Peripheral sale via SOP
 - SPED (Urgency indicator)
 - STKU (Stockroom usage code)
 - B - Service spares stockroom
 - W - Service workshop: company stock
 - X - Service workshop: customer stock

Major Data

- Contract Records (optional) - one for each active service contract, with details of each piece of equipment covered and service schedules.
- Customer Account Records - one for each customer account.
- Customer Additional Details (Workshop Management) Records - one for each customer site.
- Installation Address Records - one for each site address, within a customer account.
- Installation Records - for each customer site, details of each piece of equipment at the site.